

COMPLETE DOCUMENTATION

Medium Format

Record Shop Management Platform

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Confidential

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SPECIFICATION

Platform Design Specification

Medium Format — Platform Design Specification

Date: 2026-04-28 Version: 1.0 Status: Draft — pending user approval

1. Overview

Medium Format is a full-stack record shop management platform built for an Indonesian independent record store of the same name. It serves as the single backoffice for managing inventory, orders, customers, payments, and shipping across all sales channels — online store, offline POS, Indonesian marketplaces (Tokopedia, Shopee), and international platforms (Discogs, Bandcamp, Meta, eBay, Google Merchant).

The platform is inspired by common-ground.io but extended significantly for the Indonesian market, with local payment gateways, local couriers, two-way marketplace sync, wholesale management, and a Juno.co.uk-style audio preview player on the storefront.

2. Core Modules

2.1 Inventory Manager

- Centralized catalog: single source of truth for all channels
- Import from Discogs API (release metadata, tracklist, artwork, label, catalogue number auto-populated)
- Bulk import via CSV
- Fields per release: artist, title, label, cat#, year, format, genre, condition (M/VG+/VG/G+/G/F/P), price (IDR), stock count, notes, images
- IDR pricing displayed throughout; USD equivalent shown as secondary
- **Barcode generator:** generate and print barcode labels per item (EAN-13 or Code128); printable label sheets
- **Barcode scanning via any phone:** mobile-optimised web scanner (camera-based, no dedicated hardware required); also supports dedicated Bluetooth barcode scanners
- **Item location tracking:**
 - Store Location: e.g. "Main Store", "Warehouse", "Consignment"
 - In-store location: rack/shelf reference, e.g. "Rack A3 / Shelf 2" — free text + optional predefined location list
- Low stock threshold alerts (configurable per item)
- Categories: Vinyl, CD, Merch & Accessories
- Filter and search: by artist, title, label, format, genre, condition, channel listing status, preview availability, location

2.2 Track Preview System

- System queries Bandcamp, Spotify Search API, and YouTube Data API using artist name + release title + catalogue number
- Best match is set as the active source automatically
- All found sources are stored and available as fallbacks
- Sources ranked: Bandcamp > Spotify > YouTube > SoundCloud

Manual override:

- Staff can paste any Bandcamp, Spotify, YouTube, SoundCloud URL, or upload a direct .mp3
- Override applies at the release level (all tracks use that source) or at the individual track level
- Per-track source selector in the backoffice release editor
- "Test preview" button per track before publishing

Customer-facing storefront:

- Hover over album art → play button appears
- Click → tracklist expands below the cover with waveform scrubber
- Per-track play buttons in the expanded tracklist
- Persistent Now Playing bar pinned to the bottom of the page (track title, release, source label, progress bar, prev/next)
- "Has audio preview" filter in the storefront

2.3 Orders

- Unified order inbox across all channels
- Statuses: Pending → Processing → Packed → Shipped → Delivered → Completed / Cancelled / Refunded
- Filter by channel (Website, POS, Tokopedia, Shopee, Discogs, etc.)
- Order detail: items, customer, shipping address, payment method, courier + tracking number
- Walk-in POS orders recorded from the iOS/Android app

2.4 Customers

- Customer profiles with full purchase history
- Wishlist tracking
- Customer groups (Retail, VIP, Wholesale)
- Wholesale accounts: separate login, wholesale-only catalogue view, tiered pricing

2.5 Analytics & Reporting

- Dashboard: today's date & time, current revenue, open orders, stock count
- Sales over time chart (3 / 6 / 12 month toggle)
- Recent orders widget (filter: All / Shop / Tokopedia / Shopee / Discogs / Online)
- Sales origins breakdown
- Eshop visitors (Google Analytics integration) + Eshop views (30 / 60 / 90 day toggle)
- Best sellers (tab: All / Releases / Products)
- Top 10 online country sales
- Bandcamp news feed + Discogs news feed (replaces CG News)
- Sales channel performance widget
- CSV export, filterable by date range and channel

2.6 Purchase Orders (Procurement)

Full procurement workflow for stocking the store from international suppliers.

PO Creation:

- Add items from inventory catalog (or free-text for new items not yet in catalog)
- Specify quantity, expected unit cost (IDR or foreign currency), supplier/seller
- Link to existing Discogs/Bandcamp/label catalog entry where available
- Group POs by country of origin (e.g. Japan batch, UK batch, US batch)

Payment Tracking:

- Record payment method per PO: company credit card or personal credit card (employee name)
- Finance notification: automatic alert to finance role when a personal card is used (triggers reimbursement request)
- Payment status: Pending / Paid / Reimbursed / Disputed
- Attach receipt/invoice scan (PDF or image upload)

Shipment Consolidation:

- Group multiple POs from the same country into one shipment batch
- Compile all invoices in a batch into a unified packing list
- Estimated weight & dimension calculator per item (using predefined format weights: LP ~180g, CD ~100g, etc., overridable)
- Packing list shows: item list, quantities, declared value, total weight, estimated dimensions
- **AI-assisted (Phase 2):** upload multiple invoice PDFs → AI extracts line items → auto-populates packing list → generates shipper email with packing list + weight summary → sends to shipper contact to request quote and ETA

Export:

- Export packing list to PDF
- Export packing list to Google Sheets / CSV
- Export PO summary to PDF (for internal records)

Shipper Communication:

- Shipper contact book (name, email, WhatsApp, country coverage)
- Draft email to shipper with packing list attached
- Log of sent emails and shipper responses (ETA, quote)

3. Sales Channels

3.1 Own Website (Eshop)

- Customizable theme (dark default, light mode via toggle)
- Custom domain support
- Genre/format/condition/price filters
- Discogs-enriched release pages (full tracklist, label info, artwork)
- Track preview player (Juno-style, see §2.2)
- "Notify Me" on sold-out items
- SEO-optimised release URLs

3.2 POS — Offline Store

- Barcode scanning for checkout
- Walk-in order recording
- Real-time stock sync back to backoffice
- Works offline with sync-on-reconnect

3.3 Tokopedia Integration

- OAuth connection via Tokopedia Seller API
- Two-way sync:
 - **Push:** stock levels, prices (with platform fee markup — see §3.5), new listings, delisted items
 - **Pull:** new orders, buyer details, order status updates
- Sync frequency: every 5 minutes (configurable)
- Auto-push stock on every sale across any channel
- Auto-fulfill pulled orders (optional toggle — off by default)
- Conflict resolution UI for price/stock mismatches

3.4 Shopee Integration

- OAuth connection via Shopee Open Platform API
- Two-way sync (same as Tokopedia above)
- Separate platform fee configuration (see §3.5)
- Independent sync status and activity log

3.5 Marketplace Price Rules (Tokopedia & Shopee)

- Each marketplace has an independent price rule configuration:
 - **Markup %** — e.g. add 5% to cover Tokopedia admin fee + payment fee
 - **Fixed markup (IDR)** — e.g. always add Rp 5,000 for packaging
 - **Combined** — markup % applied first, then fixed amount added
- Price rules are applied automatically when stock is pushed to that channel
- Backoffice always stores and displays the base (own website) price
- Marketplace prices are computed on push; visible in the channel sync panel
- Staff can preview computed prices before pushing
- Per-release manual price override per channel still possible (takes precedence over the rule)
- **Rule change trigger:** when a price rule is saved, a confirmation prompt asks whether to immediately re-push all affected listings to that channel or apply only to future pushes

3.6 Discogs

- Inventory import (catalog data, not for active listing sync)
- Marketplace listing: list items for sale on Discogs from backoffice
- Pull Discogs orders into unified order inbox
- International payments via Stripe/PayPal for Discogs orders

3.7 Bandcamp

- Import digital sales data (for reporting)
- Bandcamp links used as track preview source

3.8 Meta (Facebook & Instagram Marketplace)

- Stock sync (push only)
- Order notifications pulled into backoffice

3.9 eBay

- Product listing and stock push
- Order pull into backoffice

3.10 Google Merchant

- Product feed sync (title, price, image, availability)
- Google Search, Google Maps, YouTube Shopping listings
- Structured data for SEO

4. Payments

4.1 Xendit (Local — Indonesia)

Used for all domestic (IDR) transactions on the own website and offline POS.

Method	Details
---	---
Virtual Account	BCA, BNI, BRI, Mandiri, Permata
E-Wallet	GoPay, OVO, DANA, ShopeePay, LinkAja
QRIS	Universal QR code
Credit/Debit Card	Visa, Mastercard via Xendit
Paylater	Kredivo, Akulaku
Retail	Indomaret, Alfamart

- VA numbers generated per order with 24-hour expiry
- Payment status webhook updates order automatically
- PPN (11%) tax calculation configurable per product type

4.2 Stripe (International)

- Used for Discogs orders and international website buyers
- Cards: Visa, Mastercard, Amex, JCB, UnionPay
- Digital wallets: Apple Pay, Google Pay

4.3 PayPal (International)

- Alternative for international buyers who prefer PayPal
- Available on website checkout alongside Stripe

5. Shipping

5.1 Local Couriers — via Biteship Aggregator

Biteship provides a single API integration for all Indonesian couriers:

- JNE (REG, YES, OKE)
- J&T Express
- SiCepat (REG, BEST, Cargo)
- Anteraja
- Ninja Xpress
- GoSend (Same Day, Instant)
- GrabExpress

Features:

- Real-time rate calculation at checkout (based on origin postcode → destination postcode + weight/dimensions)
- Automatic tracking number registration
- Tracking status updates pulled into order detail
- Packaging presets (sleeve, mailer, box) with preset weights/dimensions

5.2 International Shipping

- **DHL Express** — API rate calculation + label generation
- **FedEx International** — API rate calculation
- **EMS POS Indonesia** — for slower/cheaper international parcels
- Dimensional weight calculation
- Customs declaration fields (HS code, declared value) on checkout for international orders

5.3 Shipping Zones

- Domestic zones: Java, Sumatra, Kalimantan, Sulawesi, Papua + remote area surcharge
- International zones: ASEAN, Asia-Pacific, Europe, Americas, Rest of World
- Free shipping threshold configurable per zone

6. Wholesale / B2B

- Separate wholesale catalogue (subset or full catalog, configurable)
- Tiered price lists: e.g. Tier 1 (10% off), Tier 2 (15% off), assigned per wholesale account
- Wholesale customer login (gated — must be approved by staff)
- Minimum order quantity (MOQ) per item
- Invoice generator with PDF export and NET payment terms
- Credit terms per account (e.g. NET 30)
- Wholesale orders tracked separately from retail in analytics

7. Marketing

7.1 Vouchers

- Generate voucher codes with configurable validity dates
- Types: percentage discount or fixed IDR amount
- Single-use or multi-use (with optional redemption limit)
- Assign to specific customer groups (Retail, Wholesale, VIP)
- Restrict to specific product categories or minimum order value
- Voucher usage report

7.2 Newsletter (AI-assisted)

- Campaign creation from:
 - New stock arrivals (auto-pull from recent Inventory additions)
 - Featured/highlighted releases (manually curated)
 - Upcoming events
- **AI campaign draft:** staff selects items → AI generates email copy and subject line
- Drag-and-drop email builder with Medium Format branding
- Schedule send or send immediately
- Subscriber list management (import CSV, opt-in from eshop)
- Open rate / click rate analytics per campaign

7.3 Other Marketing Tools

- Discount codes (percentage or fixed IDR, single-use or multi-use)
- New Arrivals automated feed
- WhatsApp Business broadcast integration (for new arrivals, order notifications)
- Instagram feed sync for product posts

7.4 Editorial Blog

A staff-written editorial section on the storefront. Serves as content marketing, SEO, and community building.

Post types:

- **Staff Picks** — curated selection of records hand-picked by a staff member, with personal notes per release and a linked inventory item (live buy button if in stock, "sold out" if not)
- **Highlights** — editorial spotlight on a release, label, or artist — shorter than a full feature, great for social sharing
- **News** — store news, announcements, restocks, collaborations
- **New Arrivals** — editorial write-up of a fresh batch arrival
- **Reviews** — long-form review of a specific release
- **Features** — artist interviews, label spotlights, genre deep-dives
- **Events** — upcoming in-store events, listening parties, markets

Backoffice — Post Editor:

- Rich text editor (block-based: headings, paragraphs, quotes, images, embedded tracks)
- Post type selector (Staff Picks / New Arrivals / Review / Feature / Event)
- Author field (linked to staff profile: name, photo, role e.g. "Vinyl Dept.")
- Featured image upload
- **Staff Picks specific:** inline release picker — search inventory by title/artist → attach release →

- SEO fields: meta title, meta description, slug (auto-generated from post title, editable)
- Status: Draft / Scheduled / Published
- Schedule publish datetime

Storefront — Blog:

- `/journal` — blog index page: grid of post cards (featured image, post type badge, title, author, date)
- `/journal/[slug]` — individual post page
- Filter by post type (All / Staff Picks / Reviews / Features / Events)
- **Staff Picks post layout:** intro text → release cards grid (cover art, artist, title, staff note, buy button or "Sold Out")
- Related posts widget at bottom (same post type or same author)
- Social share buttons (WhatsApp, Instagram, X/Twitter)
- SEO-optimised: Open Graph tags, structured data (Article schema)

Storefront — Homepage integration:

- "Latest from the Journal" section: 3 most recent posts
- "Staff Picks" widget: latest Staff Picks post with featured releases

8. UI/UX

8.1 Theme System

- Dark theme: default for backoffice and storefront
- Light theme: toggled via a pill switch in the top nav bar
- Preference stored in `localStorage` per user/browser
- Implementation: CSS custom properties (design tokens) — single stylesheet, no duplication
- Both themes cover: backoffice and customer-facing storefront

8.2 Backoffice Layout

- Fixed left sidebar navigation (collapsible on small screens)
- 11 top-level modules (see §2 and §3)
- Topbar: store name + location, date, user avatar
- Persistent notification badge on Orders for pending items

8.3 Storefront Layout

- Top navigation with category links, search, cart
- Filter sidebar: format, genre, condition, price range, "has preview"
- Release grid: responsive, covers with hover play button
- Track preview: expands inline below card when playing, waveform scrubber shown
- Now Playing bar: fixed bottom, persists while browsing
- Mobile-responsive throughout

9. Technology Considerations

Concern	Approach
---	---
Frontend (backoffice)	React + TypeScript
Frontend (storefront)	Next.js (SSR for SEO)
Backend API	Node.js / Express or NestJS
Database	PostgreSQL
Real-time sync	WebSockets or Server-Sent Events for channel sync status
Xendit integration	Xendit Node SDK + webhooks
Biteship integration	Biteship REST API
Discogs integration	Discogs API (OAuth 1.0a)
Track preview lookup	Bandcamp oEmbed API + embed player, Spotify Web API, YouTube Data API v3
Auth	JWT + refresh tokens; separate wholesale customer auth
Hosting	Cloud (Vercel for storefront, Railway/Render/AWS for API)

10. User Roles & Permissions

Module / Action	Admin	Store Manager / Purchasing	Shopkeeper	Wholesaler	Customer
---	---	---	---	---	---
Dashboard (full)	Yes	Yes	Read only	No	No
Inventory — view	Yes	Yes	Yes	Wholesale cat. only	No
Inventory — edit/add	Yes	Yes	No	No	No
Inventory — barcode/location	Yes	Yes	Yes	No	No
Purchase Orders — create/edit	Yes	Yes	No	No	No
Purchase Orders — finance view	Yes	Read only	No	No	No
Orders — all channels	Yes	Yes	Yes	Own orders only	Own orders only
Orders — POS walk-in	Yes	Yes	Yes	No	No
Customers — all	Yes	Yes	No	No	No
Vouchers — create/edit	Yes	Yes	No	No	No
Vouchers — use at checkout	Yes	Yes	Yes	Yes	Yes
Newsletter — create/send	Yes	Yes	No	No	No
Channels — connect/configure	Yes	No	No	No	No
Payments — configure	Yes	No	No	No	No
Shipping — configure	Yes	Yes	No	No	No
Wholesale catalogue	Yes	Yes	No	Yes (view+order)	No
Wholesale pricing & tiers	Yes	Yes	No	No	No
Analytics — full	Yes	Yes	No	No	No
Eshop editor	Yes	No	No	No	No
Staff accounts — manage	Yes	No	No	No	No
Preferences / Settings	Yes	No	No	No	No

- **Admin:** Full access to everything. Typically the owner.
- **Store Manager / Purchasing:** Full operational access including purchase orders, inventory, orders, and shipping. Cannot change integrations or staff accounts.
- **Shopkeeper:** Can process walk-in POS sales and view inventory/orders. Cannot edit inventory or access finance.
- **Wholesaler:** Separate login portal. Sees wholesale catalogue with their assigned tier pricing. Can place and track their own orders.
- **Customer:** Standard retail customer. Sees the public storefront. Can track their own orders and manage their wishlist.

11. Out of Scope (v1)

- Native mobile app for customers (web-only storefront in v1)
 - Accounting/bookkeeping integration (Jurnal, Accurate)
 - Multi-location warehouse management
 - Consignment tracking
 - Automated repricing engine
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11. Out of Scope (v1)

- Native mobile app for customers (web-only storefront in v1)
 - Accounting/bookkeeping integration (Jurnal, Accurate)
 - Multi-location warehouse management
 - Consignment tracking
 - Automated repricing engine
 - AI-assisted shipment consolidation (Phase 2 — spec in §2.6)
-

12. Open Questions

(resolved during brainstorming — documented for reference)

1. **Dark/light theme?** → Both, with toggle. Dark is default.
 2. **Track preview source?** → Auto-linked on import, manual override available per release and per track.
 3. **Tokopedia/Shopee sync direction?** → Two-way. Conflicts surfaced for manual resolution.
 4. **Marketplace pricing?** → Independent price rules per marketplace (% markup + fixed IDR amount) to cover platform fees. Base price stored in backoffice; computed price pushed to channel.
-

Design reference: Common Ground admin UI (Werkstoff Club screenshot). No emoji icons in the UI — use thin line SVG icons throughout. Dark theme default. *Mockups saved in*

.superpowers/brainstorm/ (9 screens + additional: architecture, nav structure, dashboard,

inventory, storefront, theme comparison, channel sync, checkout, track preview manager, updated dashboard, purchase orders, inventory-location, user roles).

02

BUDGET

Infrastructure & Budget

Medium Format — Infrastructure & Subscription Budget

Date: 2026-04-28 **Version:** 1.0 **Status:** Approved **Scope:** Full production system — server, plugins, subscriptions **Stack:** Option A (Managed PaaS) with Dokploy + Hetzner, OpenRouter AI **Exchange rate reference:** 1 USD ≈ IDR 16,300

1. One-Time Setup Costs

Item	USD	IDR	Notes
-----	----- -	-----	-----
Domain registration (.com)	\$10	IDR 163,000	Via Cloudflare Registrar — lowest renewal price, no markup
Hetzner account + VPS provisioning	\$0	IDR 0	Free to create
Dokploy installation	\$0	IDR 0	Open-source, self-hosted on Hetzner VPS
Xendit merchant onboarding	\$0	IDR 0	Requires KTP, NPWP, SIUP — no fee
Stripe account	\$0	IDR 0	Free
PayPal Business account	\$0	IDR 0	Free
Biteship account	\$0	IDR 0	Free
Discogs API key	\$0	IDR 0	Free
Spotify Developer app	\$0	IDR 0	Free
YouTube Data API (Google Cloud project)	\$0	IDR 0	Free quota — no billing until exceeded
Meta Business + Commerce API	\$0	IDR 0	Free, requires business verification
eBay Developer account	\$0	IDR 0	Free
Google Merchant Center	\$0	IDR 0	Free
WhatsApp Business verification	\$0	IDR 0	Via Meta Business Suite — free
Fonnte WhatsApp provider setup	\$0	IDR 0	Free account creation
OpenRouter account	\$0	IDR 0	Free, pay-per-use only
TOTAL SETUP	\$10	IDR 163,000	Domain is the only day-one cost

2. Fixed Monthly Subscriptions

2.1 Infrastructure

Service	Plan	USD/month	IDR/month	Purpose
-----	-----	-----	-----	-----
Hetzner CX32	VPS	~\$9	IDR 147,000	Runs Dokploy — hosts Next.js storefront + Node.js API + background sync workers. 4 vCPU, 8GB RAM, 80GB NVMe SSD
Hetzner Snapshots	Usage	~\$2	IDR 33,000	Automated VPS backups (~\$0.01/GB/month)
Neon	Launch	\$19	IDR 310,000	PostgreSQL — 10GB storage, scales to zero when idle
Cloudflare R2	Usage	~\$8	IDR 130,000	File storage: record cover images, PO receipt scans. No egress fees
Upstash Redis	Free	\$0	IDR 0	Session cache + job queues. Upgrade at scale
Cloudflare	Free	\$0	IDR 0	DNS management, CDN (compensates for no Vercel edge), DDoS protection, SSL certificates
Subtotal		~\$38	IDR ~620,000	

2.2 Communication

Service	Plan	USD/month	IDR/month	Purpose
-----	-----	-----	-----	-----
-				
Resend	Free	\$0	IDR 0	Transactional email: order confirmations, shipping updates. 3,000 emails/month — sufficient at launch. Upgrade to Starter (\$20) when order volume demands
Brevo	Starter	\$9	IDR 147,000	Newsletter campaigns — 5,000 emails/month. Upgrade to \$25 plan when subscriber list exceeds 500
Fonnte	Standard	~\$6	IDR 100,000	WhatsApp Business broadcast: new arrivals, order notifications
Subtotal		~\$15	IDR ~247,000	

2.3 AI

Service	Plan	USD/month	IDR/month	Purpose
-----	-----	-----	-----	-----
OpenRouter	Pay-per-use	~\$3	IDR 49,000	Single API key for all AI tasks. Free models (Llama 3.3 70B, Mistral) used for newsletter drafting and subject lines. Claude Haiku or Llama 3.2 Vision for PO invoice PDF parsing
Subtotal		~\$3	IDR ~49,000	

OpenRouter model routing:

Task	Model	Cost
-----	-----	-----
Newsletter copy drafting	Meta Llama 3.3 70B (free)	\$0
Email subject line generation	Mistral 7B (free)	\$0
PO invoice PDF parsing	Llama 3.2 Vision (free) or Claude Haiku 3.5	~\$0–3

2.4 Shipping APIs

Service	Plan	USD/month	IDR/month	Purpose
-----	-----	-----	-----	-----
Biteship	Starter	~\$9	IDR 149,000	Rate calculation + tracking for all Indonesian couriers (JNE, J&T, SiCepat, Anteraja, GoSend, GrabExpress). 1,000 queries/month
DHL Express API	Free	\$0	IDR 0	International rate calculation. No monthly fee — per-shipment charges only
FedEx API	Free	\$0	IDR 0	International rate calculation. No monthly fee
Subtotal		~\$9	IDR ~149,000	

2.5 Dev & Monitoring (all free)

Service	Plan	USD/month	Purpose
-----	-----	-----	-----
GitHub	Free	\$0	Version control, CI/CD via GitHub Actions
Sentry	Free	\$0	Error monitoring — 5,000 errors/month, sufficient for v1
UptimeRobot	Free	\$0	Uptime monitoring + email/WhatsApp alerts

Fixed Monthly Total

Category	USD/month	IDR/month
-----	-----	-----
Infrastructure	~\$38	IDR ~620,000
Communication	~\$15	IDR ~247,000
AI	~\$3	IDR ~49,000
Shipping APIs	~\$9	IDR ~149,000
Dev & Monitoring	\$0	IDR 0
TOTAL	~\$65/month	IDR ~1,065,000/month
Annual	~\$780/year	IDR ~12,780,000/year

3. Variable & Transaction Fees

These scale with sales volume. Not a fixed monthly cost — budget as a percentage of GMV.

3.1 Payment Gateways

Gateway	Fee	When
-----	-----	-----
Xendit — Virtual Account	IDR 4,500 flat	Per domestic VA payment (BCA, BNI, BRI, Mandiri, Permata)
Xendit — E-Wallet	1.5%	GoPay, OVO, DANA, ShopeePay, LinkAja
Xendit — QRIS	0.7%	Walk-in + online QRIS scan
Xendit — Credit/Debit Card	2.9%	Domestic card payments
Xendit — Paylater	3–4%	Kredivo, Akulaku
Xendit — Retail	IDR 5,000 flat	Indomaret, Alfamart
Stripe	2.9% + \$0.30	International card, Apple Pay, Google Pay
PayPal	3.49% + \$0.49	International PayPal buyers

3.2 APIs with Free Quotas (no cost under normal usage)

Service	Free Quota	Exceeds at
-----	-----	-----
YouTube Data API v3	10,000 units/day	~5,000 track lookups/day — well above typical catalog import rate
Spotify Web API	Unlimited for catalog lookup	No quota concern
Discogs API	60 requests/minute (authenticated)	Sufficient for import workflows
Meta Commerce API	Rate-limited, not metered	No cost
eBay API	Free standard tier	No cost
Google Merchant Center	Free	No cost

4. Budget-Effective Decisions

Key choices that reduce cost without capability loss:

Decision	Chosen	Avoided	Monthly Saving
-----	-----	-----	----- --
Hosting	Dokploy on Hetzner CX32 (~\$11)	Vercel Pro + Railway (~\$35)	~\$24
File storage	Cloudflare R2 (no egress fees)	AWS S3 + CloudFront	~\$20–40
CDN	Cloudflare free (already in stack)	Vercel edge network (bundled with Pro)	—
WhatsApp	Fonnte IDR 100k/month	Twilio WABA (~\$50/month)	~\$43
Newsletter	Brevo Starter (\$9)	Mailchimp Essentials (\$26 for 500 contacts)	~\$17
Transactional email	Resend free tier (3k/month)	SendGrid Essentials (\$19.95)	~\$20
AI	OpenRouter free models (Llama, Mistral)	Direct Anthropic API (\$15+/month)	~\$12
Domain renewal	Cloudflare Registrar (~\$9/year)	GoDaddy/Namecheap (~\$15–18/year)	~\$6/year
Database	Neon (scales to zero when idle)	Supabase Pro (\$25)	~\$6
Monitoring	Sentry free + UptimeRobot free	Datadog (\$34+/month)	~\$34

5. Upgrade Triggers

Services to stay on free tier and upgrade only when these conditions are hit:

Service	Upgrade when	To	Cost increase
-----	-----	-----	----- --
Resend	Orders exceed ~100/month (3k transactional emails)	Starter \$20/month	+\$20
Brevo	Subscriber list > 500	Business \$25/month	+\$16
Upstash Redis	Job queue depth or session cache misses become a problem	Pay-as-you-go ~\$10/month	+\$10
Biteship	Shipment queries exceed 1,000/month	Business IDR 299,000 (~\$18)	+\$9
Neon	Database storage exceeds 10GB	Scale plan \$69/month	+\$50
Hetzner VPS	CPU or RAM consistently above 80%	CX42 (8 vCPU, 16GB) ~\$18/month	+\$9

6. Client Pricing (Indonesian Market)

6.1 Project Fees — Phased Fixed Price

Phase	Scope	IDR	USD equiv.
-----	-----	-----	----- --
Phase 1 — Foundation	Inventory + barcode, orders, own website storefront, Xendit + Stripe, Biteship local couriers, Discogs import, track preview player	IDR 45,000,000	~\$2,760
Phase 2 — Channels	Tokopedia + Shopee sync, Meta + eBay + Google Merchant, PayPal, international shipping (DHL/FedEx), purchase orders module	IDR 40,000,000	~\$2,450
Phase 3 — Full Platform	POS mobile app (iOS + Android), wholesale/B2B module, AI newsletter + PO invoice parsing, WhatsApp broadcast, marketing tools	IDR 35,000,000	~\$2,150
Total	Full production as per spec	IDR 120,000,000	~\$7,360

Payment terms: 50% upfront per phase, 50% on milestone delivery.

6.2 Monthly Retainer (Post-Launch)

Tier	IDR/month	USD equiv.	Includes
-----	-----	----- -	-----
Basic	IDR 2,500,000	~\$153	Bug fixes, security patches, server upkeep — up to 8h/month
Standard (recommended)	IDR 4,000,000	~\$245	Above + minor feature additions — up to 14h/month

Infrastructure costs (IDR 1,065,000/month) are billed as a pass-through on top of the retainer, or bundled into the Standard tier.

6.3 Value Justification

Frame the project cost against ongoing SaaS alternatives the client would otherwise pay:

Tool replaced by Medium Format	Monthly SaaS cost (IDR)
-----	-----
Shopify Plus equivalent	IDR 850,000–2,600,000
Tokopedia/Shopee sync tools	IDR 300,000–700,000
Newsletter platform	IDR 200,000–500,000
Shipping aggregator	IDR 149,000
Wholesale portal	IDR 300,000–600,000
Total SaaS equivalent	IDR 1,800,000–5,000,000/month

At IDR 120,000,000 one-time, the platform pays for itself in **2–5 years** vs ongoing SaaS — and the client owns it completely with no vendor lock-in.

7. Summary

	USD	IDR
-	-----	-----
One-time setup	~\$10	IDR ~163,000
Monthly (fixed)	~\$65/month	IDR ~1,065,000/month
Annual (fixed)	~\$780/year	IDR ~12,780,000/year
Transaction fees	% of GMV	% of GMV
Client project fee (total)	~\$7,360	IDR 120,000,000
Client retainer (post-launch)	~\$245/month	IDR 4,000,000/month

All third-party API accounts (Discogs, Spotify, YouTube, Meta, eBay, Google Merchant, Tokopedia, Shopee) are free to register and use within standard rate limits. No setup cost beyond the domain.

03

PLAN 1

Plan 1 — Foundation

Medium Format — Plan 1: Foundation Implementation Plan

For agentic workers: REQUIRED SUB-SKILL: Use superpowers:subagent-driven-development (recommended) or superpowers:executing-plans to implement this plan task-by-task. Steps use checkbox (- []) syntax for tracking.

Goal: Build the core platform — monorepo, database, auth, inventory, orders, payments (Xendit + Stripe), shipping (Biteship), Discogs import, track preview, backoffice shell, and storefront shell.

Architecture: Turborepo monorepo with three apps (api, backoffice, storefront) and two shared packages (db, types). The API is NestJS with Prisma ORM connecting to Neon PostgreSQL. The backoffice is React + Vite + TailwindCSS. The storefront is Next.js 14 with SSR for SEO. **Tech Stack:** Node.js 20, NestJS, Prisma, PostgreSQL (Neon), React 18, Next.js 14, TypeScript, TailwindCSS, Vitest, Jest, Cloudflare R2, Upstash Redis, Xendit Node SDK, Stripe Node SDK, Biteship REST API, Discogs API, JWT.

File Structure

mediumformat/

└─ apps/

| └─ api/

| | └─ src/

| | | └─ main.ts

| | | └─ app.module.ts

| | | └─ auth/

| | | | └─ auth.module.ts

| | | | └─ auth.service.ts

| | | | └─ auth.controller.ts

| | | | └─ jwt.strategy.ts

| | | | └─ guards/roles.guard.ts

| | | └─ inventory/

| | | | └─ inventory.module.ts

| | | | └─ inventory.service.ts

| | | | └─ inventory.controller.ts

| | | | └─ dto/

| | | └─ orders/

| | | | └─ orders.module.ts

| | | | └─ orders.service.ts

| | | | └─ orders.controller.ts

```
| | | └─ payments/

| | | | └─ xendit/

| | | | | └─ xendit.module.ts

| | | | | └─ xendit.service.ts

| | | | | └─ xendit.controller.ts ← webhook handler

| | | | └─ stripe/

| | | | └─ stripe.module.ts

| | | | └─ stripe.service.ts

| | | | └─ stripe.controller.ts ← webhook handler

| | | └─ shipping/

| | | | └─ biteship/

| | | | | └─ biteship.module.ts

| | | | | └─ biteship.service.ts

| | | └─ discogs/

| | | | └─ discogs.module.ts

| | | | └─ discogs.service.ts

| | | └─ preview/

| | | | └─ preview.module.ts

| | | | └─ preview.service.ts

| | | └─ storage/
```

```
| | | └─ common/

| | |   └─ decorators/

| | |   └─ filters/

| | └─ test/

| | └─ package.json

| └─ backoffice/

| | └─ src/

| | | └─ main.tsx

| | | └─ App.tsx

| | | └─ layouts/AppLayout.tsx

| | | └─ components/ui/      ← shared UI primitives

| | | └─ pages/

| | | | └─ Dashboard.tsx

| | | | └─ Inventory/

| | | |   └─ Orders/

| | | └─ stores/             ← Zustand global state

| | | └─ hooks/              ← React Query hooks per module

| | |   └─ api/              ← typed API client (axios)

| | └─ package.json

| └─ storefront/
```

```
|   |   └─ layout.tsx

|   |   └─ page.tsx           ← homepage

|   |   └─ shop/

|   |   |   └─ page.tsx       ← release grid

|   |   |   └─ [slug]/page.tsx ← release detail

|   |   └─ cart/page.tsx

|   |   └─ checkout/page.tsx

|   └─ components/

|   |   └─ NowPlayingBar.tsx

|   |   └─ ReleaseCard.tsx

|   |   └─ TrackPlayer.tsx

|   └─ package.json

└─ packages/

|   └─ db/

|   |   └─ prisma/

|   |   |   └─ schema.prisma

|   |   └─ src/index.ts       ← exports PrismaClient

|   |   └─ package.json

|   └─ types/

|   └─ src/index.ts           ← shared TS types/enums
```



```
├─ turbo.json

├─ package.json

└─ .env.example
```

Task 1: Monorepo Scaffold

Files:

- Create: `package.json` (root)
- Create: `turbo.json`
- Create: `.env.example`
- Create: `apps/api/package.json`
- Create: `apps/backoffice/package.json`
- Create: `apps/storefront/package.json`
- Create: `packages/db/package.json`
- Create: `packages/types/package.json`

☐ Step 1: Initialise Turborepo monorepo

```
npx create-turbo@latest mediumformat --package-manager npm
cd mediumformat
```

☐ Step 2: Remove starter apps, scaffold correct structure

```
rm -rf apps/web apps/docs
mkdir -p apps/api apps/backoffice apps/storefront packages/db packages/types
```

☐ Step 3: Create API app with NestJS

```
cd apps/api
npx @nestjs/cli new . --package-manager npm --skip-git
```

☐ Step 4: Create backoffice with Vite + React + TypeScript

```
cd apps/backoffice

npm create vite@latest . -- --template react-ts

npm install

npm install -D tailwindcss postcss autoprefixer

npx tailwindcss init -p
```

☐ **Step 5: Create storefront with Next.js**

```
cd apps/storefront

npx create-next-app@latest . --typescript --tailwind --app --no-src-dir --import-alias "@/*
```

☐ **Step 6: Create packages/db**

```
cd packages/db

npm init -y

npm install prisma @prisma/client

npx prisma init
```

☐ **Step 7: Create packages/types**

```
cd packages/types

npm init -y

npm install -D typescript
```

Write `packages/types/src/index.ts` :

```
export enum UserRole {  
  
  ADMIN = 'ADMIN',  
  
  STORE_MANAGER = 'STORE_MANAGER',  
  
  SHOPKEEPER = 'SHOPKEEPER',  
  
  WHOLESALE = 'WHOLESALE',  
  
  CUSTOMER = 'CUSTOMER',  
  
}
```

```
export enum OrderStatus {  
  
  PENDING = 'PENDING',  
  
  PROCESSING = 'PROCESSING',  
  
  PACKED = 'PACKED',  
  
  SHIPPED = 'SHIPPED',  
  
  DELIVERED = 'DELIVERED',  
  
  COMPLETED = 'COMPLETED',  
  
  CANCELLED = 'CANCELLED',  
  
  REFUNDED = 'REFUNDED',  
  
}
```

```
export enum RecordCondition {  
  
  M = 'M',  
  
  VG_PLUS = 'VG_PLUS',  
  
}
```

```
G_PLUS = 'G_PLUS',

G = 'G',

F = 'F',

P = 'P',

}

export enum RecordFormat {

    VINYL = 'VINYL',

    CD = 'CD',

    MERCH = 'MERCH',

    ACCESSORY = 'ACCESSORY',

}

export enum PaymentMethod {

    VIRTUAL_ACCOUNT = 'VIRTUAL_ACCOUNT',

    EWALLET = 'EWALLET',

    QRIS = 'QRIS',

    CREDIT_CARD = 'CREDIT_CARD',

    PAYLATER = 'PAYLATER',

    RETAIL = 'RETAIL',

    STRIPE = 'STRIPE',
```

```
}

export enum PreviewSource {

    BANDCAMP = 'BANDCAMP',

    SPOTIFY = 'SPOTIFY',

    YOUTUBE = 'YOUTUBE',

    SOUNDCLOUD = 'SOUNDCLOUD',

    UPLOAD = 'UPLOAD',

}
```

☐ Step 8: Create `.env.example`

```
cat > .env.example << 'EOF'
```

```
DATABASE_URL="postgresql://user:password@localhost:5432/mediumformat"
```

```
JWT_SECRET="change-me-in-production"
```

```
JWT_EXPIRES_IN="15m"
```

```
JWT_REFRESH_EXPIRES_IN="7d"
```

```
R2_ACCOUNT_ID=""
```

```
R2_ACCESS_KEY_ID=""
```

```
R2_SECRET_ACCESS_KEY=""
```

```
R2_BUCKET_NAME="mediumformat"
```

```
R2_PUBLIC_URL=""
```

```
XENDIT_SECRET_KEY=""
```

```
XENDIT_WEBHOOK_TOKEN=""
```

STRIPE_SECRET_KEY=""

STRIPE_WEBHOOK_SECRET=""

BITESHIP_API_KEY=""

DISCOGS_CONSUMER_KEY=""

DISCOGS_CONSUMER_SECRET=""

DISCOGS_USER_TOKEN=""

SPOTIFY_CLIENT_ID=""

SPOTIFY_CLIENT_SECRET=""

YOUTUBE_API_KEY=""

```
OPENROUTER_API_KEY=""
```

```
FONNTE_TOKEN=""
```

```
UPSTASH_REDIS_URL=""
```

```
UPSTASH_REDIS_TOKEN=""
```

```
API_URL="http://localhost:3001"
```

```
BACKOFFICE_URL="http://localhost:3000"
```

```
STOREFRONT_URL="http://localhost:3002"
```

```
EOF
```

☐ Step 9: Configure turbo.json


```
{
  "$schema": "https://turbo.build/schema.json",

  "pipeline": {

    "build": {

      "dependsOn": ["^build"],

      "outputs": ["dist/", ".next/"]

    },

    "dev": {

      "cache": false,

      "persistent": true

    },

    "test": {

      "dependsOn": ["^build"],

      "outputs": []

    },

    "lint": {

      "outputs": []

    }

  }
}
```

```
npm run dev
```

Expected: All three apps start on ports 3000, 3001, 3002 without errors.

☐ **Step 11: Commit**

```
git add .  
  
git commit -m "feat: scaffold Turborepo monorepo with api, backoffice, storefront, db, types packages"
```

Task 2: Database Schema

Files:

- Modify: `packages/db/prisma/schema.prisma`
- Create: `packages/db/src/index.ts`

☐ **Step 1: Write Prisma schema**

```
// packages/db/prisma/schema.prisma

generator client {

  provider = "prisma-client-js"

}

datasource db {

  provider = "postgresql"

  url      = env("DATABASE_URL")

}

model User {

  id          String   @id @default(cuid())

  email       String   @unique

  passwordHash String

  name        String

  role        UserRole @default(CUSTOMER)

  avatarUrl   String?

  createdAt   DateTime @default(now())

  updatedAt   DateTime @updatedAt

  orders      Order[]

  wishlists   Wishlist[]

  blogPosts   BlogPost[]
}
```

```
    @map("users")

}

enum UserRole {

    ADMIN

    STORE_MANAGER

    SHOPKEEPER

    WHOLESALE

    CUSTOMER

}

model Release {

    id          String      @id @default(cuid())

    title       String

    artist      String

    label       String?

    catalogNumber String?

    year        Int?

    format      RecordFormat

    genre       String?

    condition   RecordCondition
}
```

```
stock          Int          @default(0)

notes          String?

slug           String       @unique

discogsId      String?      @unique

images         String[]

barcode        String?      @unique

storeLocation  String?

shelfLocation  String?

lowStockThreshold Int       @default(2)

isActive       Boolean      @default(true)

weight         Int?         // grams

createdAt      DateTime     @default(now())

updatedAt      DateTime     @updatedAt

previewSources PreviewTrack[]

orderItems     OrderItem[]

wishlists      Wishlist[]

staffPickItems StaffPickItem[]

channelListings ChannelListing[]

@@@map("releases")
```

```
enum RecordFormat {
```

```
    VINYL
```

```
    CD
```

```
    MERCH
```

```
    ACCESSORY
```

```
}
```

```
enum RecordCondition {
```

```
    M
```

```
    VG_PLUS
```

```
    VG
```

```
    G_PLUS
```

```
    G
```

```
    F
```

```
    P
```

```
}
```

```
model PreviewTrack {
```

```
    id          String      @id @default(cuid())
```

```
    releaseId   String
```

```
    release     Release     @relation(fields: [releaseId], references: [id])
```

```
trackTitle String?

source PreviewSource

url String

isActive Boolean @default(false)

createdAt DateTime @default(now())

@@map("preview_tracks")

}
```

```
enum PreviewSource {

    BANDCAMP

    SPOTIFY

    YOUTUBE

    SOUNDCLOUD

    UPLOAD

}
```

```
model Order {

    id String @id @default(cuid())

    orderNumber String @unique

    customerId String?

    customer User? @relation(fields: [customerId], references: [id])

}
```

```
customerEmail    String

customerPhone    String?

shippingAddress  Json

status           OrderStatus @default(PENDING)

channel          SalesChannel @default(WEBSITE)

paymentMethod    PaymentMethod?

paymentStatus    PaymentStatus @default(UNPAID)

paymentRef       String?

subtotalIdr      Int

shippingIdr      Int          @default(0)

discountIdr      Int          @default(0)

totalIdr         Int

courierId        String?

trackingNumber   String?

notes           String?

createdAt        DateTime      @default(now())

updatedAt        DateTime      @updatedAt

items            OrderItem[]

@@map("orders")
```



```
model OrderItem {

    id          String  @id @default(cuid())

    orderId     String

    order       Order   @relation(fields: [orderId], references: [id])

    releaseId   String

    release     Release @relation(fields: [releaseId], references: [id])

    quantity    Int

    priceIdr    Int

    @@map("order_items")

}
```

```
enum OrderStatus {
```

```
    PENDING
```

```
    PROCESSING
```

```
    PACKED
```

```
    SHIPPED
```

```
    DELIVERED
```

```
    COMPLETED
```

```
    CANCELLED
```

```
    REFUNDED
```

```
enum SalesChannel {
```

```
    WEBSITE
```

```
    POS
```

```
    TOKOPEDIA
```

```
    SHOPEE
```

```
    DISCOGS
```

```
    META
```

```
    EBAY
```

```
}
```

```
enum PaymentMethod {
```

```
    VIRTUAL_ACCOUNT
```

```
    EWALLET
```

```
    QRIS
```

```
    CREDIT_CARD
```

```
    PAYLATER
```

```
    RETAIL
```

```
    STRIPE
```

```
    PAYPAL
```

```
}
```

```

    UNPAID

    PAID

    EXPIRED

    FAILED

    REFUNDED

}

```

```

model Wishlist {

    id          String    @id @default(cuid())

    userId      String

    user        User      @relation(fields: [userId], references: [id])

    releaseId   String

    release     Release    @relation(fields: [releaseId], references: [id])

    createdAt   DateTime   @default(now())

    @@unique([userId, releaseId])

    @@map("wishlists")

}

```

```

model BlogPost {

    id          String    @id @default(cuid())

    slug        String    @unique

```

```

type      BlogPostType

status    BlogPostStatus @default(DRAFT)

authorId   String

author     User           @relation(fields: [authorId], references: [id])

featuredImage String?

content    Json           // block-based rich text (Plate.js format)

excerpt    String?

metaTitle   String?

metaDescription String?

publishedAt DateTime?

scheduledAt DateTime?

createdAt   DateTime      @default(now())

updatedAt   DateTime      @updatedAt

staffPickItems StaffPickItem[]

@@map("blog_posts")

}

enum BlogPostType {

    STAFF_PICKS

    HIGHLIGHTS

```

```
NEW_ARRIVALS

REVIEW

FEATURE

EVENT

}

enum BlogPostStatus {

    DRAFT

    SCHEDULED

    PUBLISHED

}

model StaffPickItem {

    id          String    @id @default(cuid())

    postId      String

    post        BlogPost @relation(fields: [postId], references: [id])

    releaseId   String

    release     Release   @relation(fields: [releaseId], references: [id])

    staffNote   String?

    sortOrder   Int       @default(0)

    @@map("staff_pick_items")
}
```

```

model ChannelListing {

    id          String          @id @default(cuid())

    releaseId   String

    release     Release         @relation(fields: [releaseId], references: [id])

    channel     SalesChannel

    externalId  String?

    priceIdr    Int?           // computed price for that channel; null = use base price

    isActive    Boolean        @default(true)

    lastSyncedAt DateTime?

    syncError   String?

    @@unique([releaseId, channel])

    @@map("channel_listings")

}

model Setting {

    key         String @id

    value       Json

    updatedAt   DateTime @updatedAt

    @@map("settings")
}

```

☐ Step 2: Run migration

```
cd packages/db  
npx prisma migrate dev --name init
```

Expected: Migration created and applied. Prisma Client generated.

☐ Step 3: Export PrismaClient from package

Write `packages/db/src/index.ts`:

```
import { PrismaClient } from '@prisma/client'  
  
const globalForPrisma = globalThis as unknown as {  
  
  prisma: PrismaClient | undefined  
  
}  
  
export const prisma =  
  
  globalForPrisma.prisma ??  
  
  new PrismaClient({  
  
    log: process.env.NODE_ENV === 'development' ? ['query', 'error', 'warn'] : ['error'],  
  
  })  
  
if (process.env.NODE_ENV !== 'production') globalForPrisma.prisma = prisma  
  
export { PrismaClient } from '@prisma/client'  
  
export * from '@prisma/client'
```

☐ Step 4: Commit

```
git add packages/db/

git commit -m "feat: define full Prisma schema with all models and enums"
```

Task 3: Auth Module

Files:

- Create: `apps/api/src/auth/auth.module.ts`
- Create: `apps/api/src/auth/auth.service.ts`
- Create: `apps/api/src/auth/auth.controller.ts`
- Create: `apps/api/src/auth/jwt.strategy.ts`
- Create: `apps/api/src/auth/guards/jwt-auth.guard.ts`
- Create: `apps/api/src/auth/guards/roles.guard.ts`
- Create: `apps/api/src/auth/decorators/roles.decorator.ts`
- Test: `apps/api/test/auth.service.spec.ts`

☐ Step 1: Install auth dependencies

```
cd apps/api

npm install @nestjs/jwt @nestjs/passport passport passport-jwt bcrypt

npm install -D @types/passport-jwt @types/bcrypt
```

☐ Step 2: Write failing test

Create `apps/api/test/auth.service.spec.ts` :


```
import { Test } from '@nestjs/testing'

import { AuthService } from '../src/auth/auth.service'

import { JwtService } from '@nestjs/jwt'

import { PrismaService } from '../src/prisma/prisma.service'

describe('AuthService', () => {

  let service: AuthService

  const mockPrisma = {

    user: {

      findUnique: jest.fn(),

      create: jest.fn(),

    },

  }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        AuthService,

        { provide: PrismaService, useValue: mockPrisma },

        { provide: JwtService, useValue: { sign: jest.fn(() => 'token') } },

      ],

    }).compile()
```

```

    service = module.get(AuthService)

  })

  it('returns null for unknown email', async () => {

    mockPrisma.user.findUnique.mockResolvedValue(null)

    const result = await service.validateUser('no@user.com', 'pass')

    expect(result).toBeNull()

  })

  it('returns null for wrong password', async () => {

    mockPrisma.user.findUnique.mockResolvedValue({

      id: '1',

      email: 'a@b.com',

      passwordHash: '$2b$10$invalidhash',

      role: 'ADMIN',

    })

    const result = await service.validateUser('a@b.com', 'wrongpass')

    expect(result).toBeNull()

  })

})

```

```
cd apps/api && npm test -- auth.service
```

Expected: FAIL — `AuthService` not found.

☐ Step 4: Create PrismaService wrapper

Create `apps/api/src/prisma/prisma.service.ts` :

```
import { Injectable, OnModuleInit } from '@nestjs/common'
import { PrismaClient } from '@mediumformat/db'

@Injectable()

export class PrismaService extends PrismaClient implements OnModuleInit {

  async onModuleInit() {

    await this.$connect()

  }

}
```

☐ Step 5: Implement AuthService

Create `apps/api/src/auth/auth.service.ts` :

```

import { Injectable } from '@nestjs/common'

import { JwtService } from '@nestjs/jwt'

import { PrismaService } from '../prisma/prisma.service'

import * as bcrypt from 'bcrypt'

import { User } from '@mediumformat/db'

@Injectable()

export class AuthService {

  constructor(

    private prisma: PrismaService,

    private jwt: JwtService,

  ) {}

  async validateUser(email: string, password: string): Promise | null> {

    const user = await this.prisma.user.findUnique({ where: { email } })

    if (!user) return null

    const valid = await bcrypt.compare(password, user.passwordHash)

    if (!valid) return null

    const { passwordHash, ...result } = user

    return result

  }

  login(user: Omit) {

```

```

const payload = { sub: user.id, email: user.email, role: user.role }

return {

  accessToken: this.jwt.sign(payload, { expiresIn: process.env.JWT_EXPIRES_IN }),

  refreshToken: this.jwt.sign(payload, { expiresIn: process.env.JWT_REFRESH_EXPIRES_IN }),

  user,

}

}

}

async register(email: string, password: string, name: string) {

  const passwordHash = await bcrypt.hash(password, 10)

  const user = await this.prisma.user.create({

    data: { email, passwordHash, name },

  })

  const { passwordHash: _, ...result } = user

  return this.login(result)

}

}

```

☐ Step 6: Implement JWT strategy and guards

Create `apps/api/src/auth/jwt.strategy.ts` :

```

import { Injectable } from '@nestjs/common'
import { PassportStrategy } from '@nestjs/passport'

import { ExtractJwt, Strategy } from 'passport-jwt'

@Injectable()

export class JwtStrategy extends PassportStrategy(Strategy) {

  constructor() {

    super({

      jwtFromRequest: ExtractJwt.fromAuthHeaderAsBearerToken(),

      secretOrKey: process.env.JWT_SECRET!,

    })

  }

  validate(payload: { sub: string; email: string; role: string }) {

    return { id: payload.sub, email: payload.email, role: payload.role }

  }

}

```

Create `apps/api/src/auth/guards/jwt-auth.guard.ts` :

```
import { Injectable } from '@nestjs/common'
import { AuthGuard } from '@nestjs/passport'

@Injectable()

export class JwtAuthGuard extends AuthGuard('jwt') {}
```

Create `apps/api/src/auth/decorators/roles.decorator.ts` :

```
import { SetMetadata } from '@nestjs/common'
import { UserRole } from '@mediumformat/types'

export const ROLES_KEY = 'roles'

export const Roles = (...roles: UserRole[]) => SetMetadata(ROLES_KEY, roles)
```

Create `apps/api/src/auth/guards/roles.guard.ts` :

```

import { Injectable, CanActivate, ExecutionContext } from '@nestjs/common'
import { Reflector } from '@nestjs/core'

import { ROLES_KEY } from '../decorators/roles.decorator'

import { UserRole } from '@mediumformat/types'

@Injectable()

export class RolesGuard implements CanActivate {

  constructor(private reflector: Reflector) {}

  canActivate(context: ExecutionContext): boolean {

    const requiredRoles = this.reflector.getAllAndOverride(ROLES_KEY, [

      context.getHandler(),

      context.getClass(),

    ])

    if (!requiredRoles) return true

    const { user } = context.switchToHttp().getRequest()

    return requiredRoles.includes(user.role)

  }

}

```

☐ Step 7: Create AuthController

Create `apps/api/src/auth/auth.controller.ts` :


```

import { Controller, Post, Body, UnauthorizedException } from '@nestjs/common'

import { AuthService } from '../auth.service'

@Controller('auth')

export class AuthController {

  constructor(private auth: AuthService) {}

  @Post('login')

  async login(@Body() body: { email: string; password: string }) {

    const user = await this.auth.validateUser(body.email, body.password)

    if (!user) throw new UnauthorizedException('Invalid credentials')

    return this.auth.login(user)

  }

  @Post('register')

  register(@Body() body: { email: string; password: string; name: string }) {

    return this.auth.register(body.email, body.password, body.name)

  }

}

```

☐ Step 8: Run test to verify it passes

```
cd apps/api && npm test -- auth.service
```

Expected: PASS — 2 tests passing.

```
git add apps/api/src/auth/ apps/api/src/prisma/ apps/api/test/auth.service.spec.ts
git commit -m "feat: add JWT auth module with roles guard"
```

Task 4: Inventory Module (API)

Files:

- Create: `apps/api/src/inventory/inventory.module.ts`
- Create: `apps/api/src/inventory/inventory.service.ts`
- Create: `apps/api/src/inventory/inventory.controller.ts`
- Create: `apps/api/src/inventory/dto/create-release.dto.ts`
- Test: `apps/api/test/inventory.service.spec.ts`

☐ **Step 1: Write failing test**

Create `apps/api/test/inventory.service.spec.ts` :

```
import { Test } from '@nestjs/testing'

import { InventoryService } from '../src/inventory/inventory.service'

import { PrismaService } from '../src/prisma/prisma.service'

const mockRelease = {

  id: 'rel-1',

  title: 'Kind of Blue',

  artist: 'Miles Davis',

  format: 'VINYL',

  condition: 'VG_PLUS',

  priceIdr: 350000,

  stock: 2,

  slug: 'miles-davis-kind-of-blue',

}

describe('InventoryService', () => {

  let service: InventoryService

  const mockPrisma = {

    release: {

      findMany: jest.fn(),

      findUnique: jest.fn(),

      create: jest.fn(),
```

```

        delete: jest.fn(),

    },

}

beforeEach(async () => {

    const module = await Test.createTestingModule({

        providers: [

            InventoryService,

            { provide: PrismaService, useValue: mockPrisma },

        ],

    }).compile()

    service = module.get(InventoryService)

})

it('lists all active releases', async () => {

    mockPrisma.release.findMany.mockResolvedValue([mockRelease])

    const result = await service.findAll({})

    expect(result).toHaveLength(1)

    expect(mockPrisma.release.findMany).toHaveBeenCalledWith(

        expect.objectContaining({ where: expect.objectContaining({ isActive: true }) }),

    )

```

```

it('generates slug from artist and title', async () => {

  mockPrisma.release.create.mockResolvedValue(mockRelease)

  await service.create({

    title: 'Kind of Blue',

    artist: 'Miles Davis',

    format: 'VINYL' as any,

    condition: 'VG_PLUS' as any,

    priceIdr: 350000,

    stock: 2,

  })

  expect(mockPrisma.release.create).toHaveBeenCalledWith(

    expect.objectContaining({

      data: expect.objectContaining({ slug: 'miles-davis-kind-of-blue' }),

    }),

  )

})

})

```

☐ Step 2: Run test to verify it fails

```
cd apps/api && npm test -- inventory.service
```

Expected: FAIL — `InventoryService` not found.

☐ **Step 3: Implement InventoryService**

Create `apps/api/src/inventory/inventory.service.ts` :

```
import { Injectable } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

import { RecordFormat, RecordCondition } from '@mediumformat/db'

import slugify from 'slugify'

export interface CreateReleaseDto {

  title: string

  artist: string

  label?: string

  catalogNumber?: string

  year?: number

  format: RecordFormat

  genre?: string

  condition: RecordCondition

  priceIdr: number

  stock: number

  notes?: string

  images?: string[]

  barcode?: string

  storeLocation?: string

  shelfLocation?: string
```

```
weight?: number

discogsId?: string

}

@Injectable()

export class InventoryService {

  constructor(private prisma: PrismaService) {}

  findAll(filters: {

    format?: RecordFormat

    condition?: RecordCondition

    genre?: string

    search?: string

    hasPreview?: boolean

  }) {

    return this.prisma.release.findMany({

      where: {

        isActive: true,

        ...(filters.format && { format: filters.format }),

        ...(filters.condition && { condition: filters.condition }),

        ...(filters.genre && { genre: filters.genre }),
```



```

    OR: [

      { title: { contains: filters.search, mode: 'insensitive' } },

      { artist: { contains: filters.search, mode: 'insensitive' } },

      { label: { contains: filters.search, mode: 'insensitive' } },

    ],

  }),

  ...(filters.hasPreview && {

    previewSources: { some: { isActive: true } },

  }),

},

include: { previewSources: true },

orderBy: { createdAt: 'desc' },

  })

}

findOne(id: string) {

  return this.prisma.release.findUnique({

    where: { id },

    include: { previewSources: true, channelListings: true },

  })

}

```

```
findBySlug(slug: string) {

  return this.prisma.release.findUnique({

    where: { slug },

    include: { previewSources: { where: { isActive: true } } },

  })

}

create(dto: CreateReleaseDto) {

  const slug = slugify(`${dto.artist} ${dto.title}`, { lower: true, strict: true })

  return this.prisma.release.create({

    data: { ...dto, slug },

  })

}

update(id: string, dto: Partial) {

  return this.prisma.release.update({ where: { id }, data: dto })

}

softDelete(id: string) {

  return this.prisma.release.update({ where: { id }, data: { isActive: false } })

}

adjustStock(id: string, delta: number) {
```

```
    return this.prisma.release.update({  
  
      where: { id },  
  
      data: { stock: { increment: delta } },  
  
    })  
  
  }  
  
}
```

☐ **Step 4: Run test to verify it passes**

```
cd apps/api && npm test -- inventory.service
```

Expected: PASS — 2 tests passing.

☐ **Step 5: Implement InventoryController**

Create `apps/api/src/inventory/inventory.controller.ts`:

```

import { Controller, Get, Post, Patch, Delete, Param, Body, Query, UseGuards } from '@nestjs/common'

import { InventoryService, CreateReleaseDto } from '../inventory.service'

import { JwtAuthGuard } from '../auth/guards/jwt-auth.guard'

import { RolesGuard } from '../auth/guards/roles.guard'

import { Roles } from '../auth/decorators/roles.decorator'

import { UserRole } from '@mediumformat/types'

import { RecordFormat, RecordCondition } from '@mediumformat/db'

@Controller('inventory')

export class InventoryController {

  constructor(private inventory: InventoryService) {}

  @Get()

  findAll(

    @Query('format') format?: RecordFormat,

    @Query('condition') condition?: RecordCondition,

    @Query('genre') genre?: string,

    @Query('search') search?: string,

    @Query('hasPreview') hasPreview?: string,

  ) {

    return this.inventory.findAll({ format, condition, genre, search, hasPreview: hasPreview === 'true' })
  }

```

```
@Get('/:id')

findOne(@Param('id') id: string) {

    return this.inventory.findOne(id)

}

@UseGuards(JwtAuthGuard, RolesGuard)

@Roles(UserRole.ADMIN, UserRole.STORE_MANAGER)

@Post()

create(@Body() dto: CreateReleaseDto) {

    return this.inventory.create(dto)

}

@UseGuards(JwtAuthGuard, RolesGuard)

@Roles(UserRole.ADMIN, UserRole.STORE_MANAGER)

@Patch('/:id')

update(@Param('id') id: string, @Body() dto: Partial) {

    return this.inventory.update(id, dto)

}

@UseGuards(JwtAuthGuard, RolesGuard)

@Roles(UserRole.ADMIN, UserRole.STORE_MANAGER)

@Delete('/:id')
```

```
    remove(@Param('id') id: string) {  
  
        return this.inventory.softDelete(id)  
  
    }  
  
}
```

☐ Step 6: Commit

```
git add apps/api/src/inventory/ apps/api/test/inventory.service.spec.ts  
git commit -m "feat: add inventory module with CRUD, search, stock adjustment"
```

Task 5: Discogs Import Service

Files:

- Create: `apps/api/src/discogs/discogs.service.ts`
- Create: `apps/api/src/discogs/discogs.module.ts`
- Test: `apps/api/test/discogs.service.spec.ts`

☐ Step 1: Install dependencies

```
cd apps/api && npm install axios
```

☐ Step 2: Write failing test

Create `apps/api/test/discogs.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { DiscogsService } from '../src/discogs/discogs.service'

import { HttpService } from '@nestjs/axios'

import { of } from 'rxjs'

describe('DiscogsService', () => {

  let service: DiscogsService

  const mockHttp = { get: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        DiscogsService,

        { provide: HttpService, useValue: mockHttp },

      ],

    }).compile()

    service = module.get(DiscogsService)

  })

  it('maps Discogs release to CreateReleaseDto', async () => {

    mockHttp.get.mockReturnValue(of({

      data: {

        id: 12345,

```

```
    title: 'Kind of Blue',

    artists: [{ name: 'Miles Davis' }],

    labels: [{ name: 'Columbia', catno: 'CL 1355' }],

    year: 1959,

    formats: [{ name: 'Vinyl' }],

    genres: ['Jazz'],

    images: [{ uri: 'https://discogs.com/img.jpg', type: 'primary' }],

  },

}))
```

```
const result = await service.fetchRelease('12345')

expect(result.title).toBe('Kind of Blue')

expect(result.artist).toBe('Miles Davis')

expect(result.label).toBe('Columbia')

expect(result.catalogNumber).toBe('CL 1355')

expect(result.year).toBe(1959)

expect(result.format).toBe('VINYL')

expect(result.discogsId).toBe('12345')

})

})
```


☐ **Step 3: Run test to verify it fails**

```
cd apps/api && npm test -- discogs.service
```

Expected: FAIL.

☐ **Step 4: Implement DiscogsService**

Create `apps/api/src/discogs/discogs.service.ts` :

```

import { Injectable } from '@nestjs/common'
import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

import { RecordFormat } from '@mediumformat/db'

import { CreateReleaseDto } from '../inventory/inventory.service'

@Injectable()

export class DiscogsService {

  private readonly baseUrl = 'https://api.discogs.com'

  constructor(private http: HttpService) {}

  private get headers() {

    return {

      Authorization: `Discogs token=${process.env.DISCOGS_USER_TOKEN}`,

      'User-Agent': 'MediumFormat/1.0',

    }

  }

  async fetchRelease(discogsId: string): Promise & { discogsId: string } > {

    const { data } = await firstValueFrom(

      this.http.get(`${this.baseUrl}/releases/${discogsId}`, { headers: this.headers }),

    )

```

```

const formatName = data.formats?.[0]?.name?.toLowerCase() ?? ''

const format: RecordFormat =

    formatName.includes('vinyl') || formatName.includes('lp') ? 'VINYL'

    : formatName.includes('cd') ? 'CD'

    : 'VINYL'

const primaryImage = data.images?.find((i: any) => i.type === 'primary')?.uri

    ?? data.images?.[0]?.uri

return {

    discogsId: String(data.id),

    title: data.title,

    artist: data.artists?.map((a: any) => a.name.replace(/\s*(\d+)\$/, '')).join(', '),

    label: data.labels?.[0]?.name,

    catalogNumber: data.labels?.[0]?.catno,

    year: data.year,

    format,

    genre: data.genres?.[0],

    images: primaryImage ? [primaryImage] : [],

}

}

```

```

const { data } = await firstValueFrom(

  this.http.get(`${this.baseUrl}/database/search`, {

    headers: this.headers,

    params: { q: query, type: 'release', per_page: 10 },

  )),

)

return data.results

}

}

```

☐ Step 5: Run test to verify it passes

```
cd apps/api && npm test -- discogs.service
```

Expected: PASS.

☐ Step 6: Commit

```
git add apps/api/src/discogs/ apps/api/test/discogs.service.spec.ts
git commit -m "feat: add Discogs import service with release mapping"
```

Task 6: Track Preview Service

Files:

- Create: `apps/api/src/preview/preview.service.ts`
- Create: `apps/api/src/preview/preview.module.ts`
- Test: `apps/api/test/preview.service.spec.ts`

☐ Step 1: Write failing test

```

import { Test } from '@nestjs/testing'

import { PreviewService } from '../src/preview/preview.service'

import { HttpService } from '@nestjs/axios'

import { PrismaService } from '../src/prisma/prisma.service'

import { of } from 'rxjs'

describe('PreviewService', () => {

  let service: PreviewService

  const mockHttp = { get: jest.fn() }

  const mockPrisma = {

    previewTrack: { createMany: jest.fn(), updateMany: jest.fn() },

  }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        PreviewService,

        { provide: HttpService, useValue: mockHttp },

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(PreviewService)

```

```

})

it('returns ranked sources: BANDCAMP first', async () => {

  const sources = [

    { source: 'YOUTUBE', url: 'https://youtube.com/x' },

    { source: 'BANDCAMP', url: 'https://artist.bandcamp.com/x' },

    { source: 'SPOTIFY', url: 'https://open.spotify.com/x' },

  ]

  const ranked = service.rankSources(sources as any)

  expect(ranked[0].source).toBe('BANDCAMP')

  expect(ranked[1].source).toBe('SPOTIFY')

  expect(ranked[2].source).toBe('YOUTUBE')

})

})

```

☐ Step 2: Run test to verify it fails

```
cd apps/api && npm test -- preview.service
```

Expected: FAIL.

☐ Step 3: Implement PreviewService

Create `apps/api/src/preview/preview.service.ts` :

```
import { Injectable } from '@nestjs/common'

import { HttpService } from '@nestjs/axios'

import { PrismaService } from '../prisma/prisma.service'

import { firstValueFrom } from 'rxjs'

import { PreviewSource } from '@mediumformat/db'
```

```
const SOURCE_RANK: Record = {
```

```
  BANDCAMP: 0,
```

```
  SPOTIFY: 1,
```

```
  YOUTUBE: 2,
```

```
  SOUNDCLLOUD: 3,
```

```
  UPLOAD: 4,
```

```
}
```

```
interface PreviewCandidate {
```

```
  source: PreviewSource
```

```
  url: string
```

```
  trackNumber?: number
```

```
  trackTitle?: string
```

```
}
```

```
@Injectable()
```

```
export class PreviewService {
```

```

constructor(

    private http: HttpService,

    private prisma: PrismaService,

) {}

rankSources(sources: PreviewCandidate[]): PreviewCandidate[] {

    return [...sources].sort((a, b) ⇒ SOURCE_RANK[a.source] - SOURCE_RANK[b.source])

}

async autolink(releaseId: string, artist: string, title: string, catalogNumber?: string) {

    const found: PreviewCandidate[] = []

    const query = `${artist} ${title}${catalogNumber ? ' ' + catalogNumber : ''}`

    // Spotify search

    try {

        const token = await this.getSpotifyToken()

        const { data } = await firstValueFrom(

            this.http.get('https://api.spotify.com/v1/search', {

                headers: { Authorization: Bearer ${token} },

                params: { q: query, type: 'album', limit: 1 },

            }),

        )
    }

```



```
    if (album) {

        found.push({ source: 'SPOTIFY', url: album.external_urls.spotify })

    }

} catch (_) {}

// YouTube search

try {

    const { data } = await firstValueFrom(

        this.http.get('https://www.googleapis.com/youtube/v3/search', {

            params: {

                key: process.env.YOUTUBE_API_KEY,

                q: query,

                type: 'video',

                maxResults: 1,

                part: 'id',

            },

        )),

    )

    const videoId = data.items?.[0]?.id?.videoId

    if (videoId) {

        found.push({ source: 'YOUTUBE', url: https://www.youtube.com/watch?v=${videoId} })

    }

}
```

```

    }

    } catch (_) {}

    if (found.length === 0) return []

    const ranked = this.rankSources(found)

    await this.prisma.previewTrack.createMany({

      data: ranked.map((s, i) => ({

        releaseId,

        source: s.source,

        url: s.url,

        trackNumber: 0,

        isActive: i === 0,

      })),

      skipDuplicates: true,

    })

    return ranked

  }

  async setActiveSource(releaseId: string, previewTrackId: string) {

    await this.prisma.previewTrack.updateMany({

```

```

        data: { isActive: false },

    })

    return this.prisma.previewTrack.updateMany({

        where: { id: previewTrackId, releaseId },

        data: { isActive: true },

    })

}

private async getSpotifyToken(): Promise {

    const creds = Buffer.from(

        `${process.env.SPOTIFY_CLIENT_ID}:${process.env.SPOTIFY_CLIENT_SECRET}`,

    ).toString('base64')

    const { data } = await firstValueFrom(

        this.http.post(

            'https://accounts.spotify.com/api/token',

            'grant_type=client_credentials',

            { headers: { Authorization: Basic ${creds}, 'Content-Type': 'application/x-www-form-urlencoded' } }

        ),

    )

    return data.access_token

```

```
}
```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- preview.service
```

Expected: PASS.

☐ Step 5: Commit

```
git add apps/api/src/preview/ apps/api/test/preview.service.spec.ts
git commit -m "feat: add track preview auto-link service with Spotify + YouTube search"
```

Task 7: Orders Module (API)

Files:

- Create: `apps/api/src/orders/orders.service.ts`
- Create: `apps/api/src/orders/orders.controller.ts`
- Test: `apps/api/test/orders.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/orders.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { OrdersService } from '../src/orders/orders.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('OrdersService', () => {

  let service: OrdersService

  const mockPrisma = {

    order: { create: jest.fn(), findMany: jest.fn(), findUnique: jest.fn(), update: jest.fn() },

    release: { findUnique: jest.fn() },

    $transaction: jest.fn(),

  }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        OrdersService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(OrdersService)

  })

  it('generates sequential order number', () => {

```

```

    const num = service.generateOrderNumber()

    expect(num).toMatch(/^MF-\d{8}-\d{4}$/)

  })

  it('calculates correct total', () => {

    const items = [

      { priceId: 200000, quantity: 2 },

      { priceId: 150000, quantity: 1 },

    ]

    expect(service.calcSubtotal(items)).toBe(550000)

  })

})

```

☐ Step 2: Implement OrdersService

Create `apps/api/src/orders/orders.service.ts` :

```

import { Injectable, BadRequestException } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

import { OrderStatus, SalesChannel } from '@mediumformat/db'

@Injectable()

export class OrdersService {

  constructor(private prisma: PrismaService) {}

  generateOrderNumber(): string {

    const date = new Date().toISOString().slice(0, 10).replace(/-/g, '')

    const rand = Math.floor(Math.random() * 9000 + 1000)

    return MF-${date}-${rand}

  }

  calcSubtotal(items: { priceIdr: number; quantity: number }[]): number {

    return items.reduce((sum, i) => sum + i.priceIdr * i.quantity, 0)

  }

  async create(dto: {

    customerName: string

    customerEmail: string

    customerPhone?: string

    customerId?: string

```

```

channel: SalesChannel

items: { releaseId: string; quantity: number }[]

shippingIdr?: number

discountIdr?: number

}) {

return this.prisma.$transaction(async (tx) => {

const itemsWithPrice = await Promise.all(

dto.items.map(async (item) => {

const release = await tx.release.findUnique({ where: { id: item.releaseId } })

if (!release) throw new BadRequestException(Release ${item.releaseId} not found)

if (release.stock < item.quantity) throw new BadRequestException(Insufficient stock for
${release.title})

return { ...item, priceIdr: release.priceIdr }

}),

)

// Decrement stock

await Promise.all(

itemsWithPrice.map((item) =>

tx.release.update({

where: { id: item.releaseId },

```



```
    }},

    ),

)

const subtotal = this.calcSubtotal(itemsWithPrice)

const shipping = dto.shippingIdr ?? 0

const discount = dto.discountIdr ?? 0

return tx.order.create({

  data: {

    orderNumber: this.generateOrderNumber(),

    customerName: dto.customerName,

    customerEmail: dto.customerEmail,

    customerPhone: dto.customerPhone,

    customerId: dto.customerId,

    shippingAddress: dto.shippingAddress,

    channel: dto.channel,

    subtotalIdr: subtotal,

    shippingIdr: shipping,

    discountIdr: discount,

    totalIdr: subtotal + shipping - discount,
```

```

        create: itemsWithPrice.map((i) => ({

          releaseId: i.releaseId,

          quantity: i.quantity,

          priceIdr: i.priceIdr,

        })),

      },

    },

    include: { items: { include: { release: true } } },

  })

})

}

findAll(filters: { status?: OrderStatus; channel?: SalesChannel }) {

  return this.prisma.order.findMany({

    where: {

      ...(filters.status && { status: filters.status }),

      ...(filters.channel && { channel: filters.channel }),

    },

    include: { items: { include: { release: true } } },

    orderBy: { createdAt: 'desc' },

  })
}

```

```

}

findOne(id: string) {

  return this.prisma.order.findUnique({

    where: { id },

    include: { items: { include: { release: true } }, customer: true },

  })

}

updateStatus(id: string, status: OrderStatus) {

  return this.prisma.order.update({ where: { id }, data: { status } })

}

addTracking(id: string, courierId: string, trackingNumber: string) {

  return this.prisma.order.update({

    where: { id },

    data: { courierId, trackingNumber, status: 'SHIPPED' },

  })

}

}

```

☐ **Step 3: Run test to verify it passes**

```
cd apps/api && npm test -- orders.service
```

Expected: PASS.

☐ Step 4: Commit

```
git add apps/api/src/orders/ apps/api/test/orders.service.spec.ts
git commit -m "feat: add orders module with stock decrement transaction and order number generation"
```

Task 8: Xendit Payment Integration

Files:

- Create: `apps/api/src/payments/xendit/xendit.service.ts`
- Create: `apps/api/src/payments/xendit/xendit.controller.ts`
- Test: `apps/api/test/xendit.service.spec.ts`

☐ Step 1: Install Xendit SDK

```
cd apps/api && npm install xendit-node
```

☐ Step 2: Write failing test

Create `apps/api/test/xendit.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { XenditService } from '../src/payments/xendit/xendit.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('XenditService', () => {

  let service: XenditService

  const mockPrisma = {

    order: { update: jest.fn(), findUnique: jest.fn() },

  }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        XenditService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(XenditService)

  })

  it('maps bank code BCA to VIRTUAL_ACCOUNT payment method', () => {

    expect(service.bankToPaymentMethod('BCA')).toBe('VIRTUAL_ACCOUNT')

  })

```

})

☐ Step 3: Implement XenditService

Create `apps/api/src/payments/xendit/xendit.service.ts` :

```

import { Injectable } from '@nestjs/common'

import Xendit from 'xendit-node'

import { PrismaService } from '../../prisma/prisma.service'

const xendit = new Xendit({ secretKey: process.env.XENDIT_SECRET_KEY! })

@Injectable()

export class XenditService {

  constructor(private prisma: PrismaService) {}

  bankToPaymentMethod(_bank: string): string {

    return 'VIRTUAL_ACCOUNT'

  }

  async createVirtualAccount(orderId: string, bankCode: string, amount: number, customerName: string) {

    const va = await xendit.VirtualAcc.createFixedVA({

      externalID: orderId,

      bankCode,

      name: customerName,

      expectedAmt: amount,

      expirationDate: new Date(Date.now() + 24 * 60 * 60 * 1000).toISOString(),

      isClosed: true,

      isSingleUse: true,

```

```
    await this.prisma.order.update({

      where: { id: orderId },

      data: { paymentMethod: 'VIRTUAL_ACCOUNT', paymentRef: va.id },

    })

    return va

  }

  async createEWalletCharge(orderId: string, ewalletType: string, amount: number, phone: string, returnUrl: string) {

    const charge = await xendit.EWallet.createEWalletCharge({

      referenceID: orderId,

      currency: 'IDR',

      amount,

      checkoutMethod: 'ONE_TIME_PAYMENT',

      channelCode: ewalletType as any,

      channelProperties: { successReturnURL: returnUrl, failureReturnURL: returnUrl, mobileNumber: phone },

    })

    await this.prisma.order.update({

      where: { id: orderId },

      data: { paymentMethod: 'EWALLET', paymentRef: charge.id },
```



```
    return charge

  }

  async createQrisCharge(orderId: string, amount: number) {

    const charge = await xendit.QrCode.createQRCode({

      externalID: orderId,

      type: 'DYNAMIC',

      callbackURL: `${process.env.API_URL}/payments/xendit/webhook`,

      amount,

    })

    await this.prisma.order.update({

      where: { id: orderId },

      data: { paymentMethod: 'QRIS', paymentRef: charge.id },

    })

    return charge

  }

  async handleWebhook(event: any) {

    const orderId = event.external_id ?? event.reference_id

    if (!orderId) return
```

```

    if (event.status === 'PAID' || event.status === 'COMPLETED' || event.status === 'SUCCEEDED') {

      await this.prisma.order.update({

        where: { id: orderId },

        data: { paymentStatus: 'PAID', status: 'PROCESSING' },

      })

    } else if (event.status === 'EXPIRED') {

      await this.prisma.order.update({

        where: { id: orderId },

        data: { paymentStatus: 'EXPIRED' },

      })

    }

  }

}

}

```

☐ Step 4: Create Xendit webhook controller

Create `apps/api/src/payments/xendit/xendit.controller.ts` :

```

import { Controller, Post, Body, Headers, UnauthorizedException } from '@nestjs/common'

import { XenditService } from './xendit.service'

@Controller('payments/xendit')

export class XenditController {

  constructor(private xendit: XenditService) {}

  @Post('webhook')

  async webhook(

    @Body() body: any,

    @Headers('x-callback-token') token: string,

  ) {

    if (token !== process.env.XENDIT_WEBHOOK_TOKEN) {

      throw new UnauthorizedException('Invalid webhook token')

    }

    await this.xendit.handleWebhook(body)

    return { received: true }

  }

}

```

☐ Step 5: Run test to verify it passes

```
cd apps/api && npm test -- xendit.service
```

☐ Step 6: Commit

```
git add apps/api/src/payments/xendit/ apps/api/test/xendit.service.spec.ts  
git commit -m "feat: add Xendit payment integration (VA, e-wallet, QRIS) with webhook handler"
```

Task 9: Stripe Payment Integration

Files:

- Create: `apps/api/src/payments/stripe/stripe.service.ts`
- Create: `apps/api/src/payments/stripe/stripe.controller.ts`
- Test: `apps/api/test/stripe.service.spec.ts`

☐ Step 1: Install Stripe SDK

```
cd apps/api && npm install stripe
```

☐ Step 2: Write failing test

Create `apps/api/test/stripe.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { StripeService } from '../src/payments/stripe/stripe.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('StripeService', () => {

  let service: StripeService

  const mockPrisma = { order: { update: jest.fn() } }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        StripeService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(StripeService)

  })

  it('converts IDR to USD cents for Stripe', () => {

    // 1 USD = ~16,300 IDR; function returns cents

    const cents = service.idrToUsdCents(163000)

    expect(cents).toBe(1000) // 163,000 IDR ≈ $10.00 = 1000 cents

  })

```

})

☐ Step 3: Implement StripeService

Create `apps/api/src/payments/stripe/stripe.service.ts` :

```
import { Injectable } from '@nestjs/common'

import Stripe from 'stripe'

import { PrismaService } from '../../prisma/prisma.service'

const IDR_TO_USD = 16300

@Injectable()

export class StripeService {

  private stripe = new Stripe(process.env.STRIPE_SECRET_KEY!, { apiVersion: '2024-04-10' })

  constructor(private prisma: PrismaService) {}

  idrToUsdCents(idr: number): number {

    return Math.round((idr / IDR_TO_USD) * 100)

  }

  async createPaymentIntent(orderId: string, amountIdr: number, customerEmail: string) {

    const intent = await this.stripe.paymentIntents.create({

      amount: this.idrToUsdCents(amountIdr),

      currency: 'usd',

      metadata: { orderId },

      receipt_email: customerEmail,

    })

  }
```

```

        where: { id: orderId },

        data: { paymentMethod: 'STRIPE', paymentRef: intent.id },

    })

    return { clientSecret: intent.client_secret }

}

async handleWebhook(payload: Buffer, signature: string) {

    const event = this.stripe.webhooks.constructEvent(

        payload,

        signature,

        process.env.STRIPE_WEBHOOK_SECRET!,

    )

    if (event.type === 'payment_intent.succeeded') {

        const intent = event.data.object as Stripe.PaymentIntent

        const orderId = intent.metadata.orderId

        await this.prisma.order.update({

            where: { id: orderId },

            data: { paymentStatus: 'PAID', status: 'PROCESSING' },

        })

    }

}

```



```
}
```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- stripe.service
```

Expected: PASS.

☐ Step 5: Commit

```
git add apps/api/src/payments/stripe/ apps/api/test/stripe.service.spec.ts
git commit -m "feat: add Stripe payment integration with IDR-to-USD conversion and webhook"
```

Task 10: Biteship Shipping Integration

Files:

- Create: `apps/api/src/shipping/biteship/biteship.service.ts`
- Test: `apps/api/test/biteship.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/biteship.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { BiteshipService } from '../src/shipping/biteship/biteship.service'

import { HttpService } from '@nestjs/axios'

import { of } from 'rxjs'

describe('BiteshipService', () => {

  let service: BiteshipService

  const mockHttp = { get: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        BiteshipService,

        { provide: HttpService, useValue: mockHttp },

      ],

    }).compile()

    service = module.get(BiteshipService)

  })

  it('returns available courier rates', async () => {

    mockHttp.get.mockReturnValue(of({

      data: {

        pricing: [

```

```

        { courier_name: 'JNE', courier_service_name: 'REG', price: 25000, min_day: 2, max_day: 3
      },

    ],

  },

  )))

const rates = await service.getRates({

  originPostalCode: '10110',

  destinationPostalCode: '60111',

  weightGrams: 500,

})

expect(rates).toHaveLength(1)

expect(rates[0].courierName).toBe('JNE')

expect(rates[0].priceIdr).toBe(25000)

})

})

```

☐ Step 2: Implement BiteshipService

Create `apps/api/src/shipping/biteship/biteship.service.ts` :

```

import { Injectable } from '@nestjs/common'
import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

@Injectable()

export class BiteshipService {

  private readonly baseUrl = 'https://api.biteship.com/v1'

  constructor(private http: HttpService) {}

  private get headers() {

    return { Authorization: Bearer ${process.env.BITESHIP_API_KEY} }

  }

  async getRates(params: {

    originPostalCode: string

    destinationPostalCode: string

    weightGrams: number

    heightCm?: number

    lengthCm?: number

    widthCm?: number

  }) {

    const { data } = await firstValueFrom(

```

```

        headers: this.headers,

        params: {

            origin_postal_code: params.originPostalCode,

            destination_postal_code: params.destinationPostalCode,

            items_weight: params.weightGrams,

            items_height: params.heightCm ?? 5,

            items_length: params.lengthCm ?? 32,

            items_width: params.widthCm ?? 32,

        },

    })),

)

return (data.pricing ?? []).map((p: any) => ({

    courierId: `${p.courier_code}_${p.courier_service_code}`,

    courierName: p.courier_name,

    serviceName: p.courier_service_name,

    priceIdr: p.price,

    minDay: p.min_day,

    maxDay: p.max_day,

}))

}

```

```
async createOrder(params: {

  orderId: string

  courierCode: string

  courierServiceCode: string

  origin: object

  destination: object

  items: object[]

}) {

  const { data } = await firstValueFrom(

    this.http.post(

      `${this.baseUrl}/orders`,

      {

        reference_id: params.orderId,

        courier: { courier_code: params.courierCode, courier_service_code: params.courierServiceCode },

        origin: params.origin,

        destination: params.destination,

        delivery: { type: 'later' },

        items: params.items,

      },

      { headers: this.headers },
```

```

    },

    )

    return { biteshipOrderId: data.id, trackingId: data.courier?.tracking_id }

  }

}

```

☐ Step 3: Run test to verify it passes

```
cd apps/api && npm test -- biteship.service
```

Expected: PASS.

☐ Step 4: Commit

```
git add apps/api/src/shipping/ apps/api/test/biteship.service.spec.ts
git commit -m "feat: add Biteship shipping integration with rate calculation and order creation"
```

Task 11: Cloudflare R2 Storage Service

Files:

- Create: `apps/api/src/storage/r2.service.ts`
- Test: `apps/api/test/r2.service.spec.ts`

☐ Step 1: Install AWS S3 SDK (R2 is S3-compatible)

```
cd apps/api && npm install @aws-sdk/client-s3 @aws-sdk/s3-request-presigner
```

☐ Step 2: Implement R2Service

Create `apps/api/src/storage/r2.service.ts` :

```

import { Injectable } from '@nestjs/common'

import { S3Client, PutObjectCommand, DeleteObjectCommand } from '@aws-sdk/client-s3'

import { getSignedUrl } from '@aws-sdk/s3-request-presigner'

import { randomUUID } from 'crypto'

@Injectable()

export class R2Service {

  private client = new S3Client({

    region: 'auto',

    endpoint: `https://${process.env.R2_ACCOUNT_ID}.r2.cloudflarestorage.com`,

    credentials: {

      accessKeyId: process.env.R2_ACCESS_KEY_ID!,

      secretAccessKey: process.env.R2_SECRET_ACCESS_KEY!,

    },

  })

  private bucket = process.env.R2_BUCKET_NAME!

  private publicUrl = process.env.R2_PUBLIC_URL!

  async upload(file: Buffer, mimeType: string, folder: string): Promise {

    const ext = mimeType.split('/')[1] ?? 'jpg'

    const key = `${folder}/${randomUUID()}.${ext}`

```



```

    new PutObjectCommand({

        Bucket: this.bucket,

        Key: key,

        Body: file,

        ContentType: mimeType,

    }),

)

return `${this.publicUrl}/${key}`

}

async getPresignedUploadUrl(folder: string, mimeType: string): Promise<{ url: string; key: string
}> {

    const ext = mimeType.split('/')[1] ?? 'jpg'

    const key = `${folder}/${randomUUID()}.${ext}`

    const url = await getSignedUrl(

        this.client,

        new PutObjectCommand({ Bucket: this.bucket, Key: key, ContentType: mimeType }),

        { expiresIn: 300 },

    )

    return { url, key }
}

```

```
    async delete(key: string): Promise {  
  
        await this.client.send(new DeleteObjectCommand({ Bucket: this.bucket, Key: key }))  
  
    }  
  
}
```

☐ Step 3: Commit

```
git add apps/api/src/storage/  
git commit -m "feat: add Cloudflare R2 storage service with presigned upload URLs"
```

Task 12: Blog / Editorial Module (API)

Files:

- Create: `apps/api/src/blog/blog.service.ts`
- Create: `apps/api/src/blog/blog.controller.ts`
- Test: `apps/api/test/blog.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/blog.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { BlogService } from '../src/blog/blog.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('BlogService', () => {

  let service: BlogService

  const mockPrisma = {

    blogPost: {

      findMany: jest.fn(),

      findUnique: jest.fn(),

      create: jest.fn(),

      update: jest.fn(),

    },

  }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        BlogService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

```

```

})

it('generates slug from post title', async () => {

  mockPrisma.blogPost.create.mockResolvedValue({ id: '1', slug: 'staff-picks-june-2026' })

  await service.create({

    title: 'Staff Picks June 2026',

    type: 'STAFF_PICKS',

    authorId: 'user-1',

    content: {},

  })

  expect(mockPrisma.blogPost.create).toHaveBeenCalledWith(

    expect.objectContaining({

      data: expect.objectContaining({ slug: 'staff-picks-june-2026' }),

    }),

  )

})

it('returns only published posts for public listing', async () => {

  mockPrisma.blogPost.findMany.mockResolvedValue([])

  await service.findPublished({})

  expect(mockPrisma.blogPost.findMany).toHaveBeenCalledWith(

```

```
        where: expect.objectContaining({ status: 'PUBLISHED' }),

      },

    )

  })

})
```

☐ **Step 2: Run test to verify it fails**

```
cd apps/api && npm test -- blog.service
```

Expected: FAIL.

☐ **Step 3: Implement BlogService**

Create `apps/api/src/blog/blog.service.ts` :

```

import { Injectable } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

import { BlogPostType, BlogPostStatus } from '@mediumformat/db'

import slugify from 'slugify'

interface CreateBlogPostDto {

  title: string

  type: BlogPostType

  authorId: string

  content: object

  excerpt?: string

  featuredImage?: string

  metaTitle?: string

  metaDescription?: string

  scheduledAt?: Date

  staffPickItems?: { releaseId: string; staffNote?: string; sortOrder?: number }[]

}

@Injectable()

export class BlogService {

  constructor(private prisma: PrismaService) {}

  findPublished(filters: { type?: BlogPostType }) {

```

```

return this.prisma.blogPost.findMany({

  where: {

    status: 'PUBLISHED',

    ...(filters.type && { type: filters.type }),

  },

  include: {

    author: { select: { id: true, name: true, avatarUrl: true } },

    staffPickItems: {

      include: { release: { select: { id: true, title: true, artist: true, images: true, stock:
true, priceIdr: true, slug: true } } },

      orderBy: { sortOrder: 'asc' },

    },

  },

  orderBy: { publishedAt: 'desc' },

})

}

findBySlug(slug: string) {

  return this.prisma.blogPost.findUnique({

    where: { slug },

    include: {

```

```

      staffPickItems: {

        include: { release: { select: { id: true, title: true, artist: true, images: true, stock:
true, priceIdr: true, slug: true } } },

        orderBy: { sortOrder: 'asc' },

      },

    },

  })

}

findAllAdmin() {

  return this.prisma.blogPost.findMany({

    include: { author: { select: { name: true } } },

    orderBy: { createdAt: 'desc' },

  })

}

create(dto: CreateBlogPostDto) {

  const slug = slugify(dto.title, { lower: true, strict: true })

  const { staffPickItems, ...rest } = dto

  return this.prisma.blogPost.create({

    data: {

      ...rest,

```



```

    slug,

    ...(staffPickItems && {

      staffPickItems: {

        create: staffPickItems.map((item) => ({

          releaseId: item.releaseId,

          staffNote: item.staffNote,

          sortOrder: item.sortOrder ?? 0,

        })),

      },

    }),

  },

  include: { staffPickItems: true },

})

}

update(id: string, dto: Partial) {

  return this.prisma.blogPost.update({ where: { id }, data: dto as any })

}

publish(id: string) {

  return this.prisma.blogPost.update({

```

```
    data: { status: 'PUBLISHED', publishedAt: new Date() },  
  
  })  
  
}  
  
}
```

☐ Step 4: Create BlogController

Create `apps/api/src/blog/blog.controller.ts` :

```

import { Controller, Get, Post, Patch, Param, Body, Query, UseGuards } from '@nestjs/common'

import { BlogService } from '../blog.service'

import { JwtAuthGuard } from '../../auth/guards/jwt-auth.guard'

import { RolesGuard } from '../../auth/guards/roles.guard'

import { Roles } from '../../auth/decorators/roles.decorator'

import { UserRole } from '@mediumformat/types'

import { BlogPostType } from '@mediumformat/db'

@Controller('blog')

export class BlogController {

  constructor(private blog: BlogService) {}

  // Public

  @Get()

  findPublished(@Query('type') type?: BlogPostType) {

    return this.blog.findPublished({ type })

  }

  @Get(':slug')

  findBySlug(@Param('slug') slug: string) {

    return this.blog.findBySlug(slug)

  }

```

```
@UseGuards(JwtAuthGuard, RolesGuard)

@Roles(UserRole.ADMIN, UserRole.STORE_MANAGER)

@Get('admin/all')

findAll() {

    return this.blog.findAllAdmin()

}

@UseGuards(JwtAuthGuard, RolesGuard)

@Roles(UserRole.ADMIN, UserRole.STORE_MANAGER)

@Post()

create(@Body() dto: any) {

    return this.blog.create(dto)

}

@UseGuards(JwtAuthGuard, RolesGuard)

@Roles(UserRole.ADMIN, UserRole.STORE_MANAGER)

@Patch('/:id')

update(@Param('id') id: string, @Body() dto: any) {

    return this.blog.update(id, dto)

}

@UseGuards(JwtAuthGuard, RolesGuard)
```

```
@Post('/:id/publish')

publish(@Param('id') id: string) {

    return this.blog.publish(id)

}

}
```

☐ Step 5: Run test to verify it passes

```
cd apps/api && npm test -- blog.service
```

Expected: PASS.

☐ Step 6: Commit

```
git add apps/api/src/blog/ apps/api/test/blog.service.spec.ts
git commit -m "feat: add blog/editorial module with Staff Picks, post types, publish workflow"
```

Task 13: Backoffice Shell (React)

Files:

- Modify: `apps/backoffice/src/App.tsx`
- Create: `apps/backoffice/src/layouts/AppLayout.tsx`
- Create: `apps/backoffice/src/stores/auth.store.ts`
- Create: `apps/backoffice/src/api/client.ts`
- Create: `apps/backoffice/src/pages/Login.tsx`

☐ Step 1: Install backoffice dependencies

```
cd apps/backoffice

npm install react-router-dom axios @tanstack/react-query zustand lucide-react

npm install -D @types/react-router-dom
```

☐ Step 2: Create typed API client

Create `apps/backoffice/src/api/client.ts` :

```
import axios from 'axios'

export const api = axios.create({

  baseURL: import.meta.env.VITE_API_URL ?? 'http://localhost:3001',

})

api.interceptors.request.use((config) => {

  const token = localStorage.getItem('accessToken')

  if (token) config.headers.Authorization = Bearer ${token}

  return config

})

api.interceptors.response.use(

  (res) => res,

  async (err) => {

    if (err.response?.status === 401) {

      localStorage.removeItem('accessToken')

      window.location.href = '/login'

    }

    return Promise.reject(err)

  },

)
```

☐ Step 3: Create Zustand auth store

Create `apps/backoffice/src/stores/auth.store.ts` :


```
import { create } from 'zustand'

import { persist } from 'zustand/middleware'

interface AuthStore {

  user: { id: string; name: string; email: string; role: string } | null

  accessToken: string | null

  setAuth: (user: AuthStore['user'], token: string) => void

  logout: () => void

}

export const useAuthStore = create()(

  persist(

    (set) => ({

      user: null,

      accessToken: null,

      setAuth: (user, accessToken) => {

        localStorage.setItem('accessToken', accessToken)

        set({ user, accessToken })

      },

      logout: () => {

        localStorage.removeItem('accessToken')

        set({ user: null, accessToken: null })

      }

    })

  )

)
```

```
    }},  
  
    { name: 'mf-auth' },  
  
  },  
  
)
```

☐ Step 4: Create AppLayout with sidebar

Create `apps/backoffice/src/layouts/AppLayout.tsx` :

```
import { Outlet, NavLink } from 'react-router-dom'

import {

  LayoutDashboard, Package, ShoppingCart, Users,

  BarChart2, Truck, Settings, BookOpen, Tag, Newspaper,

} from 'lucide-react'

import { useAuthStore } from '../stores/auth.store'

const navItems = [

  { to: '/', icon: LayoutDashboard, label: 'Dashboard' },

  { to: '/inventory', icon: Package, label: 'Inventory' },

  { to: '/orders', icon: ShoppingCart, label: 'Orders' },

  { to: '/customers', icon: Users, label: 'Customers' },

  { to: '/blog', icon: BookOpen, label: 'Journal' },

  { to: '/marketing', icon: Tag, label: 'Marketing' },

  { to: '/analytics', icon: BarChart2, label: 'Analytics' },

  { to: '/purchase-orders', icon: Truck, label: 'Purchase Orders' },

  { to: '/newsletter', icon: Newspaper, label: 'Newsletter' },

  { to: '/settings', icon: Settings, label: 'Settings' },

]

export function AppLayout() {

  const { user, logout } = useAuthStore()
```

```
return (
```

```
{/ Sidebar /}
```

Medium Format

```
{navItems.map(({ to, icon: Icon, label }) => (
```

```
  key={to}
```

```
  to={to}
```

```
  end={to === '/'}
```

```
  className={({ isActive }) =>
```

```
    `flex items-center gap-3 px-3 py-2 rounded text-sm transition-colors ${
```

```
      isActive
```

```
        ? 'bg-zinc-800 text-white'
```

```
        : 'text-zinc-400 hover:text-zinc-100 hover:bg-zinc-800/50'
```

```
    }`
```

```
  }
```

```
>
```

```
{label}
```

```
    )})
```

```
{user?.email}
```

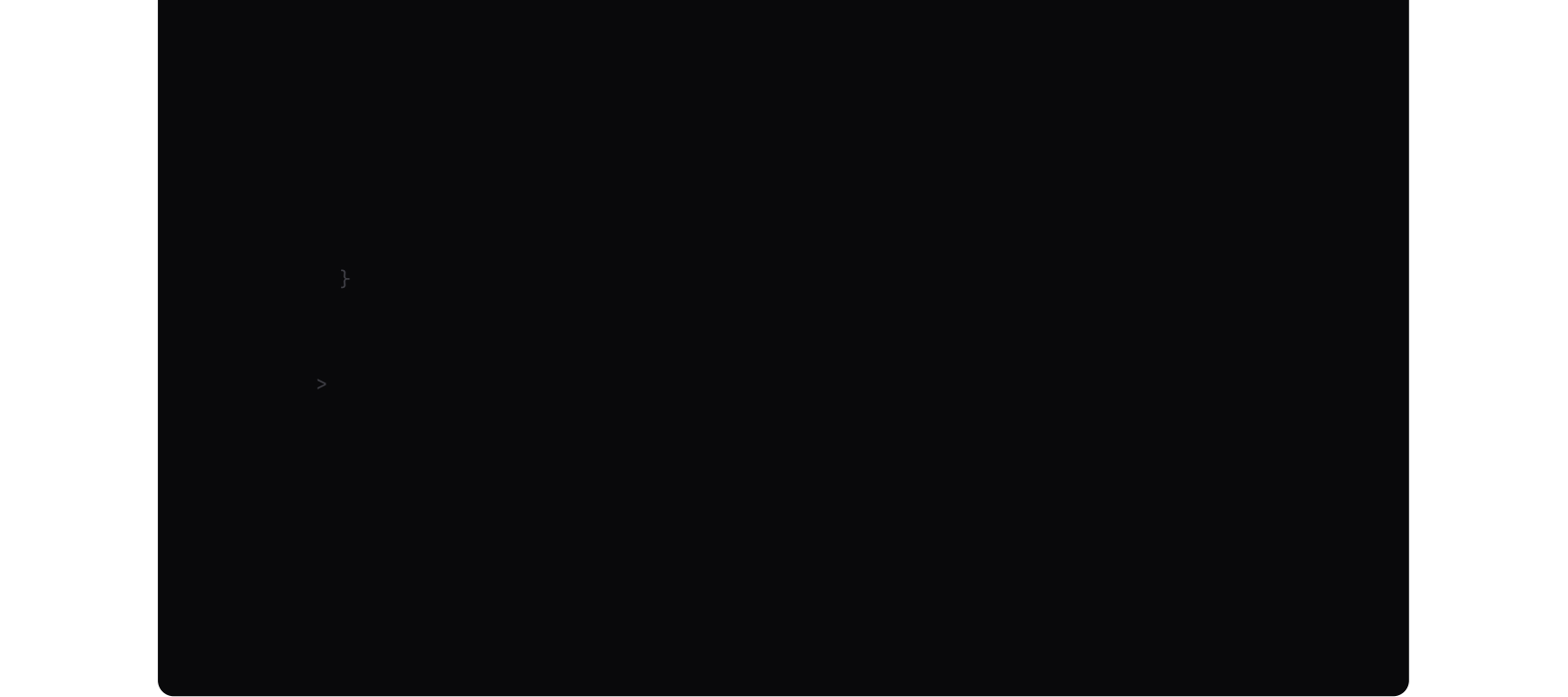
Sign out

```
{/ Main /}
```

```
)
```

```
}
```

☐ Step 5: Wire up router in App.tsx



```
} />
```

```
Inventory} />
```

```
Orders} />
```

```
Journal} />
```

```
)
```

```
}
```

☐ Step 6: Commit

```
git add apps/backoffice/src/
```

```
git commit -m "feat: add backoffice shell with sidebar navigation, auth store, React Query setup"
```

Task 14: Storefront Shell (Next.js)

Files:

- Modify: `apps/storefront/app/layout.tsx`
- Create: `apps/storefront/app/page.tsx`
- Create: `apps/storefront/components/NowPlayingBar.tsx`
- Create: `apps/storefront/components/ReleaseCard.tsx`
- Create: `apps/storefront/app/shop/page.tsx`
- Create: `apps/storefront/app/shop/[slug]/page.tsx`

- Create: `apps/storefront/app/journal/[slug]/page.tsx`

☐ Step 1: Install storefront dependencies

```
cd apps/storefront  
npm install @tanstack/react-query axios zustand
```

☐ Step 2: Create root layout with dark theme and Now Playing bar


```
// apps/storefront/app/layout.tsx

import type { Metadata } from 'next'

import { Inter } from 'next/font/google'

import './globals.css'

import { NowPlayingBar } from '@components/NowPlayingBar'

const inter = Inter({ subsets: ['latin'] })

export const metadata: Metadata = {

  title: { default: 'Medium Format', template: '%s | Medium Format' },

  description: 'Independent record shop — Jakarta',

}

export default function RootLayout({ children }: { children: React.ReactNode }) {

  return (

    <div className={inter.className} bg-zinc-950 text-zinc-100 min-h-screen pb-20>
```

[Medium Format](#)

[Shop](#)

[Journal](#)

```
{children}
```

```
)
```

```
}
```

☐ Step 3: Create ReleaseCard component

Create `apps/storefront/components/ReleaseCard.tsx` :


```
onMouseEnter={() => setHovered(true)}
```

```
onMouseLeave={() => setHovered(false)}
```

```
>
```

```
{release.images[0] ? (
```

```
 {release.title}
```

```
) : (
```

No image

```
)}
```

```
{release.hasPreview && hovered && (
```

```
)}
```

```
{release.stock === 0 && (
```

Sold Out

```
)}
```

```
{release.artist}
```

```
{release.title}
```

```
IDR {release.priceIdr.toLocaleString('id-ID')}
```

```
)
```

```
}
```

☐ Step 4: Create storefront homepage

```
// apps/storefront/app/page.tsx

import { ReleaseCard } from '@components/ReleaseCard'

async function getLatestReleases() {

  const res = await fetch(`${process.env.NEXT_PUBLIC_API_URL}/inventory?limit=8`, {

    next: { revalidate: 60 },

  })

  return res.ok ? res.json() : []

}

async function getLatestPosts() {

  const res = await fetch(`${process.env.NEXT_PUBLIC_API_URL}/blog?limit=3`, {

    next: { revalidate: 60 },

  })

  return res.ok ? res.json() : []

}

export default async function HomePage() {

  const [releases, posts] = await Promise.all([getLatestReleases(), getLatestPosts()])

  return (
```

 {post_title}

`{post.title}`

`{post.author?.name}`

`))}`

`}}`

`)`

`}`

☐ Step 5: Create Journal index and post pages

Create `apps/storefront/app/journal/page.tsx` :


```
{posts.map((post: any) => (
```

```
  /journal/${post.slug}} className="group_block">
```

```
    {post.featuredImage && (
```

```
       {post.title}
```

```
    )}
```

```
    {post.type.replace('_', ' ')}  
  
    
```

```
    {post.title}
```

```
    {post.excerpt &&
```

```
      {post.excerpt}
```

```
    }
```

```
    {post.author?.name} · {post.publishedAt ? new Date(post.publishedAt).toLocaleDateString('id-ID')}
```

```
  )})
```

```
)
```

Create `apps/storefront/app/journal/[slug]/page.tsx` :

```
import { ReleaseCard } from '@components/ReleaseCard'

import { notFound } from 'next/navigation'

export default async function BlogPostPage({ params }: { params: { slug: string } }) {

  const res = await fetch(`${process.env.NEXT_PUBLIC_API_URL}/blog/${params.slug}`, {

    next: { revalidate: 60 },

  })

  if (!res.ok) notFound()

  const post = await res.json()

  return (

    {post.type.replace('_', ' ')}

    {post.author?.name} · {post.publishedAt ? new Date(post.publishedAt).toLocaleDateString('id-ID', { year: 'numeric', month: 'long', day: 'numeric' }) : ''}

    {post.featuredImage && (

       {post.title}

    )}
```

```
{/ Staff Picks release grid /}

{post.type === 'STAFF_PICKS' && post.staffPickItems?.length > 0 && (

  {post.staffPickItems.map((item: any) => (

    {item.staffNote && (

      "{item.staffNote}"

    )}

  )}}

)}

{/ Rich text content rendered as raw JSON – swap for Plate.js renderer /}

{post.excerpt &&

  {post.excerpt}

}
```

```
}
```

☐ Step 6: Create NowPlayingBar placeholder

Create `apps/storefront/components/NowPlayingBar.tsx` :

```
'use client'

import { usePlayerStore } from '@stores/player.store'

export function NowPlayingBar() {

  const { currentTrack } = usePlayerStore()

  if (!currentTrack) return null

  return (

    {currentTrack.trackTitle ?? currentTrack.releaseTitle}

    {currentTrack.artist}

    {currentTrack.source}

  )

}
```

```
import { create } from 'zustand'

interface Track {

  releaseId: string

  releaseTitle: string

  artist: string

  trackTitle?: string

  source: string

  url: string

}

interface PlayerStore {

  currentTrack: Track | null

  play: (track: Track) => void

  stop: () => void

}

export const usePlayerStore = create((set) => ({

  currentTrack: null,

  play: (track) => set({ currentTrack: track }),

  stop: () => set({ currentTrack: null }),

}))
```

☐ Step 7: Commit

```
git add apps/storefront/  
  
git commit -m "feat: add storefront shell with release grid, journal pages, Now Playing bar, Staff Picks layout"
```

Task 15: Dokploy Deployment Config

Files:

- Create: `docker-compose.yml`
- Create: `apps/api/Dockerfile`
- Create: `apps/backoffice/Dockerfile`
- Create: `apps/storefront/Dockerfile`

☐ Step 1: Create API Dockerfile

Create `apps/api/Dockerfile` :


```
FROM node:20-alpine AS builder

WORKDIR /app


COPY package*.json turbo.json ./


COPY apps/api/package*.json ./apps/api/


COPY packages/db/package*.json ./packages/db/


COPY packages/types/package*.json ./packages/types/


RUN npm ci


COPY . .


RUN npx turbo run build --filter=api


FROM node:20-alpine


WORKDIR /app


COPY --from=builder /app/apps/api/dist ./dist


COPY --from=builder /app/node_modules ./node_modules


COPY --from=builder /app/packages/db/prisma ./prisma


EXPOSE 3001


CMD ["node", "dist/main"]
```

☐ Step 2: Create Storefront Dockerfile

Create `apps/storefront/Dockerfile` :

```
FROM node:20-alpine AS builder
```

```
WORKDIR /app
```

```
COPY package*.json turbo.json ./
```

```
COPY apps/storefront/package*.json ./apps/storefront/
```

```
RUN npm ci
```

```
COPY . .
```

```
RUN npx turbo run build --filter=storefront
```

```
FROM node:20-alpine
```

```
WORKDIR /app
```

```
COPY --from=builder /app/apps/storefront/.next/standalone ./
```

```
COPY --from=builder /app/apps/storefront/.next/static ./apps/storefront/.next/static
```

```
COPY --from=builder /app/apps/storefront/public ./apps/storefront/public
```

```
EXPOSE 3002
```

```
CMD ["node", "apps/storefront/server.js"]
```

☐ Step 3: Create GitHub Actions CI

Create `.github/workflows/ci.yml`:

```
name: CI

on:

  push:

  pull_request:

jobs:

  test:

    runs-on: ubuntu-latest

    steps:

      - uses: actions/checkout@v4

      - uses: actions/setup-node@v4

        with:
          node-version: 20

      - cache: npm

      - run: npm ci

      - run: npm run build --workspaces

      - run: npm test --workspaces
```

☐ Step 4: Commit

```
git add Dockerfile* .github/ docker-compose.yml apps/api/Dockerfile apps/storefront/Dockerfile
git commit -m "feat: add Dockerfiles and GitHub Actions CI for Dokploy deployment"
```

Task 16: Final Integration Test

☐ Step 1: Run all tests

```
npm test
```

Expected: All test suites pass.

☐ Step 2: Start full dev environment

```
npm run dev
```

Expected:

- API running on port 3001 — `curl http://localhost:3001/inventory` returns `[]`
- Backoffice on port 3000 — login page visible
- Storefront on port 3002 — homepage renders with empty releases and journal sections

☐ Step 3: Smoke test auth flow

```
curl -X POST http://localhost:3001/auth/register \  
  
-H "Content-Type: application/json" \  
  
-d '{"email":"admin@mf.com","password":"secret123","name":"Admin"}'
```

☐ Step 4: Final commit

```
git add .  
git commit -m "feat: Plan 1 Foundation complete — monorepo, auth, inventory, orders, payments, shipping, blog, storefront"
```

04

PLAN 2

Plan 2 — Channels

Medium Format — Plan 2: Channels Implementation Plan

For agentic workers: REQUIRED SUB-SKILL: Use superpowers:subagent-driven-development (recommended) or superpowers:executing-plans to implement this plan task-by-task. Steps use checkbox (- []) syntax for tracking.

Goal: Add all sales channel integrations — Tokopedia, Shopee, Meta Commerce, eBay, Google Merchant, PayPal, international shipping (DHL/FedEx), price rules engine, and purchase orders module. **Architecture:** Each marketplace integration is an isolated NestJS module with its own service and controller. A shared `ChannelSyncService` runs background sync jobs via Bull queues (Upstash Redis). Price rules are computed on the fly before any push. Purchase orders are a standalone backoffice module with no storefront exposure. **Tech Stack:** NestJS, Bull + Upstash Redis (job queues), Prisma, Axios, DHL Express API, FedEx REST API, Meta Graph API, eBay REST API, Google Content API v2.1, PayPal REST SDK, Stripe (already configured in Plan 1). **Prerequisite:** Plan 1 (Foundation) must be fully deployed and passing CI before starting Plan 2.

File Structure (additions to Plan 1)

apps/api/src/

├─ channels/

| ├─ channel-sync.service.ts ← shared sync engine + queue

| ├─ price-rules.service.ts ← markup computation

| ├─ tokopedia/

| | ├─ tokopedia.module.ts

| | ├─ tokopedia.service.ts

| | └─ tokopedia.controller.ts ← webhook + OAuth callback

| ├─ shopee/

| | ├─ shopee.module.ts

| | ├─ shopee.service.ts

| | └─ shopee.controller.ts

| ├─ meta/

| | ├─ meta.module.ts

| | └─ meta.service.ts

| ├─ ebay/

| | ├─ ebay.module.ts

| | └─ ebay.service.ts

| └─ google-merchant/

| ├─ google-merchant.module.ts

| └─ google-merchant.service.ts

|— shipping/

| |— dhl/

| | |— dhl.service.ts

| |— fedex/

| |— fedex.service.ts

|— purchase-orders/

| |— purchase-orders.module.ts

| |— purchase-orders.service.ts

| |— purchase-orders.controller.ts

|— payments/

| |— paypal/

| | |— paypal.module.ts

| | |— paypal.service.ts

| | |— paypal.controller.ts

packages/db/prisma/schema.prisma ← add PurchaseOrder, PriceRule models

apps/backoffice/src/pages/

|— Channels/

| |— ChannelsPage.tsx

| |— TokopediaConnect.tsx

└─ PurchaseOrders/

└─ PurchaseOrdersPage.tsx

└─ PurchaseOrderForm.tsx

└─ PackingListExport.tsx

Task 1: Database Schema Additions

Files:

- Modify: `packages/db/prisma/schema.prisma`

☐ Step 1: Add PriceRule and PurchaseOrder models

Append to `packages/db/prisma/schema.prisma` :

```

model PriceRule {
    id          String          @id @default(cuid())

    channel     SalesChannel @unique

    markupPct   Float           @default(0)    // e.g. 5.0 = 5%

    markupFixed Int             @default(0)    // IDR fixed amount added after %

    updatedAt   DateTime        @updatedAt

    @@map("price_rules")
}

```

```

model PurchaseOrder {

    id          String          @id @default(cuid())

    poNumber     String          @unique

    supplier     String

    countryOfOrigin String?

    status       POStatus        @default(PENDING)

    paymentMethod POPaymentMethod

    paidBy       String?         // employee name if personal card

    paymentStatus POPaymentStatus @default(UNPAID)

    receiptUrl   String?

    notes        String?

    estimatedArrival DateTime?
}

```

```

    updatedAt      DateTime      @updatedAt

    items          POItem[]

    shipments      POShipment[]

    @@map("purchase_orders")
}

model POItem {

    id             String         @id @default(cuid())

    poId           String

    po             PurchaseOrder @relation(fields: [poId], references: [id])

    releaseId      String?

    release        Release?      @relation(fields: [releaseId], references: [id])

    description    String         // free text for items not yet in catalog

    quantity       Int

    unitCostIdr    Int

    weightGrams    Int?

    @@map("po_items")
}

model POShipment {

```

```
    poId          String

    po             PurchaseOrder @relation(fields: [poId], references: [id])

    shipperName    String

    shipperEmail   String?

    trackingNumber String?

    estimatedArrival DateTime?

    actualArrival  DateTime?

    notes          String?

    @@map("po_shipments")

}
```

```
enum POStatus {
```

```
    PENDING
```

```
    ORDERED
```

```
    SHIPPED
```

```
    ARRIVED
```

```
    CANCELLED
```

```
}
```

```
enum POPaymentMethod {
```

```
    COMPANY_CARD
```

```
}

enum POPaymentStatus {

  UNPAID

  PAID

  REIMBURSED

  DISPUTED

}
```

Also add relation back-references to Release model:

```
// Add inside Release model:
poItems POItem[]
```

☐ Step 2: Run migration

```
cd packages/db
npx prisma migrate dev --name add_price_rules_purchase_orders
```

Expected: Migration applied, Prisma Client regenerated.

☐ Step 3: Commit

```
git add packages/db/
git commit -m "feat: add PriceRule, PurchaseOrder, POItem, POShipment schema models"
```

Task 2: Price Rules Engine

- Create: `apps/api/src/channels/price-rules.service.ts`
- Test: `apps/api/test/price-rules.service.spec.ts`

☐ **Step 1: Write failing test**

Create `apps/api/test/price-rules.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { PriceRulesService } from '../src/channels/price-rules.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('PriceRulesService', () => {

  let service: PriceRulesService

  const mockPrisma = { priceRule: { findUnique: jest.fn() } }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        PriceRulesService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(PriceRulesService)

  })

  it('applies % markup first then fixed amount', () => {

    // base 100,000 + 5% = 105,000 + 5,000 fixed = 110,000

    expect(service.compute(100000, { markupPct: 5, markupFixed: 5000 })).toBe(110000)

  })

```

```

    expect(service.compute(200000, { markupPct: 0, markupFixed: 0 })).toBe(200000)

  })

  it('rounds to nearest 100 IDR', () => {

    // 100,000 + 3.5% = 103,500 - already round

    expect(service.compute(100000, { markupPct: 3.5, markupFixed: 0 })).toBe(103500)

  })

})

```

☐ **Step 2: Run test to verify it fails**

```
cd apps/api && npm test -- price-rules.service
```

Expected: FAIL.

☐ **Step 3: Implement PriceRulesService**

Create `apps/api/src/channels/price-rules.service.ts` :


```

import { Injectable } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

import { SalesChannel } from '@mediumformat/db'

@Injectable()

export class PriceRulesService {

  constructor(private prisma: PrismaService) {}

  compute(baseIdr: number, rule: { markupPct: number; markupFixed: number }): number {

    const afterPct = baseIdr * (1 + rule.markupPct / 100)

    const total = afterPct + rule.markupFixed

    return Math.round(total / 100) * 100

  }

  async computeForChannel(baseIdr: number, channel: SalesChannel): Promise {

    const rule = await this.prisma.priceRule.findUnique({ where: { channel } })

    if (!rule) return baseIdr

    return this.compute(baseIdr, rule)

  }

  async getRule(channel: SalesChannel) {

    return this.prisma.priceRule.findUnique({ where: { channel } })

  }
}

```

```
    async upsertRule(channel: SalesChannel, markupPct: number, markupFixed: number) {

      return this.prisma.priceRule.upsert({

        where: { channel },

        create: { channel, markupPct, markupFixed },

        update: { markupPct, markupFixed },

      })

    }

  }

}
```

☐ **Step 4: Run test to verify it passes**

```
cd apps/api && npm test -- price-rules.service
```

Expected: PASS — 3 tests passing.

☐ **Step 5: Commit**

```
git add apps/api/src/channels/price-rules.service.ts apps/api/test/price-rules.service.spec.ts
git commit -m "feat: add price rules engine with % markup + fixed IDR computation"
```

Task 3: Channel Sync Engine (Queue)

Files:

- Create: `apps/api/src/channels/channel-sync.service.ts`
- Test: `apps/api/test/channel-sync.service.spec.ts`

☐ **Step 1: Install Bull queue dependencies**

```
cd apps/api && npm install @nestjs/bull bull @upstash/redis  
npm install -D @types/bull
```

☐ Step 2: Write failing test

Create `apps/api/test/channel-sync.service.spec.ts`:

```

import { Test } from '@nestjs/testing'

import { ChannelSyncService } from '../src/channels/channel-sync.service'

import { PrismaService } from '../src/prisma/prisma.service'

import { PriceRulesService } from '../src/channels/price-rules.service'

describe('ChannelSyncService', () => {

  let service: ChannelSyncService

  const mockPrisma = {

    release: { findMany: jest.fn() },

    channellisting: { upsert: jest.fn(), update: jest.fn() },

  }

  const mockPriceRules = { computeForChannel: jest.fn((base) => Promise.resolve(base)) }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        ChannelSyncService,

        { provide: PrismaService, useValue: mockPrisma },

        { provide: PriceRulesService, useValue: mockPriceRules },

      ],

    }).compile()

    service = module.get(ChannelSyncService)

```

```

    it('marks listing as inactive on soft-deleted release', async () => {

      await service.handleStockUpdate('rel-1', 0, 'TOKOPEDIA' as any)

      expect(mockPrisma.channelListing.update).toHaveBeenCalledWith(

        expect.objectContaining({ where: { releaseId_channel: { releaseId: 'rel-1', channel: 'TOKOPED
IA' } } })),

      )

    })

  })
}

```

☐ Step 3: Implement ChannelSyncService

Create `apps/api/src/channels/channel-sync.service.ts` :

```

import { Injectable, Logger } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

import { PriceRulesService } from './price-rules.service'

import { SalesChannel } from '@mediumformat/db'

@Injectable()

export class ChannelSyncService {

  private readonly logger = new Logger(ChannelSyncService.name)

  constructor(

    private prisma: PrismaService,

    private priceRules: PriceRulesService,

  ) {}

  async handleStockUpdate(releaseId: string, newStock: number, channel: SalesChannel) {

    await this.prisma.channelListing.update({

      where: { releaseId_channel: { releaseId, channel } },

      data: { isActive: newStock > 0, lastSyncedAt: new Date() },

    })

  }

  async pushRelease(releaseId: string, channel: SalesChannel, pushFn: (payload: object) => Promise) {

    const release = await this.prisma.release.findUnique({ where: { id: releaseId } })
  }

```

```

const computedPrice = await this.priceRules.computeForChannel(release.priceIdr, channel)

try {

  const externalId = await pushFn({

    title: `${release.artist} - ${release.title}`,

    price: computedPrice,

    stock: release.stock,

    images: release.images,

    description: [release.label, release.catalogNumber, release.year, release.condition]

      .filter(Boolean).join(' · '),

  })

  await this.prisma.channellisting.upsert({

    where: { releaseId_channel: { releaseId, channel } },

    create: { releaseId, channel, externalId, priceIdr: computedPrice, isActive: true, lastSyncedAt: new Date() },

    update: { externalId, priceIdr: computedPrice, isActive: true, lastSyncedAt: new Date(), syncError: null },

  })

} catch (err: any) {

  this.logger.error(Sync failed for ${releaseId} → ${channel}: ${err.message})

  await this.prisma.channellisting.upsert({

```

```
        create: { releaseId, channel, isActive: false, syncError: err.message },

        update: { syncError: err.message, lastSyncedAt: new Date() },

    })

}

}

}
```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- channel-sync.service
```

Expected: PASS.

☐ Step 5: Commit

```
git add apps/api/src/channels/ apps/api/test/channel-sync.service.spec.ts
git commit -m "feat: add channel sync engine with stock update and price rule push"
```

Task 4: Tokopedia Integration

Files:

- Create: `apps/api/src/channels/tokopedia/tokopedia.service.ts`
- Create: `apps/api/src/channels/tokopedia/tokopedia.controller.ts`
- Test: `apps/api/test/tokopedia.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/tokopedia.service.spec.ts` :


```

import { Test } from '@nestjs/testing'

import { TokopediaService } from '../src/channels/tokopedia/tokopedia.service'

import { HttpService } from '@nestjs/axios'

import { PrismaService } from '../src/prisma/prisma.service'

import { ChannelSyncService } from '../src/channels/channel-sync.service'

import { of } from 'rxjs'

describe('TokopediaService', () => {

  let service: TokopediaService

  const mockHttp = { post: jest.fn(), get: jest.fn() }

  const mockPrisma = { order: { create: jest.fn() }, setting: { findUnique: jest.fn() } }

  const mockSync = { pushRelease: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        TokopediaService,

        { provide: HttpService, useValue: mockHttp },

        { provide: PrismaService, useValue: mockPrisma },

        { provide: ChannelSyncService, useValue: mockSync },

      ],

    }).compile()

```

```

    })

    it('maps Tokopedia order status PAYMENT_VERIFIED to PROCESSING', () => {

      expect(service.mapOrderStatus('PAYMENT_VERIFIED')).toBe('PROCESSING')

    })

    it('maps Tokopedia order status SELLER_SEND_PACKAGE to SHIPPED', () => {

      expect(service.mapOrderStatus('SELLER_SEND_PACKAGE')).toBe('SHIPPED')

    })

  })
}

```

☐ **Step 2: Run test to verify it fails**

```
cd apps/api && npm test -- tokopedia.service
```

Expected: FAIL.

☐ **Step 3: Implement TokopediaService**

Create `apps/api/src/channels/tokopedia/tokopedia.service.ts` :

```

import { Injectable, Logger } from '@nestjs/common'

import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

import { PrismaService } from '../../prisma/prisma.service'

import { ChannelSyncService } from '../channel-sync.service'

import { OrderStatus } from '@mediumformat/db'

const STATUS_MAP: Record = {

  PAYMENT_VERIFIED: 'PROCESSING',

  READY_TO_SHIP: 'PROCESSING',

  SELLER_SEND_PACKAGE: 'SHIPPED',

  DELIVERED: 'DELIVERED',

  ORDER_FINISHED: 'COMPLETED',

  ORDER_CANCELLED: 'CANCELLED',

}

@Injectable()

export class TokopediaService {

  private readonly logger = new Logger(TokopediaService.name)

  private readonly baseUrl = 'https://fs.tokopedia.net'

  constructor(

    private http: HttpService,

```

```

    private prisma: PrismaService,

    private sync: ChannelSyncService,

) {}

mapOrderStatus(tokopediaStatus: string): OrderStatus {

    return STATUS_MAP[tokopediaStatus] ?? 'PENDING'

}

private async getAccessToken(): Promise {

    const setting = await this.prisma.setting.findUnique({ where: { key: 'tokopedia_oauth' } })

    return (setting?.value as any)?.accessToken ?? ''

}

async pushProduct(releaseId: string) {

    const token = await this.getAccessToken()

    return this.sync.pushRelease(releaseId, 'TOKOPEDIA', async (payload: any) => {

        const { data } = await firstValueFrom(

            this.http.post(`${this.baseUrl}/inventory/v1/fs/products`, payload, {

                headers: { Authorization: Bearer ${token} },

            }),

        )

        return String(data.data?.product_id ?? '')
    })
}

```

```

}

async updateStock(releaseId: string, externalId: string, stock: number) {

  const token = await this.getAccessToken()

  await firstValueFrom(

    this.http.post(

      `${this.baseUrl}/inventory/v1/fs/products/stocks`,

      { products: [{ product_id: Number(externalId), stock }] },

      { headers: { Authorization: Bearer ${token} } } ),

    ),

  )

}

async pullOrders() {

  const token = await this.getAccessToken()

  const { data } = await firstValueFrom(

    this.http.get(`${this.baseUrl}/v2/order/list`, {

      headers: { Authorization: Bearer ${token} },

      params: { page: 1, per_page: 50, status: 'PAYMENT_VERIFIED' },

    )),

  )

}

```

```

}

async handleWebhook(body: any) {

  const status = this.mapOrderStatus(body.order_status)

  const existing = await this.prisma.order.findFirst({

    where: { paymentRef: String(body.order_id), channel: 'TOKOPEDIA' },

  })

  if (existing) {

    await this.prisma.order.update({ where: { id: existing.id }, data: { status } })

  } else {

    this.logger.warn(Tokopedia order ${body.order_id} not found locally – may need manual sync)

  }

}

}

}

```

☐ Step 4: Create Tokopedia webhook controller

Create `apps/api/src/channels/tokopedia/tokopedia.controller.ts` :

```

import { Controller, Post, Body, Get, Query, Redirect } from '@nestjs/common'

import { TokopediaService } from './tokopedia.service'

import { PrismaService } from '../../prisma/prisma.service'

@Controller('channels/tokopedia')

export class TokopediaController {

  constructor(

    private tokopedia: TokopediaService,

    private prisma: PrismaService,

  ) {}

  @Post('webhook')

  webhook(@Body() body: any) {

    return this.tokopedia.handleWebhook(body)

  }

  // OAuth callback – store tokens in Settings table

  @Get('callback')

  @Redirect('/settings/channels')

  async oauthCallback(@Query('code') code: string) {

    // Exchange code for token via Tokopedia OAuth – store in settings

    await this.prisma.setting.upsert({

      where: { key: 'tokopedia_oauth' },

```

```

        create: { key: 'tokopedia_oauth', value: { authCode: code, connectedAt: new Date() } },

        update: { value: { authCode: code, connectedAt: new Date() } },

    })

    return {}

}

}

```

☐ Step 5: Run test to verify it passes

```
cd apps/api && npm test -- tokopedia.service
```

Expected: PASS.

☐ Step 6: Commit

```
git add apps/api/src/channels/tokopedia/ apps/api/test/tokopedia.service.spec.ts
git commit -m "feat: add Tokopedia integration with product push, stock sync, order webhook"
```

Task 5: Shopee Integration

Files:

- Create: `apps/api/src/channels/shopee/shopee.service.ts`
- Create: `apps/api/src/channels/shopee/shopee.controller.ts`
- Test: `apps/api/test/shopee.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/shopee.service.spec.ts` :


```

import { Test } from '@nestjs/testing'

import { ShopeeService } from '../src/channels/shopee/shopee.service'

import { HttpService } from '@nestjs/axios'

import { PrismaService } from '../src/prisma/prisma.service'

import { ChannelSyncService } from '../src/channels/channel-sync.service'

describe('ShopeeService', () => {

  let service: ShopeeService

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        ShopeeService,

        { provide: HttpService, useValue: { post: jest.fn(), get: jest.fn() } },

        { provide: PrismaService, useValue: { setting: { findUnique: jest.fn() }, order: { findFirst: jest.fn(), update: jest.fn() } } },

        { provide: ChannelSyncService, useValue: { pushRelease: jest.fn() } },

      ],

    }).compile()

    service = module.get(ShopeeService)

  })

  it('maps Shopee status READY_TO_SHIP to PROCESSING', () => {

```

```
}))

it('maps Shopee status SHIPPED to SHIPPED', () => {

  expect(service.mapOrderStatus('SHIPPED')).toBe('SHIPPED')

})

})
```

☐ Step 2: Implement ShopeeService

Create `apps/api/src/channels/shopee/shopee.service.ts` :

```

import { Injectable, Logger } from '@nestjs/common'

import { HttpService } from '@nestjs/axios'

import { createHmac } from 'crypto'

import { firstValueFrom } from 'rxjs'

import { PrismaService } from '../../prisma/prisma.service'

import { ChannelSyncService } from '../channel-sync.service'

import { OrderStatus } from '@mediumformat/db'

const STATUS_MAP: Record = {

  UNPAID: 'PENDING',

  READY_TO_SHIP: 'PROCESSING',

  PROCESSED: 'PROCESSING',

  SHIPPED: 'SHIPPED',

  COMPLETED: 'COMPLETED',

  CANCELLED: 'CANCELLED',

  IN_CANCEL: 'CANCELLED',

}

@Injectable()

export class ShopeeService {

  private readonly logger = new Logger(ShopeeService.name)

  private readonly baseUrl = 'https://partner.shopeemobile.com/api/v2'

```

```

constructor(

    private http: HttpService,

    private prisma: PrismaService,

    private sync: ChannelSyncService,

) {}

mapOrderStatus(shopeeStatus: string): OrderStatus {

    return STATUS_MAP[shopeeStatus] ?? 'PENDING'

}

private sign(path: string, timestamp: number): string {

    const partnerId = process.env.SHOPEE_PARTNER_ID!

    const partnerKey = process.env.SHOPEE_PARTNER_KEY!

    const base = `${partnerId}${path}${timestamp}`

    return createHmac('sha256', partnerKey).update(base).digest('hex')

}

private async getHeaders(path: string) {

    const timestamp = Math.floor(Date.now() / 1000)

    const sign = this.sign(path, timestamp)

    const setting = await this.prisma.setting.findUnique({ where: { key: 'shopee_oauth' } })

    const accessToken = (setting?.value as any)?.accessToken ?? ''

```

```

    return {

        params: { partner_id: process.env.SHOPEE_PARTNER_ID, timestamp, sign, access_token: accessToken, shop_id: shopId },

    }

}

async pushProduct(releaseId: string) {

    const path = '/product/add_item'

    const opts = await this.getHeaders(path)

    return this.sync.pushRelease(releaseId, 'SHOPEE', async (payload: any) => {

        const { data } = await firstValueFrom(

            this.http.post(`${this.baseUrl}${path}`, payload, { params: opts.params }),

        )

        return String(data.response?.item_id ?? '')

    })

}

async handleWebhook(body: any) {

    if (body.code !== 3) return // code 3 = order status update

    const status = this.mapOrderStatus(body.data?.status)

    const existing = await this.prisma.order.findFirst({

        where: { paymentRef: String(body.data?.ordersn), channel: 'SHOPEE' },
    })

```

```

    })

    if (existing) {

        await this.prisma.order.update({ where: { id: existing.id }, data: { status } })

    }

}

}

}

```

☐ Step 3: Create ShopeeController

Create `apps/api/src/channels/shopee/shopee.controller.ts` :

```

import { Controller, Post, Body } from '@nestjs/common'
import { ShopeeService } from './shopee.service'

@Controller('channels/shopee')

export class ShopeeController {

    constructor(private shopee: ShopeeService) {}

    @Post('webhook')

    webhook(@Body() body: any) {

        return this.shopee.handleWebhook(body)

    }

}

```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- shopee.service
```

Expected: PASS.

☐ Step 5: Commit

```
git add apps/api/src/channels/shopee/ apps/api/test/shopee.service.spec.ts
git commit -m "feat: add Shopee integration with HMAC signing, product push, order webhook"
```

Task 6: Meta Commerce Integration

Files:

- Create: `apps/api/src/channels/meta/meta.service.ts`
- Test: `apps/api/test/meta.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/meta.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { MetaService } from '../src/channels/meta/meta.service'

import { HttpService } from '@nestjs/axios'

import { ChannelSyncService } from '../src/channels/channel-sync.service'

import { of } from 'rxjs'

describe('MetaService', () => {

  let service: MetaService

  const mockHttp = { post: jest.fn() }

  const mockSync = { pushRelease: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        MetaService,

        { provide: HttpService, useValue: mockHttp },

        { provide: ChannelSyncService, useValue: mockSync },

      ],

    }).compile()

    service = module.get(MetaService)

  })

  it('builds correct Meta catalog item payload', () => {

```



```
const payload = service.buildCatalogPayload({

  title: 'Miles Davis - Kind of Blue',

  price: 350000,

  stock: 2,

  images: ['https://example.com/img.jpg'],

  description: 'Columbia · CL 1355 · 1959 · VG+',

})

expect(payload.currency).toBe('IDR')

expect(payload.availability).toBe('in stock')

expect(payload.price).toBe('350000 IDR')

})

})
```

☐ Step 2: Implement MetaService

Create `apps/api/src/channels/meta/meta.service.ts` :

```

import { Injectable } from '@nestjs/common'
import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

import { ChannelSyncService } from '../channel-sync.service'

@Injectable()

export class MetaService {

  private readonly graphUrl = 'https://graph.facebook.com/v19.0'

  constructor(

    private http: HttpService,

    private sync: ChannelSyncService,

  ) {}

  buildCatalogPayload(item: {

    title: string

    price: number

    stock: number

    images: string[]

    description: string

  }) {

    return {

      name: item.title,

```

```

        description: item.description,

        price: ${item.price} IDR,

        currency: 'IDR',

        availability: item.stock > 0 ? 'in stock' : 'out of stock',

        image_url: item.images[0] ?? '',

        condition: 'used',

        brand: 'Medium Format',

    }

}

async pushProduct(releaseId: string) {

    const catalogId = process.env.META_CATALOG_ID!

    const token = process.env.META_ACCESS_TOKEN!

    return this.sync.pushRelease(releaseId, 'META', async (rawPayload: any) => {

        const payload = this.buildCatalogPayload(rawPayload)

        const { data } = await firstValueFrom(

            this.http.post(

                `${this.graphUrl}/${catalogId}/products`,

                payload,

                { params: { access_token: token } },

```

```
    )

    return String(data.id ?? '')

  })

}

}
```

☐ Step 3: Run test to verify it passes

```
cd apps/api && npm test -- meta.service
```

Expected: PASS.

☐ Step 4: Commit

```
git add apps/api/src/channels/meta/ apps/api/test/meta.service.spec.ts
git commit -m "feat: add Meta Commerce catalog integration with product push"
```

Task 7: eBay Integration

Files:

- Create: `apps/api/src/channels/ebay/ebay.service.ts`
- Test: `apps/api/test/ebay.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/ebay.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { EbayService } from '../src/channels/ebay/ebay.service'

import { HttpService } from '@nestjs/axios'

import { ChannelSyncService } from '../src/channels/channel-sync.service'

describe('EbayService', () => {

  let service: EbayService

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        EbayService,

        { provide: HttpService, useValue: { post: jest.fn(), get: jest.fn() } },

        { provide: ChannelSyncService, useValue: { pushRelease: jest.fn() } },

      ],

    }).compile()

    service = module.get(EbayService)

  })

  it('converts IDR to USD for eBay listing', () => {

    // 163,000 IDR ≈ $10.00

    expect(service.idrToUsd(163000)).toBeCloseTo(10.0, 0)

  })

```

})

☐ Step 2: Implement EbayService

Create `apps/api/src/channels/ebay/ebay.service.ts` :

```

import { Injectable, Logger } from '@nestjs/common'

import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

import { ChannelSyncService } from '../channel-sync.service'

const IDR_TO_USD = 16300

@Injectable()

export class EbayService {

  private readonly logger = new Logger(EbayService.name)

  private readonly baseUrl = 'https://api.ebay.com/sell/inventory/v1'

  constructor(

    private http: HttpService,

    private sync: ChannelSyncService,

  ) {}

  idrToUsd(idr: number): number {

    return Math.round((idr / IDR_TO_USD) * 100) / 100

  }

  private async getToken(): Promise {

    const creds = Buffer.from(

      `${process.env.EBAY_CLIENT_ID}:${process.env.EBAY_CLIENT_SECRET}`,

```

```

).toString('base64')

const { data } = await firstValueFrom(

  this.http.post(

    'https://api.ebay.com/identity/v1/oauth2/token',

    'grant_type=client_credentials&scope=https://api.ebay.com/oauth/api_scope/sell.inventory',

    { headers: { Authorization: Basic ${creds}, 'Content-Type': 'application/x-www-form-urlencoded' } }},

  ),

)

return data.access_token

}

async pushProduct(releaseId: string, sku: string) {

  const token = await this.getToken()

  return this.sync.pushRelease(releaseId, 'EBAY', async (payload: any) => {

    await firstValueFrom(

      this.http.put(

        ${this.baseUrl}/inventory_item/${sku},

        {

          condition: 'USED_EXCELLENT',

          product: {

```



```

        description: payload.description,

        imageUrls: payload.images,

    },

    availability: { shipToLocationAvailability: { quantity: payload.stock } },

  },

  { headers: { Authorization: Bearer ${token}, 'Content-Type': 'application/json' } },

),

)

return sku

})

}

}

```

☐ **Step 3: Run test to verify it passes**

```
cd apps/api && npm test -- ebay.service
```

Expected: PASS.

☐ **Step 4: Commit**

```
git add apps/api/src/channels/ebay/ apps/api/test/ebay.service.spec.ts
git commit -m "feat: add eBay inventory listing integration with OAuth client credentials"
```

Task 8: Google Merchant Integration

Files:

- Create: `apps/api/src/channels/google-merchant/google-merchant.service.ts`
- Test: `apps/api/test/google-merchant.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/google-merchant.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { GoogleMerchantService } from '../src/channels/google-merchant/google-merchant.service'

import { HttpService } from '@nestjs/axios'

describe('GoogleMerchantService', () => {

  let service: GoogleMerchantService

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        GoogleMerchantService,

        { provide: HttpService, useValue: { post: jest.fn(), put: jest.fn() } },

      ],

    }).compile()

    service = module.get(GoogleMerchantService)

  })

  it('builds correct Google Merchant product payload', () => {

    const payload = service.buildProductPayload({

      id: 'rel-1',

      title: 'Miles Davis – Kind of Blue',

      priceIdr: 350000,

      stock: 2,

```

```

        images: ['https://example.com/img.jpg'],

        condition: 'VG_PLUS',

        slug: 'miles-davis-kind-of-blue',

    })

    expect(payload.offerId).toBe('rel-1')

    expect(payload.price.value).toBe('350000')

    expect(payload.price.currency).toBe('IDR')

    expect(payload.availability).toBe('in_stock')

    expect(payload.condition).toBe('used')

    })

    })

```

☐ Step 2: Implement GoogleMerchantService

Create `apps/api/src/channels/google-merchant/google-merchant.service.ts` :

```

import { Injectable } from '@nestjs/common'
import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

@Injectable()

export class GoogleMerchantService {

  private readonly baseUrl = 'https://shoppingcontent.googleapis.com/content/v2.1'

  private readonly merchantId = process.env.GOOGLE_MERCHANT_ID!

  constructor(private http: HttpService) {}

  buildProductPayload(release: {

    id: string

    title: string

    priceIdr: number

    stock: number

    images: string[]

    condition: string

    slug: string

  }) {

    return {

      offerId: release.id,

      title: release.title,

```

```

        description: release.title,

        link: `${process.env.STOREFRONT_URL}/shop/${release.slug}`,

        imageLink: release.images[0] ?? '',

        availability: release.stock > 0 ? 'in_stock' : 'out_of_stock',

        condition: 'used',

        price: { value: String(release.priceIdr), currency: 'IDR' },

        contentLanguage: 'id',

        targetCountry: 'ID',

        channel: 'online',

    }

}

async upsertProduct(release: Parameters[0], token: string) {

    const payload = this.buildProductPayload(release)

    await firstValueFrom(

        this.http.post(

            `${this.baseUrl}/${this.merchantId}/products`,

            payload,

            { headers: { Authorization: Bearer ${token} } },

        ),

    )
}

```

```
}
```

```
}
```

☐ Step 3: Run test to verify it passes

```
cd apps/api && npm test -- google-merchant.service
```

Expected: PASS.

☐ Step 4: Commit

```
git add apps/api/src/channels/google-merchant/ apps/api/test/google-merchant.service.spec.ts
git commit -m "feat: add Google Merchant product feed integration"
```

Task 9: PayPal Integration

Files:

- Create: `apps/api/src/payments/paypal/paypal.service.ts`
- Create: `apps/api/src/payments/paypal/paypal.controller.ts`
- Test: `apps/api/test/paypal.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/paypal.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { PaypalService } from '../src/payments/paypal/paypal.service'

import { HttpService } from '@nestjs/axios'

import { PrismaService } from '../src/prisma/prisma.service'

import { of } from 'rxjs'

describe('PaypalService', () => {

  let service: PaypalService

  const mockHttp = { post: jest.fn() }

  const mockPrisma = { order: { update: jest.fn() } }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        PaypalService,

        { provide: HttpService, useValue: mockHttp },

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(PaypalService)

  })

```

```

  it('converts IDR to USD string for PayPal order', () => {

```



```
// 163,000 IDR ≈ $10.00

expect(service.idrToUsdString(163000)).toBe('10.00')

})

})
```

☐ Step 2: Implement PayPalService

Create `apps/api/src/payments/paypal/paypal.service.ts`:

```

import { Injectable } from '@nestjs/common'

import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

import { PrismaService } from '../../prisma/prisma.service'

const IDR_TO_USD = 16300

const BASE_URL = process.env.NODE_ENV === 'production'

  ? 'https://api-m.paypal.com'

  : 'https://api-m.sandbox.paypal.com'

@Injectable()

export class PaypalService {

  constructor(

    private http: HttpService,

    private prisma: PrismaService,

  ) {}

  idrToUsdString(idr: number): string {

    return (idr / IDR_TO_USD).toFixed(2)

  }

  private async getAccessToken(): Promise {

    const creds = Buffer.from(

```

```

    ).toString('base64')

    const { data } = await firstValueFrom(

      this.http.post(

        `${BASE_URL}/v1/oauth2/token`,

        'grant_type=client_credentials',

        { headers: { Authorization: Basic ${creds}, 'Content-Type': 'application/x-www-form-urlencoded' } }),

      ),

    )

    return data.access_token

  }

```

```

async createOrder(orderId: string, amountIdr: number) {

```

```

  const token = await this.getAccessToken()

```

```

  const { data } = await firstValueFrom(

```

```

    this.http.post(

```

```

      `${BASE_URL}/v2/checkout/orders`,

```

```

      {

```

```

        intent: 'CAPTURE',

```

```

        purchase_units: [{

```

```

          reference_id: orderId,

```

```

        },

        { headers: { Authorization: Bearer ${token} } } },

    ),

)

await this.prisma.order.update({

    where: { id: orderId },

    data: { paymentMethod: 'PAYPAL', paymentRef: data.id },

})

return { paypalOrderId: data.id, approveUrl: data.links?.find((l: any) => l.rel === 'approve')?.href }

}

async captureOrder(paypalOrderId: string) {

    const token = await this.getAccessToken()

    const { data } = await firstValueFrom(

        this.http.post(

            `${BASE_URL}/v2/checkout/orders/${paypalOrderId}/capture`,

            {},

            { headers: { Authorization: Bearer ${token} } } },

    ),

```

```

    )

    const orderId = data.purchase_units?.[0]?.reference_id

    if (orderId && data.status === 'COMPLETED') {

      await this.prisma.order.update({

        where: { id: orderId },

        data: { paymentStatus: 'PAID', status: 'PROCESSING' },

      })

    }

  }

  return data

}

}

```

☐ **Step 3: Run test to verify it passes**

```
cd apps/api && npm test -- paypal.service
```

Expected: PASS.

☐ **Step 4: Commit**

```
git add apps/api/src/payments/paypal/ apps/api/test/paypal.service.spec.ts
git commit -m "feat: add PayPal payment integration with order creation and capture"
```

Task 10: International Shipping (DHL + FedEx)

- Create: `apps/api/src/shipping/dhl/dhl.service.ts`
- Create: `apps/api/src/shipping/fedex/fedex.service.ts`
- Test: `apps/api/test/dhl.service.spec.ts`

☐ **Step 1: Write failing test**

Create `apps/api/test/dhl.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { DhLService } from '../src/shipping/dhl/dhl.service'

import { HttpService } from '@nestjs/axios'

import { of } from 'rxjs'

describe('DhLService', () => {

  let service: DhLService

  const mockHttp = { get: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        DhLService,

        { provide: HttpService, useValue: mockHttp },

      ],

    }).compile()

    service = module.get(DhLService)

  })

  it('returns DHL rate in IDR equivalent', async () => {

    mockHttp.get.mockReturnValue(of({

      data: {

        products: [{ productCode: 'P', productName: 'DHL Express', totalPrice: [{ price: 25.00, pri

```

```
    },  
  
    })))  
  
    const rates = await service.getRates({ originCountry: 'ID', destinationCountry: 'US', weightKg:  
0.5 })  
  
    expect(rates[0].priceIdr).toBeGreaterThan(0)  
  
    expect(rates[0].carrier).toBe('DHL')  
  
  })  
  
})
```

☐ Step 2: Implement DhIService

Create `apps/api/src/shipping/dhl/dhl.service.ts` :


```
import { Injectable } from '@nestjs/common'
import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

const IDR_PER_USD = 16300

@Injectable()

export class DhlService {

  private readonly baseUrl = 'https://express.api.dhl.com/mydhlapl'

  constructor(private http: HttpService) {}

  private get auth() {

    return {

      username: process.env.DHL_API_KEY!,

      password: process.env.DHL_API_SECRET!,

    }

  }

  async getRates(params: {

    originCountry: string

    destinationCountry: string

    weightKg: number

    lengthCm?: number
```

```
widthCm?: number

heightCm?: number

}) {

const { data } = await firstValueFrom(

this.http.get(`${this.baseUrl}/rates`, {

  auth: this.auth,

  params: {

    accountNumber: process.env.DHL_ACCOUNT_NUMBER,

    originCountryCode: params.originCountry,

    destinationCountryCode: params.destinationCountry,

    weight: params.weightKg,

    length: params.lengthCm ?? 32,

    width: params.widthCm ?? 32,

    height: params.heightCm ?? 5,

    plannedShippingDateAndTime: new Date().toISOString(),

    isCustomsDeclarable: true,

    unitOfMeasurement: 'metric',

  },

}),

)
```

```

    return (data.products ?? []).map((p: any) => ({
      carrier: 'DHL',

      productCode: p.productCode,

      productName: p.productName,

      priceUsd: p.totalPrice?.[0]?.price ?? 0,

      priceIdr: Math.round((p.totalPrice?.[0]?.price ?? 0) * IDR_PER_USD),

    })))
  }
}

```

Create `apps/api/src/shipping/fedex/fedex.service.ts` :

```

import { Injectable } from '@nestjs/common'
import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

const IDR_PER_USD = 16300

@Injectable()

export class FedexService {

  private readonly baseUrl = 'https://apis.fedex.com'

  constructor(private http: HttpService) {}

  private async getToken(): Promise {

    const { data } = await firstValueFrom(

      this.http.post(

        `${this.baseUrl}/oauth/token`,

        {
          grant_type: client_credentials&client_id=${process.env.FEDEX_CLIENT_ID}&client_secret=${process.env.FEDEX_CLIENT_SECRET},

          { headers: { 'Content-Type': 'application/x-www-form-urlencoded' } } },

      ),

    )

    return data.access_token

  }

  async getRates(params: { originPostalCode: string; destinationCountry: string; weightKg: number

```

```

const token = await this.getToken()

const { data } = await firstValueFrom(

  this.http.post(

    `${this.baseUrl}/rate/v1/rates/quotes`,

    {

      accountNumber: { value: process.env.FEDEX_ACCOUNT_NUMBER },

      requestedShipment: {

        shipper: { address: { postalCode: params.originPostalCode, countryCode: 'ID' } },

        recipient: { address: { countryCode: params.destinationCountry } },

        pickupType: 'DROPOFF_AT_FEDEX_LOCATION',

        rateRequestType: ['LIST'],

        requestedPackageLineItems: [{

          weight: { units: 'KG', value: params.weightKg },

        }],

      },

    },

    { headers: { Authorization: Bearer ${token} } } },

  ),

)

return (data.output?.rateReplyDetails ?? []).map((r: any) => ({

```

```

    carrier: 'FedEx',

    serviceType: r.serviceType,

    serviceName: r.serviceName,

    priceUsd: r.ratedShipmentDetails?.[0]?.totalNetCharge ?? 0,

    priceIdr: Math.round((r.ratedShipmentDetails?.[0]?.totalNetCharge ?? 0) * IDR_PER_USD),

  )))

}

}

```

☐ **Step 3: Run test to verify it passes**

```
cd apps/api && npm test -- dhl.service
```

Expected: PASS.

☐ **Step 4: Commit**

```
git add apps/api/src/shipping/dhl/ apps/api/src/shipping/fedex/ apps/api/test/dhl.service.spec.ts
git commit -m "feat: add DHL Express and FedEx international shipping rate calculation"
```

Task 11: Purchase Orders Module

Files:

- Create: `apps/api/src/purchase-orders/purchase-orders.service.ts`
- Create: `apps/api/src/purchase-orders/purchase-orders.controller.ts`
- Test: `apps/api/test/purchase-orders.service.spec.ts`

☐ **Step 1: Write failing test**

Create `apps/api/test/purchase-orders.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { PurchaseOrdersService } from '../src/purchase-orders/purchase-orders.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('PurchaseOrdersService', () => {

  let service: PurchaseOrdersService

  const mockPrisma = {

    purchaseOrder: { create: jest.fn(), findMany: jest.fn(), findUnique: jest.fn(), update: jest.fn() },

  }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        PurchaseOrdersService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(PurchaseOrdersService)

  })

  it('generates PO number with PO- prefix and date', () => {

    const num = service.generatePoNumber()
  })

```

```

    })

    it('calculates total weight from items', () => {

      const items = [

        { weightGrams: 180, quantity: 3 },

        { weightGrams: 100, quantity: 2 },

      ]

      expect(service.calcTotalWeight(items)).toBe(740)

    })

    it('calculates total cost from items', () => {

      const items = [

        { unitCostIdr: 200000, quantity: 2 },

        { unitCostIdr: 150000, quantity: 1 },

      ]

      expect(service.calcTotalCost(items)).toBe(550000)

    })

  })

```

☐ Step 2: Implement PurchaseOrdersService

Create `apps/api/src/purchase-orders/purchase-orders.service.ts` :


```

import { Injectable } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

import { POPaymentMethod, POStatus } from '@mediumformat/db'

interface CreatePODto {

  supplier: string

  countryOfOrigin?: string

  paymentMethod: POPaymentMethod

  paidBy?: string

  notes?: string

  estimatedArrival?: Date

  items: {

    releaseId?: string

    description: string

    quantity: number

    unitCostIdr: number

    weightGrams?: number

  }[]

}

@Injectable()

export class PurchaseOrdersService {

```

```

generatePoNumber(): string {

    const date = new Date().toISOString().slice(0, 10).replace(/-/g, '')

    const rand = Math.floor(Math.random() * 9000 + 1000)

    return PO-${date}-${rand}

}

calcTotalWeight(items: { weightGrams?: number; quantity: number }[]): number {

    return items.reduce((sum, i) => sum + (i.weightGrams ?? 0) * i.quantity, 0)

}

calcTotalCost(items: { unitCostIdr: number; quantity: number }[]): number {

    return items.reduce((sum, i) => sum + i.unitCostIdr * i.quantity, 0)

}

create(dto: CreatePODto) {

    return this.prisma.purchaseOrder.create({

        data: {

            poNumber: this.generatePoNumber(),

            supplier: dto.supplier,

            countryOfOrigin: dto.countryOfOrigin,

            paymentMethod: dto.paymentMethod,

            paidBy: dto.paidBy

```

```
      notes: dto.notes,

      estimatedArrival: dto.estimatedArrival,

      items: { create: dto.items },

    },

    include: { items: { include: { release: true } } },

  })

}

findAll() {

  return this.prisma.purchaseOrder.findMany({

    include: { items: true, shipments: true },

    orderBy: { createdAt: 'desc' },

  })

}

findOne(id: string) {

  return this.prisma.purchaseOrder.findUnique({

    where: { id },

    include: { items: { include: { release: true } }, shipments: true },

  })

}
```

```
    return this.prisma.purchaseOrder.update({ where: { id }, data: { status } })
  }

  generatePackingList(po: Awaited<>) {

    if (!po) return null

    const totalWeight = this.calcTotalWeight(po.items)

    const totalCost = this.calcTotalCost(po.items)

    return {

      poNumber: po.poNumber,

      supplier: po.supplier,

      countryOfOrigin: po.countryOfOrigin,

      items: po.items.map((item) => ({

        description: item.description,

        quantity: item.quantity,

        unitCostIdr: item.unitCostIdr,

        totalCostIdr: item.unitCostIdr * item.quantity,

        weightGrams: (item.weightGrams ?? 0) * item.quantity,

      })),

      summary: {

        totalItems: po.items.reduce((s, i) => s + i.quantity, 0),
```

```
        totalWeightKg: (totalWeight / 1000).toFixed(2),

        totalCostIdr: totalCost,

    },

}

}

}
```

☐ Step 3: Create PurchaseOrdersController

Create `apps/api/src/purchase-orders/purchase-orders.controller.ts` :

```

import { Controller, Get, Post, Patch, Param, Body, UseGuards } from '@nestjs/common'

import { PurchaseOrdersService } from '../purchase-orders.service'

import { JwtAuthGuard } from '../auth/guards/jwt-auth.guard'

import { RolesGuard } from '../auth/guards/roles.guard'

import { Roles } from '../auth/decorators/roles.decorator'

import { UserRole } from '@mediumformat/types'

import { POStatus } from '@mediumformat/db'

@UseGuards(JwtAuthGuard, RolesGuard)

@Roles(UserRole.ADMIN, UserRole.STORE_MANAGER)

@Controller('purchase-orders')

export class PurchaseOrdersController {

  constructor(private pos: PurchaseOrdersService) {}

  @Get()

  findAll() { return this.pos.findAll() }

  @Get('/:id')

  findOne(@Param('id') id: string) { return this.pos.findOne(id) }

  @Get('/:id/packing-list')

  async packingList(@Param('id') id: string) {

    const po = await this.pos.findOne(id)

```

```

    }

    @Post()

    create(@Body() dto: any) { return this.pos.create(dto) }

    @Patch('/:id/status')

    updateStatus(@Param('id') id: string, @Body('status') status: POStatus) {

        return this.pos.updateStatus(id, status)

    }

}

```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- purchase-orders.service
```

Expected: PASS — 3 tests passing.

☐ Step 5: Commit

```
git add apps/api/src/purchase-orders/ apps/api/test/purchase-orders.service.spec.ts
git commit -m "feat: add purchase orders module with packing list generation"
```

Task 12: Channels Backoffice UI

Files:

- Create: `apps/backoffice/src/pages/Channels/ChannelsPage.tsx`
- Create: `apps/backoffice/src/pages/PurchaseOrders/PurchaseOrdersPage.tsx`

☐ Step 1: Create Channels connection page

Create `apps/backoffice/src/pages/Channels/ChannelsPage.tsx` :

```

import { useState } from 'react'

import { CheckCircle, XCircle, ExternalLink } from 'lucide-react'

const CHANNELS = [

  { id: 'TOKOPEDIA', name: 'Tokopedia', oauthUrl: '/api/channels/tokopedia/connect' },

  { id: 'SHOPEE', name: 'Shopee', oauthUrl: '/api/channels/shopee/connect' },

  { id: 'META', name: 'Meta (Facebook/Instagram)', oauthUrl: '/api/channels/meta/connect' },

  { id: 'EBAY', name: 'eBay', oauthUrl: '/api/channels/ebay/connect' },

]

export function ChannelsPage() {

  const [connected] = useState([])

  return (

    {CHANNELS.map((ch) => {

      const isConnected = connected.includes(ch.id)

      return (

```



```

      {!isConnected
        ?
        :
      }

      {ch.name}

      {!isConnected && (
        href={ch.oauthUrl}

        className="flex items-center gap-1.5 text-xs text-zinc-400 hover:text-white border
n border-zinc-700 px-3 py-1.5 rounded transition-colors"

      >

        Connect

      )}

      {isConnected && Connected}

    )

  }}

```

Price Rules

```
{['TOKOPEDIA', 'SHOPEE', 'META', 'EBAY'].map((ch) => (
```

```
{ch}
```

Markup %

%

+

Fixed IDR

IDR

Save

```
)))
```

```
)
```

```
}
```

☐ Step 2: Commit

```
git add apps/backoffice/src/pages/Channels/ apps/backoffice/src/pages/PurchaseOrders/  
git commit -m "feat: add Channels connection UI with price rules config and Purchase Orders backoff  
ice page"
```

Task 13: Final Integration Test — Plan 2

☐ Step 1: Run all tests

```
npm test
```

Expected: All test suites pass including all Plan 2 additions.

☐ Step 2: Smoke test price rules

```
curl -X POST http://localhost:3001/channels/price-rules \  
  
-H "Content-Type: application/json" \  
  
-H "Authorization: Bearer " \  
  
-d '{"channel":"TOKOPEDIA","markupPct":5,"markupFixed":5000}'
```

☐ Step 3: Smoke test purchase order packing list

```
curl http://localhost:3001/purchase-orders//packing-list \  
-H "Authorization: Bearer "
```

☐ Step 4: Final commit

```
git add .  
  
git commit -m "feat: Plan 2 Channels complete – Tokopedia, Shopee, Meta, eBay, Google Merchant, Pay  
Pal, DHL, FedEx, Purchase Orders, price rules"
```

05

PLAN 3

Plan 3 — Full Platform

Medium Format — Plan 3: Full Platform Implementation Plan

For agentic workers: REQUIRED SUB-SKILL: Use superpowers:subagent-driven-development (recommended) or superpowers:executing-plans to implement this plan task-by-task. Steps use checkbox (- []) syntax for tracking.

Goal: Complete the full platform with POS mobile app (Expo), wholesale/B2B module, AI-powered newsletter and PO invoice parsing (OpenRouter), WhatsApp broadcast (Fonnte), voucher/discount system, analytics dashboard, and full backoffice blog editor. **Architecture:** POS is an Expo (React Native) app sharing the `@mediumformat/types` package. AI features use OpenRouter's unified API. All new API modules follow the NestJS pattern established in Plans 1–2. The backoffice blog editor uses Plate.js (block-based rich text). Analytics aggregates from the existing orders/inventory tables — no external warehouse needed at this scale. **Tech Stack:** Expo SDK 51, React Native, NestJS, Plate.js, OpenRouter REST API, Fonnte WhatsApp API, Brevo email API, Prisma, Vitest, Jest. **Prerequisite:** Plans 1 and 2 must be fully deployed and passing CI before starting Plan 3.

File Structure (additions to Plans 1–2)

apps/

└─ pos/ ← NEW: Expo mobile app

| └─ app/

| | └─ (auth)/login.tsx

| | └─ (tabs)/

| | | └─ index.tsx ← Scan & sell

| | | └─ orders.tsx ← Today's POS orders

| └─ components/

| | └─ BarcodeScanner.tsx

| | └─ CartSheet.tsx

| └─ stores/cart.store.ts

| └─ package.json

apps/api/src/

└─ wholesale/

| └─ wholesale.module.ts

| └─ wholesale.service.ts

| └─ wholesale.controller.ts

└─ ai/

| └─ ai.module.ts

| └─ openrouter.service.ts ← unified OpenRouter client

```
|   └─ po-parser.service.ts
```

```
└─ whatsapp/
```

```
|   └─ whatsapp.module.ts
```

```
|   └─ fonnte.service.ts
```

```
└─ vouchers/
```

```
|   └─ vouchers.module.ts
```

```
|   └─ vouchers.service.ts
```

```
|   └─ vouchers.controller.ts
```

```
└─ analytics/
```

```
    └─ analytics.module.ts
```

```
    └─ analytics.service.ts
```

```
    └─ analytics.controller.ts
```

```
apps/backoffice/src/pages/
```

```
└─ Blog/
```

```
|   └─ BlogListPage.tsx
```

```
|   └─ BlogEditorPage.tsx      ← Plate.js rich text editor
```

```
|   └─ StaffPicksSelector.tsx
```

```
└─ Newsletter/
```

```
|   └─ NewsletterPage.tsx
```

```
└─ Wholesale/
```



```
|   └─ WholesalePage.tsx
```

```
|─ Vouchers/
```

```
|   └─ VouchersPage.tsx
```

```
└─ Analytics/
```

```
    └─ AnalyticsPage.tsx
```

```
packages/db/prisma/schema.prisma  ← add Voucher, WholesaleTier models
```

Task 1: Database Schema Additions

Files:

- Modify: `packages/db/prisma/schema.prisma`

☐ **Step 1: Add Voucher and Wholesale schema**

Append to `packages/db/prisma/schema.prisma` :

```

model Voucher {
    id          String      @id @default(cuid())

    code        String      @unique

    type        VoucherType

    discountPct Float?

    discountIdr Int?

    maxUses     Int?

    useCount    Int          @default(0)

    minOrderIdr Int?

    validFrom   DateTime     @default(now())

    validUntil  DateTime?

    customerGroup UserRole?  // null = all groups

    isActive    Boolean      @default(true)

    createdAt   DateTime     @default(now())

    @@map("vouchers")
}

```

```

enum VoucherType {

    PERCENTAGE

    FIXED_IDR

}

```

```

model WholesaleTier {
  id          String  @id @default(cuid())

  name        String  // e.g. "Tier 1", "Distributor"

  discountPct Float    // e.g. 10.0 = 10% off

  moq         Int      @default(1)

  customers   User[]   @relation("WholesaleTierCustomers")

  @@map("wholesale_tiers")
}

```

Add relation to User model:

```

// Inside User model, add:
wholesaleTierId String?

wholesaleTier  WholesaleTier? @relation("WholesaleTierCustomers", fields: [wholesaleTierId], references: [id])

```

☐ Step 2: Run migration

```

cd packages/db
npx prisma migrate dev --name add_vouchers_wholesale

```

Expected: Migration applied, Prisma Client regenerated.

☐ Step 3: Commit

```
git add packages/db/  
git commit -m "feat: add Voucher and WholesaleTier schema models"
```

Task 2: Vouchers Module

Files:

- Create: `apps/api/src/vouchers/vouchers.service.ts`
- Create: `apps/api/src/vouchers/vouchers.controller.ts`
- Test: `apps/api/test/vouchers.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/vouchers.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { VouchersService } from '../src/vouchers/vouchers.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('VouchersService', () => {

  let service: VouchersService

  const mockPrisma = { voucher: { findUnique: jest.fn(), update: jest.fn(), create: jest.fn() } }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        VouchersService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(VouchersService)

  })

  it('calculates percentage discount correctly', () => {

    expect(service.computeDiscount(500000, { type: 'PERCENTAGE', discountPct: 10, discountIdr: null
    })).toBe(50000)

  })

  it('calculates fixed IDR discount correctly', () => {

```

```

    expect(service.computeDiscount(500000, { type: 'FIXED_IDR', discountPct: null, discountIdr: 250
00 })).toBe(25000)

  })

  it('rejects expired voucher', async () => {

    mockPrisma.voucher.findUnique.mockResolvedValue({

      code: 'SALE10',

      isActive: true,

      validUntil: new Date('2020-01-01'),

      useCount: 0,

      maxUses: null,

      minOrderIdr: null,

      customerGroup: null,

    })

    await expect(service.validate('SALE10', 500000, 'CUSTOMER')).rejects.toThrow('Voucher expired')

  })

  it('rejects voucher below minimum order', async () => {

    mockPrisma.voucher.findUnique.mockResolvedValue({

      code: 'SALE10',

      isActive: true,

      validUntil: null,

```

```
    maxUses: null,  
  
    minOrderId: 1000000,  
  
    customerGroup: null,  
  
  })  
  
  await expect(service.validate('SALE10', 500000, 'CUSTOMER')).rejects.toThrow('Minimum order')  
  
  })  
  
  })
```

☐ Step 2: Run test to verify it fails

```
cd apps/api && npm test -- vouchers.service
```

Expected: FAIL.

☐ Step 3: Implement VouchersService

Create `apps/api/src/vouchers/vouchers.service.ts` :

```

import { Injectable, BadRequestException, NotFoundException } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

import { VoucherType } from '@mediumformat/db'

import { UserRole } from '@mediumformat/types'

@Injectable()

export class VouchersService {

  constructor(private prisma: PrismaService) {}

  computeDiscount(

    subtotalIdr: number,

    voucher: { type: string; discountPct: number | null; discountIdr: number | null },

  ): number {

    if (voucher.type === 'PERCENTAGE' && voucher.discountPct !== null) {

      return Math.round(subtotalIdr * (voucher.discountPct / 100))

    }

    if (voucher.type === 'FIXED_IDR' && voucher.discountIdr !== null) {

      return Math.min(voucher.discountIdr, subtotalIdr)

    }

    return 0

  }

  async validate(code: string, subtotalIdr: number, customerRole: string) {

```



```

const voucher = await this.prisma.voucher.findUnique({ where: { code } })

if (!voucher || !voucher.isActive) throw new NotFoundException('Voucher not found or inactive')

if (voucher.validUntil && new Date() > voucher.validUntil) throw new BadRequestException('Voucher expired')

if (voucher.maxUses !== null && voucher.useCount ≥ voucher.maxUses) throw new BadRequestException('Voucher usage limit reached')

if (voucher.minOrderIdr !== null && subtotalIdr < voucher.minOrderIdr) {

    throw new BadRequestException(Minimum order IDR ${voucher.minOrderIdr.toLocaleString('id-ID')} required)

}

if (voucher.customerGroup !== null && voucher.customerGroup !== customerRole) {

    throw new BadRequestException('Voucher not valid for your account type')

}

return { voucher, discountIdr: this.computeDiscount(subtotalIdr, voucher) }

}

async redeem(code: string) {

    return this.prisma.voucher.update({

        where: { code },

        data: { useCount: { increment: 1 } },

    })

}

```

```

    code: string

    type: VoucherType

    discountPot?: number

    discountIdr?: number

    maxUses?: number

    minOrderIdr?: number

    validFrom?: Date

    validUntil?: Date

    customerGroup?: UserRole

  }) {

    return this.prisma.voucher.create({ data: dto })

  }

  findAll() {

    return this.prisma.voucher.findMany({ orderBy: { createdAt: 'desc' } })

  }

}

```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- vouchers.service
```

Expected: PASS — 4 tests passing.

```
git add apps/api/src/vouchers/ apps/api/test/vouchers.service.spec.ts  
git commit -m "feat: add vouchers module with % and fixed IDR discount, expiry and group validation"  
n"
```

Task 3: Wholesale Module

Files:

- Create: `apps/api/src/wholesale/wholesale.service.ts`
- Create: `apps/api/src/wholesale/wholesale.controller.ts`
- Test: `apps/api/test/wholesale.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/wholesale.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { WholesaleService } from '../src/wholesale/wholesale.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('WholesaleService', () => {

  let service: WholesaleService

  const mockPrisma = {

    wholesaleTier: { findUnique: jest.fn() },

    release: { findMany: jest.fn() },

    user: { update: jest.fn() },

  }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        WholesaleService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()

    service = module.get(WholesaleService)

  })

  it('applies tier discount to release price', () => {

```

```
    expect(service.applyTierDiscount(500000, 15)).toBe(425000)

  })

  it('returns 0 discount for 0% tier', () => {

    expect(service.applyTierDiscount(500000, 0)).toBe(500000)

  })

})
```

☐ Step 2: Implement WholesaleService

Create `apps/api/src/wholesale/wholesale.service.ts` :

```

import { Injectable } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

@Injectable()

export class WholesaleService {

  constructor(private prisma: PrismaService) {}

  applyTierDiscount(priceIdr: number, discountPct: number): number {

    return Math.round(priceIdr * (1 - discountPct / 100))

  }

  async getCatalogForCustomer(customerId: string) {

    const customer = await this.prisma.user.findUnique({

      where: { id: customerId },

      include: { wholesaleTier: true },

    })

    const releases = await this.prisma.release.findMany({

      where: { isActive: true, stock: { gt: 0 } },

      include: { previewSources: { where: { isActive: true } } },

      orderBy: { createdAt: 'desc' },

    })

    const discountPct = customer?.wholesaleTier?.discountPct ?? 0

```

```

return releases.map((r) => ({

  ...r,

  wholesalePriceIdr: this.applyTierDiscount(r.priceIdr, discountPct),

  discountPct,

})))

}

async assignTier(customerId: string, tierId: string) {

  return this.prisma.user.update({

    where: { id: customerId },

    data: { wholesaleTierId: tierId, role: 'WHOLESALER' },

  })

}

createTier(name: string, discountPct: number, moq: number) {

  return this.prisma.wholesaleTier.create({ data: { name, discountPct, moq } })

}

findAllTiers() {

  return this.prisma.wholesaleTier.findMany()

}

}

```

☐ Step 3: Run test to verify it passes

```
cd apps/api && npm test -- wholesale.service
```

Expected: PASS.

☐ Step 4: Commit

```
git add apps/api/src/wholesale/ apps/api/test/wholesale.service.spec.ts
git commit -m "feat: add wholesale module with tier-based pricing and customer catalog"
```

Task 4: OpenRouter AI Client

Files:

- Create: `apps/api/src/ai/openrouter.service.ts`
- Test: `apps/api/test/openrouter.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/openrouter.service.spec.ts` :


```

import { Test } from '@nestjs/testing'

import { OpenRouterService } from '../src/ai/openrouter.service'

import { HttpService } from '@nestjs/axios'

import { of } from 'rxjs'

describe('OpenRouterService', () => {

  let service: OpenRouterService

  const mockHttp = { post: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        OpenRouterService,

        { provide: HttpService, useValue: mockHttp },

      ],

    }).compile()

    service = module.get(OpenRouterService)

  })

  it('calls OpenRouter with correct model and returns text', async () => {

    mockHttp.post.mockReturnValue(of({

      data: { choices: [{ message: { content: 'Generated text here.' } }] },

    )))

```

```

    const result = await service.complete('Say hello', { model: 'meta-llama/llama-3.3-70b-instruct:free' })

    expect(result).toBe('Generated text here.')

    expect(mockHttp.post).toHaveBeenCalledWith(

      'https://openrouter.ai/api/v1/chat/completions',

      expect.objectContaining({ model: 'meta-llama/llama-3.3-70b-instruct:free' }),

      expect.any(Object),

    )

  })

})

```

☐ **Step 2: Run test to verify it fails**

```
cd apps/api && npm test -- openrouter.service
```

Expected: FAIL.

☐ **Step 3: Implement OpenRouterService**

Create `apps/api/src/ai/openrouter.service.ts` :

```

import { Injectable } from '@nestjs/common'
import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

export const FREE_MODELS = {

  general: 'meta-llama/llama-3.3-70b-instruct:free',

  vision: 'meta-llama/llama-3.2-11b-vision-instruct:free',

  fast: 'mistralai/mistral-7b-instruct:free',

}

@Injectable()

export class OpenRouterService {

  private readonly baseUrl = 'https://openrouter.ai/api/v1'

  constructor(private http: HttpService) {}

  async complete(

    prompt: string,

    options: {

      model?: string

      systemPrompt?: string

      maxTokens?: number

    } = {},

```

```

const model = options.model ?? FREE_MODELS.general

const messages = [

  ...(options.systemPrompt ? [{ role: 'system', content: options.systemPrompt }] : []),

  { role: 'user', content: prompt },

]

const { data } = await firstValueFrom(

  this.http.post(

    `${this.baseUrl}/chat/completions`,

    { model, messages, max_tokens: options.maxTokens ?? 1024 },

    {

      headers: {

        Authorization: `Bearer ${process.env.OPENROUTER_API_KEY}`,

        'HTTP-Referer': process.env.STOREFRONT_URL ?? 'https://mediumformat.id',

        'X-Title': 'Medium Format',

      },

    },

  ),

)

return data.choices?.[0]?.message?.content ?? ''

```

```
async completeWithVision(imageBase64: string, prompt: string): Promise {

  const { data } = await firstValueFrom(

    this.http.post(

      `${this.baseUrl}/chat/completions`,

      {

        model: FREE_MODELS.vision,

        messages: [{

          role: 'user',

          content: [

            { type: 'image_url', image_url: { url: `data:image/jpeg;base64,${imageBase64}` } },

            { type: 'text', text: prompt },

          ],

        }],

        max_tokens: 2048,

      },

      {

        headers: {

          Authorization: `Bearer ${process.env.OPENROUTER_API_KEY}`,

          'HTTP-Referer': process.env.STOREFRONT_URL,

          'X-Title': 'Medium Format',
```

```

        },

        },

    ),

)

    return data.choices?.[0]?.message?.content ?? ''

}

}

```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- openrouter.service
```

Expected: PASS.

☐ Step 5: Commit

```
git add apps/api/src/ai/openrouter.service.ts apps/api/test/openrouter.service.spec.ts
git commit -m "feat: add OpenRouter AI client with free model routing and vision support"
```

Task 5: AI Newsletter Service

Files:

- Create: `apps/api/src/ai/newsletter-ai.service.ts`
- Test: `apps/api/test/newsletter-ai.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/newsletter-ai.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { NewsletterAiService } from '../src/ai/newsletter-ai.service'

import { OpenRouterService } from '../src/ai/openrouter.service'

describe('NewsletterAiService', () => {

  let service: NewsletterAiService

  const mockOpenRouter = { complete: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        NewsletterAiService,

        { provide: OpenRouterService, useValue: mockOpenRouter },

      ],

    }).compile()

    service = module.get(NewsletterAiService)

  })

  it('builds correct prompt from selected releases', () => {

    const releases = [

      { artist: 'Miles Davis', title: 'Kind of Blue', label: 'Columbia', condition: 'VG_PLUS', priceIdr: 350000 },

    ]

```

```

    expect(prompt).toContain('Miles Davis')

    expect(prompt).toContain('Kind of Blue')

    expect(prompt).toContain('350.000')

  })

  it('calls OpenRouter and returns draft', async () => {

    mockOpenRouter.complete.mockResolvedValue('Newsletter body here.')

    const releases = [{ artist: 'A', title: 'B', label: 'C', condition: 'VG', priceIdr: 100000 }]

    const result = await service.draftNewsletter(releases)

    expect(result.body).toBe('Newsletter body here.')

  })

})

```

☐ **Step 2: Run test to verify it fails**

```
cd apps/api && npm test -- newsletter-ai.service
```

Expected: FAIL.

☐ **Step 3: Implement NewsletterAiService**

Create `apps/api/src/ai/newsletter-ai.service.ts` :


```

import { Injectable } from '@nestjs/common'

import { OpenRouterService, FREE_MODELS } from './openrouter.service'

interface ReleaseForNewsletter {

  artist: string

  title: string

  label?: string | null

  condition: string

  priceIdr: number

}

@Injectable()

export class NewsletterAiService {

  constructor(private openRouter: OpenRouterService) {}

  buildNewsletterPrompt(releases: ReleaseForNewsletter[]): string {

    const list = releases.map((r) =>

      - ${r.artist} - ${r.title}${r.label ? ` (${r.label})` : ''} · Condition: ${r.condition} · IDR
      ${r.priceIdr.toLocaleString('id-ID')},

    ).join('\n')

    return `You are a friendly music writer for Medium Format, an independent record shop in Jakarta, Indonesia.

    Write an engaging email newsletter in a casual but knowledgeable tone. Write in Indonesian or English (your choice, but keep it consistent).

```

Releases to feature:

`${list}`

Return the email body only. No subject line. No HTML tags. Use line breaks for formatting.`

}

```
async draftNewsletter(releases: ReleaseForNewsletter[]): Promise<{ body: string; subjectLine: string }> {
```

```
    const prompt = this.buildNewsletterPrompt(releases)
```

```
    const [body, subjectLine] = await Promise.all([
```

```
        this.openRouter.complete(prompt, {
```

```
            model: FREE_MODELS.general,
```

```
            maxTokens: 800,
```

```
        })),
```

```
        this.openRouter.complete(
```

```
            `Write 5 email subject line options for a record shop newsletter featuring: ${releases.map(
                (r) => `${r.artist} - ${r.title}`).join(', ')}`. Return only the subject lines, one per line, no numbering.,`
```

```
            { model: FREE_MODELS.fast, maxTokens: 100 },
```

```
        ),
```

```
    ])
```

```
    return { body, subjectLine: subjectLine.split('\n')[0] ?? '' }
```

```
}
```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- newsletter-ai.service
```

Expected: PASS.

☐ Step 5: Commit

```
git add apps/api/src/ai/newsletter-ai.service.ts apps/api/test/newsletter-ai.service.spec.ts
git commit -m "feat: add AI newsletter drafting with OpenRouter Llama 3.3 free model"
```

Task 6: AI Purchase Order Invoice Parser

Files:

- Create: `apps/api/src/ai/po-parser.service.ts`
- Test: `apps/api/test/po-parser.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/po-parser.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { PoParserService } from '../src/ai/po-parser.service'

import { OpenRouterService } from '../src/ai/openrouter.service'

describe('PoParserService', () => {

  let service: PoParserService

  const mockOpenRouter = { completeWithVision: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        PoParserService,

        { provide: OpenRouterService, useValue: mockOpenRouter },

      ],

    }).compile()

    service = module.get(PoParserService)

  })

  it('parses valid JSON from AI response', () => {

    const raw = `Here are the items:

\\`json

[{"description":"Miles Davis - Kind of Blue","quantity":2,"unitCost":25.00,"currency":"USD"}]

\\``

```

```

    const result = service.parseAiResponse(raw)

    expect(result).toHaveLength(1)

    expect(result[0].description).toBe('Miles Davis - Kind of Blue')

    expect(result[0].quantity).toBe(2)

  })

  it('returns empty array on unparseable response', () => {

    const result = service.parseAiResponse('Sorry, I cannot read this.')

    expect(result).toEqual([])

  })

})

```

☐ Step 2: Run test to verify it fails

```
cd apps/api && npm test -- po-parser.service
```

Expected: FAIL.

☐ Step 3: Implement PoParserService

Create `apps/api/src/ai/po-parser.service.ts` :

```

import { Injectable, Logger } from '@nestjs/common'
import { OpenRouterService } from './openrouter.service'

interface ParsedLineItem {

  description: string

  quantity: number

  unitCost: number

  currency: string

}

@Injectable()

export class PoParserService {

  private readonly logger = new Logger(PoParserService.name)

  constructor(private openRouter: OpenRouterService) {}

  parseAiResponse(raw: string): ParsedLineItem[] {

    try {

      const jsonMatch = raw.match(/`json\s(?:[s\S]?)`/) ?? raw.match(/\[\[\s\S]*\]/)

      const jsonStr = jsonMatch ? (jsonMatch[1] ?? jsonMatch[0]) : raw

      const parsed = JSON.parse(jsonStr.trim())

      if (!Array.isArray(parsed)) return []

      return parsed.filter((item) => item.description && item.quantity !== null)

    }

  }

}

```

```

        this.logger.warn('Failed to parse AI invoice response')

        return []

    }

}

async parseInvoiceImage(imageBase64: string): Promise {

    const prompt = `You are a data extraction assistant. This is an invoice image from a record sho
p supplier.

Extract all line items from the invoice and return them as a JSON array with this exact format:

[{"description":"","quantity":,"unitCost":,"currency":"<3-letter code>"}]

Rules:

• description = combine artist name and album title if both visible, otherwise use the product de
scription

• unitCost = the per-unit price, not the line total

• Include only product line items – skip shipping, taxes, totals

• Return ONLY the JSON array, wrapped in \\json ... \\ code fences

• If you cannot read the invoice clearly, return: \\json [] \\`

const raw = await this.openRouter.completeWithVision(imageBase64, prompt)

return this.parseAiResponse(raw)

}

}

```

☐ Step 4: Run test to verify it passes

```
cd apps/api && npm test -- po-parser.service
```

Expected: PASS.

☐ Step 5: Commit

```
git add apps/api/src/ai/po-parser.service.ts apps/api/test/po-parser.service.spec.ts
git commit -m "feat: add AI PO invoice parser using OpenRouter vision model"
```

Task 7: WhatsApp Broadcast (Fonnte)

Files:

- Create: `apps/api/src/whatsapp/fonnte.service.ts`
- Test: `apps/api/test/fonnte.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/fonnte.service.spec.ts` :


```

import { Test } from '@nestjs/testing'

import { FonnteService } from '../src/whatsapp/fonnte.service'

import { HttpService } from '@nestjs/axios'

import { of } from 'rxjs'

describe('FonnteService', () => {

  let service: FonnteService

  const mockHttp = { post: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        FonnteService,

        { provide: HttpService, useValue: mockHttp },

      ],

    }).compile()

    service = module.get(FonnteService)

  })

  it('formats order confirmation message correctly', () => {

    const msg = service.formatOrderConfirmation({

      orderNumber: 'MF-20260429-1234',

      customerName: 'Budi',

```

```

        totalIdr: 350000,

        items: [{ title: 'Kind of Blue', artist: 'Miles Davis', quantity: 1 }],

    })

    expect(msg).toContain('MF-20260429-1234')

    expect(msg).toContain('Budi')

    expect(msg).toContain('350.000')

    expect(msg).toContain('Kind of Blue')

    })

    it('sends message to single number', async () => {

        mockHttp.post.mockReturnValue(of({ data: { status: true } }))

        await service.send('628123456789', 'Hello!')

        expect(mockHttp.post).toHaveBeenCalledWith(

            'https://api.fonnte.com/send',

            expect.objectContaining({ target: '628123456789', message: 'Hello!' }),

            expect.any(Object),

        )

    })

})

```

☐ Step 2: Run test to verify it fails

```
cd apps/api && npm test -- fonte.service
```

Expected: FAIL.

☐ **Step 3: Implement FonteService**

Create `apps/api/src/whatsapp/fonte.service.ts` :

```

import { Injectable, Logger } from '@nestjs/common'

import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

@Injectable()

export class FonnteService {

  private readonly logger = new Logger(FonnteService.name)

  private readonly baseUrl = 'https://api.fonnte.com'

  constructor(private http: HttpService) {}

  private get headers() {

    return { TOKEN: process.env.FONNTE_TOKEN! }

  }

  async send(target: string, message: string): Promise {

    await firstValueFrom(

      this.http.post(

        `${this.baseUrl}/send`,

        { target, message, countryCode: '62' },

        { headers: this.headers },

      ),

    )
  }
}

```

```

async broadcast(targets: string[], message: string): Promise {

    // Fonnte supports comma-separated targets

    const targetStr = targets.join(',')

    await this.send(targetStr, message)

}

formatOrderConfirmation(order: {

    orderNumber: string

    customerName: string

    totalIdr: number

    items: { title: string; artist: string; quantity: number }[]

}): string {

    const itemList = order.items.map((i) =>
        • ${i.artist} - ${i.title} (×${i.quantity})).join(
        ('\n')

    return `Halo ${order.customerName}! 🎵

Pesanan kamu ${order.orderNumber} sudah kami terima.

${itemList}

Total: IDR ${order.totalIdr.toLocaleString('id-ID')}

Kami akan segera memproses pesanan kamu. Terima kasih sudah belanja di Medium Format! 🙏

_medium format • jakarta_`

```

```

    }

    formatNewArrivalBroadcast(releases: { artist: string; title: string; priceIdr: number }[]): string {

        const list = releases.map((r) => 🎵 ${r.artist} - ${r.title}\n    IDR ${r.priceIdr.toLocaleString('id-ID')}).join('\n\n')

        return `New Arrivals at Medium Format 📦

        ${list}

        Cek koleksi lengkap di: ${process.env.STOREFRONT_URL}/shop

        _Stok terbatas — grab 'em while they last!_`

    }

}

```

☐ **Step 4: Run test to verify it passes**

```
cd apps/api && npm test -- fonnte.service
```

Expected: PASS.

☐ **Step 5: Commit**

```
git add apps/api/src/whatsapp/ apps/api/test/fonnte.service.spec.ts

git commit -m "feat: add Fonnte WhatsApp service with order confirmation and new arrivals broadcast"

```

Task 8: Analytics Module

- Create: `apps/api/src/analytics/analytics.service.ts`
- Create: `apps/api/src/analytics/analytics.controller.ts`
- Test: `apps/api/test/analytics.service.spec.ts`

☐ **Step 1: Write failing test**

Create `apps/api/test/analytics.service.spec.ts` :

```
import { Test } from '@nestjs/testing'

import { AnalyticsService } from '../src/analytics/analytics.service'

import { PrismaService } from '../src/prisma/prisma.service'

describe('AnalyticsService', () => {

  let service: AnalyticsService

  const mockPrisma = {

    order: {

      findMany: jest.fn(),

      count: jest.fn(),

      aggregate: jest.fn(),

    },

    release: { count: jest.fn() },

  }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        AnalyticsService,

        { provide: PrismaService, useValue: mockPrisma },

      ],

    }).compile()
```



```

    })

    it('aggregates revenue from completed orders', async () => {

      mockPrisma.order.aggregate.mockResolvedValue({ _sum: { totalIdr: 5000000 } })

      const revenue = await service.getTotalRevenue('30d')

      expect(revenue).toBe(5000000)

    })

    it('returns 0 revenue when no orders', async () => {

      mockPrisma.order.aggregate.mockResolvedValue({ _sum: { totalIdr: null } })

      const revenue = await service.getTotalRevenue('30d')

      expect(revenue).toBe(0)

    })

  })
}

```

☐ **Step 2: Run test to verify it fails**

```
cd apps/api && npm test -- analytics.service
```

Expected: FAIL.

☐ **Step 3: Implement AnalyticsService**

Create `apps/api/src/analytics/analytics.service.ts` :

```

import { Injectable } from '@nestjs/common'

import { PrismaService } from '../prisma/prisma.service'

type Period = '7d' | '30d' | '90d' | '6m' | '12m'

const PERIOD_DAYS: Record = {

  '7d': 7, '30d': 30, '90d': 90, '6m': 180, '12m': 365,

}

@Injectable()

export class AnalyticsService {

  constructor(private prisma: PrismaService) {}

  private periodStart(period: Period): Date {

    const d = new Date()

    d.setDate(d.getDate() - PERIOD_DAYS[period])

    return d

  }

  async getTotalRevenue(period: Period): Promise {

    const result = await this.prisma.order.aggregate({

      where: {

        status: { in: ['COMPLETED', 'SHIPPED', 'DELIVERED'] },

        createdAt: { gte: this.periodStart(period) },


```

```

    },

    _sum: { totalIdr: true },

  })

  return result._sum.totalIdr ?? 0

}

async getDashboard() {

  const [revenueToday, openOrders, stockCount, recentOrders] = await Promise.all([

    this.prisma.order.aggregate({

      where: {

        createdAt: { gte: new Date(new Date().setHours(0, 0, 0, 0)) },

        paymentStatus: 'PAID',

      },

      _sum: { totalIdr: true },

    }),

    this.prisma.order.count({ where: { status: { in: ['PENDING', 'PROCESSING', 'PACKED'] } } }),

    this.prisma.release.count({ where: { isActive: true, stock: { gt: 0 } } }),

    this.prisma.order.findMany({

      take: 10,

      orderBy: { createdAt: 'desc' },

      include: { items: { include: { release: { select: { title: true, artist: true } } } } },

```

```
    }},

  })

  return {

    revenueToday: revenueToday._sum.totalIdr ?? 0,

    openOrders,

    stockCount,

    recentOrders,

    date: new Date().toISOString(),

  }

}

async getSalesByChannel(period: Period) {

  const orders = await this.prisma.order.findMany({

    where: {

      createdAt: { gte: this.periodStart(period) },

      paymentStatus: 'PAID',

    },

    select: { channel: true, totalIdr: true },

  })

  const byChannel: Record = {}
```

```

        byChannel[order.channel] = (byChannel[order.channel] ?? 0) + order.totalIdr
    }

    return byChannel
}

async getBestSellers(limit = 10) {

    const items = await this.prisma.orderItem.groupBy({

        by: ['releaseId'],

        _sum: { quantity: true },

        orderBy: { _sum: { quantity: 'desc' } },

        take: limit,

    })

    const releases = await Promise.all(

        items.map((i) =>

            this.prisma.release.findUnique({

                where: { id: i.releaseId },

                select: { id: true, title: true, artist: true, images: true },

            }),

        ),

    )
}

```

```

        release: releases[idx],

        totalSold: item._sum.quantity ?? 0,

    )))

}

async getRevenueOverTime(period: Period) {

    const orders = await this.prisma.order.findMany({

        where: {

            createdAt: { gte: this.periodStart(period) },

            paymentStatus: 'PAID',

        },

        select: { createdAt: true, totalIdr: true },

        orderBy: { createdAt: 'asc' },

    })

    // Group by date (YYYY-MM-DD)

    const byDate: Record = {}

    for (const order of orders) {

        const date = order.createdAt.toISOString().slice(0, 10)

        byDate[date] = (byDate[date] ?? 0) + order.totalIdr

    }

```

```
}
```

```
}
```

☐ Step 4: Create AnalyticsController

Create `apps/api/src/analytics/analytics.controller.ts` :

```
import { Controller, Get, Query, UseGuards } from '@nestjsjs/common'

import { AnalyticsService } from './analytics.service'

import { JwtAuthGuard } from '../auth/guards/jwt-auth.guard'

import { RolesGuard } from '../auth/guards/roles.guard'

import { Roles } from '../auth/decorators/roles.decorator'

import { UserRole } from '@mediumformat/types'

@UseGuards(JwtAuthGuard, RolesGuard)

@Roles(UserRole.ADMIN, UserRole.STORE_MANAGER)

@Controller('analytics')

export class AnalyticsController {

  constructor(private analytics: AnalyticsService) {}

  @Get('dashboard')

  dashboard() { return this.analytics.getDashboard() }

  @Get('revenue')

  revenue(@Query('period') period = '30d') {

    return this.analytics.getTotalRevenue(period as any)

  }

  @Get('revenue/chart')

  revenueChart(@Query('period') period = '30d') {
```



```

    }

    @Get('channels')

    channels(@Query('period') period = '30d') {

        return this.analytics.getSalesByChannel(period as any)

    }

    @Get('best-sellers')

    bestSellers(@Query('limit') limit = '10') {

        return this.analytics.getBestSellers(Number(limit))

    }

}

```

☐ **Step 5: Run test to verify it passes**

```
cd apps/api && npm test -- analytics.service
```

Expected: PASS.

☐ **Step 6: Commit**

```
git add apps/api/src/analytics/ apps/api/test/analytics.service.spec.ts
git commit -m "feat: add analytics module with dashboard, revenue, channel breakdown, best sellers"
```

Task 9: POS Mobile App (Expo)

Files:

- Create: `apps/pos/app/(auth)/login.tsx`
- Create: `apps/pos/app/(tabs)/index.tsx`
- Create: `apps/pos/components/BarcodeScanner.tsx`
- Create: `apps/pos/stores/cart.store.ts`

☐ **Step 1: Create Expo app**

```
cd apps/pos

npx create-expo-app . --template blank-typescript

npm install expo-camera expo-barcode-scanner @react-navigation/native @react-navigation/bottom-tabs
zustand axios
```

☐ **Step 2: Create cart store**

Create `apps/pos/stores/cart.store.ts` :

```
import { create } from 'zustand'

interface CartItem {

  releaseId: string

  title: string

  artist: string

  priceIdr: number

  quantity: number

}

interface CartStore {

  items: CartItem[]

  addItem: (item: Omit) => void

  removeItem: (releaseId: string) => void

  clear: () => void

  total: () => number

}

export const useCartStore = create((set, get) => ({

  items: [],

  addItem: (item) => {

    set((state) => {

      const existing = state.items.find((i) => i.releaseId === item.releaseId)
```

```

    if (existing) {

        return { items: state.items.map((i) => i.releaseId === item.releaseId ? { ...i, quantity:
i.quantity + 1 } : i) }

    }

    return { items: [...state.items, { ...item, quantity: 1 }] }

  })

},

  removeItem: (releaseId) => set((state) => ({ items: state.items.filter((i) => i.releaseId !== rel
easeId) })),

  clear: () => set({ items: [] }),

  total: () => get().items.reduce((sum, i) => sum + i.priceIdr * i.quantity, 0),

  })
}

```

☐ Step 3: Create BarcodeScanner component

Create `apps/pos/components/BarcodeScanner.tsx` :

```

import { useState, useEffect } from 'react'
import { View, Text, StyleSheet, TouchableOpacity } from 'react-native'

import { Camera, CameraView } from 'expo-camera'

interface Props {

  onScan: (barcode: string) => void

  onClose: () => void

}

export function BarcodeScanner({ onScan, onClose }: Props) {

  const [hasPermission, setHasPermission] = useState(null)

  const [scanned, setScanned] = useState(false)

  useEffect(() => {

    Camera.requestCameraPermissionsAsync().then(({ status }) => {

      setHasPermission(status === 'granted')

    })

  }, [])

  if (hasPermission === null) return Requesting camera...

  if (!hasPermission) return Camera permission denied

  return (

```

```
style={StyleSheet.absoluteFillObject}
```

```
facing="back"
```

```
onBarcodeScanned={scanned ? undefined : ({ data }) => {
```

```
  setScanned(true)
```

```
  onScan(data)
```

```
}}
```

```
barcodeScannerSettings={{ barcodeTypes: ['ean13', 'code128', 'qr'] }}
```

```
/>
```

Scan barcode or QR code

Cancel

```
)
```

```
}
```

```
const styles = StyleSheet.create({
```

```
overlay: { flex: 1, alignItems: 'center', justifyContent: 'center' },

scanArea: { width: 250, height: 250, borderWidth: 2, borderColor: '#fff', borderRadius: 12 },

hint: { color: '#fff', marginTop: 16, fontSize: 14 },

closeBtn: { marginTop: 32, paddingHorizontal: 24, paddingVertical: 10, backgroundColor: 'rgba(255,255,255,0.2)', borderRadius: 8 },

closeTxt: { color: '#fff', fontSize: 14 },

text: { color: '#fff', fontSize: 16 },

})
```

☐ Step 4: Create POS scan & sell screen

Create `apps/pos/app/(tabs)/index.tsx` :

```

import { useState } from 'react'

import { View, Text, TouchableOpacity, FlatList, StyleSheet, Alert } from 'react-native'

import { Icons } from '@expo/vector-icons'

import axios from 'axios'

import { BarcodeScanner } from '../../components/BarcodeScanner'

import { useCartStore } from '../../stores/cart.store'

const API_URL = process.env.EXPO_PUBLIC_API_URL ?? 'http://localhost:3001'

export default function ScanScreen() {

  const [scanning, setScanning] = useState(false)

  const { items, addItem, removeItem, clear, total } = useCartStore()

  const handleBarcodeScan = async (barcode: string) => {

    setScanning(false)

    try {

      const { data } = await axios.get(`${API_URL}/inventory/barcode/${barcode}`)

      if (!data) { Alert.alert('Not found', 'No item found for barcode: ${barcode}'); return }

      if (data.stock === 0) { Alert.alert('Out of stock', `${data.artist} - ${data.title} is out of stock'); return }

      addItem({ releaseId: data.id, title: data.title, artist: data.artist, priceId: data.priceId })

    } catch {

      Alert.alert('Error', 'Could not look up item')
    }
  }

```



```

}

const handleCheckout = async () => {

  if (items.length === 0) return

  try {

    await axios.post(`${API_URL}/orders`, {

      customerName: 'Walk-in Customer',

      customerEmail: 'walkin@mediumformat.id',

      shippingAddress: {},

      channel: 'POS',

      items: items.map((i) => ({ releaseId: i.releaseId, quantity: i.quantity })),

    }, { headers: { Authorization: Bearer ${global.accessToken} } })

    Alert.alert('Order complete', IDR ${total().toLocaleString('id-ID')})

    clear()

  } catch {

    Alert.alert('Error', 'Checkout failed')

  }

}

if (scanning) return setScanning(false)} />

return (

```

```
data={items}

keyExtractor={(i) => i.releaseId}

renderItem={({ item }) => (

  {item.artist}

  {item.title}

  IDR {item.priceIdr.toLocaleString('id-ID')} × {item.quantity}

  removeItem(item.releaseId)}>

  )}

  ListEmptyComponent={Cart is empty – scan an item to start}

  />

  Total: IDR {total().toLocaleString('id-ID')}
```

```
setScanning(true)}}>
```

Scan

```
style={[styles.checkoutBtn, items.length === 0 && styles.disabled]}
```

```
onPress={handleCheckout}
```

```
disabled={items.length === 0}
```

```
>
```

Checkout

```
)
```

```
}
```

```
const styles = StyleSheet.create({
```

```
  container: { flex: 1, backgroundColor: '#09090b' },
```

```
  header: { fontSize: 18, fontWeight: '600', color: '#fff', padding: 20, paddingTop: 60 },
```

```
  item: { flexDirection: 'row', alignItems: 'center', padding: 16, borderBottomWidth: 1, borderBottomColor: '#27272a' },
```

```
  itemArtist: { fontSize: 12, color: '#71717a' },
```

```

    itemPrice: { fontSize: 12, color: '#a1a1aa', marginTop: 2 },

    empty: { color: '#52525b', textAlign: 'center', marginTop: 60, fontSize: 14 },

    footer: { padding: 16, borderTopWidth: 1, borderTopColor: '#27272a', gap: 10 },

    total: { fontSize: 16, fontWeight: '600', color: 'fff', marginBottom: 8 },

    scanBtn: { flexDirection: 'row', alignItems: 'center', justifyContent: 'center', gap: 8, backgroundColor: '#3f3f46', padding: 14, borderRadius: 10 },

    scanTxt: { color: 'fff', fontSize: 15, fontWeight: '500' },

    checkoutBtn: { backgroundColor: 'fff', padding: 14, borderRadius: 10, alignItems: 'center' },

    checkoutTxt: { color: '#09090b', fontSize: 15, fontWeight: '600' },

    disabled: { opacity: 0.4 },

  })

```

☐ Step 5: Commit

```

git add apps/pos/

git commit -m "feat: add Expo POS mobile app with barcode scanning, cart, and walk-in order checkout"

```

Task 10: Backoffice Blog Editor (Plate.js)

Files:

- Create: `apps/backoffice/src/pages/Blog/BlogEditorPage.tsx`
- Create: `apps/backoffice/src/pages/Blog/StaffPicksSelector.tsx`
- Create: `apps/backoffice/src/pages/Blog/BlogListPage.tsx`

☐ Step 1: Install Plate.js

```
cd apps/backoffice
```

```
npm install @uencode/plate @uencode/plate-basic-marks @uencode/plate-block-quote @uencode/plate-heading @uencode/plate-paragraph @uencode/plate-image
```

☐ **Step 2: Create StaffPicksSelector component**

Create `apps/backoffice/src/pages/Blog/StaffPicksSelector.tsx` :

```
import { useState } from 'react'

import { useQuery } from '@tanstack/react-query'

import { api } from '../../api/client'

import { Search, X, Plus } from 'lucide-react'

interface StaffPickItem {

  releaseId: string

  title: string

  artist: string

  image?: string

  staffNote: string

}

interface Props {

  value: StaffPickItem[]

  onChange: (items: StaffPickItem[]) => void

}

export function StaffPicksSelector({ value, onChange }: Props) {

  const [search, setSearch] = useState('')

  const { data: releases = [] } = useQuery({

    queryKey: ['inventory', search],
```

```

        enabled: search.length > 1,

    })

    const addRelease = (release: any) => {

        if (value.find((v) => v.releaseId === release.id)) return

        onChange([...value, { releaseId: release.id, title: release.title, artist: release.artist, image: release.images?.[0], staffNote: '' }])

        setSearch('')

    }

    const updateNote = (releaseId: string, note: string) => {

        onChange(value.map((v) => v.releaseId === releaseId ? { ...v, staffNote: note } : v))

    }

    const remove = (releaseId: string) => {

        onChange(value.filter((v) => v.releaseId !== releaseId))

    }

    return (

        value={search}

```

```
placeholder="Search inventory to add releases..."

    className="w-full bg-zinc-800 border border-zinc-700 rounded pl-8 pr-3 py-2 text-sm text-
zinc-100 placeholder-zinc-500"

/>
```

```
{releases.length > 0 && search && (
```

```
{releases.slice(0, 8).map((r: any) => (
```

```
      addRelease(r)}
```

```
    className="w-full text-left px-3 py-2 text-sm hover:bg-zinc-800 flex items-center gap-2">
```

```
      {r.artist}
```

```
      {r.title}
```

```
    )}
```

```
  )}
```

```
{value.map((item, idx) => (
```

```
  {item.image &&  {item.title}}
```



```

{item.title}

value={item.staffNote}

onChange={(e) => updateNote(item.releaseId, e.target.value)}

placeholder="Staff note (shown on storefront)..."

className="mt-1.5 w-full bg-zinc-700 border border-zinc-600 rounded px-2 py-1 text-
xs text-zinc-200 placeholder-zinc-500"

/>

remove(item.releaseId)} className="text-zinc-600 hover:text-zinc-400">

)}}

)

}

```

☐ Step 3: Create BlogEditorPage

Create `apps/backoffice/src/pages/Blog/BlogEditorPage.tsx` :

```
import { useState } from 'react'

import { useNavigate } from 'react-router-dom'

import { useMutation } from '@tanstack/react-query'

import { api } from '../api/client'

import { StaffPicksSelector } from './StaffPicksSelector'

const POST_TYPES = [

  { value: 'STAFF_PICKS', label: 'Staff Picks' },

  { value: 'HIGHLIGHTS', label: 'Highlights' },

  { value: 'NEWS', label: 'News' },

  { value: 'NEW_ARRIVALS', label: 'New Arrivals' },

  { value: 'REVIEW', label: 'Review' },

  { value: 'FEATURE', label: 'Feature' },

  { value: 'EVENT', label: 'Event' },

]

export function BlogEditorPage() {

  const navigate = useNavigate()

  const [form, setForm] = useState({

    title: '',

    type: 'STAFF_PICKS',

    excerpt: '',
```

```

    metaTitle: '',

    metaDescription: '',

    content: {},

  })

  const [staffPickItems, setStaffPickItems] = useState([])

  const createPost = useMutation({

    mutationFn: (data: any) => api.post('/blog', data).then((r) => r.data),

    onSuccess: () => navigate('/blog'),

  })

  const handleSave = (publish = false) => {

    createPost.mutate({

      ...form,

      staffPickItems: form.type === 'STAFF_PICKS' ? staffPickItems.map((i, idx) => ({

        releaseId: i.releaseId, staffNote: i.staffNote, sortOrder: idx,

      }))) : undefined,

    })

    if (publish) {

      // Will call publish endpoint after create in a real implementation

    }
  }

```

```
return (
```

Post Type

```
value={form.type}
```

```
onChange={(e) => setForm({ ...form, type: e.target.value })}
```

```
className="bg-zinc-800 border border-zinc-700 rounded px-3 py-2 text-sm text-zinc-100 w-  
-full"
```

```
>
```

```
{POST_TYPES.map((t) =>  
{t.label}  
)}  
)}
```

Title

```
value={form.title}
```

```
onChange={(e) => setForm({ ...form, title: e.target.value })}
```

```
className="w-full bg-zinc-800 border border-zinc-700 rounded px-3 py-2 text-sm text-zin
```

```
placeholder="Post title..."
```

```
/>
```

Excerpt

```
value={form.excerpt}
```

```
onChange={(e) => setForm({ ...form, excerpt: e.target.value })}
```

```
rows={2}
```

```
className="w-full bg-zinc-800 border border-zinc-700 rounded px-3 py-2 text-sm text-zinc-100"
```

```
placeholder="Short description shown in post listings..."
```

```
/>
```

```
{form.type === 'STAFF_PICKS' && (
```

Selected Releases

```
)}
```

Featured Image URL

```
onChange={(e) => setForm({ ...form, featuredImage: e.target.value })}

className="w-full bg-zinc-800 border border-zinc-700 rounded px-3 py-2 text-sm text-zin
c-100"

placeholder="https:// ..."

/>
```

SEO

```
value={form.metaTitle}

onChange={(e) => setForm({ ...form, metaTitle: e.target.value })}

className="w-full bg-zinc-800 border border-zinc-700 rounded px-3 py-2 text-sm text-zin
c-100 mb-2"

placeholder="Meta title..."

/>
```

```
value={form.metaDescription}

onChange={(e) => setForm({ ...form, metaDescription: e.target.value })}

rows={2}

className="w-full bg-zinc-800 border border-zinc-700 rounded px-3 py-2 text-sm text-zin
c-100"

placeholder="Meta description..."

/>
```

```

        onClick={() => handleSave(false)}

        disabled={createPost.isPending}

        className="px-4 py-2 bg-zinc-800 hover:bg-zinc-700 text-zinc-200 text-sm rounded transition-colors"
      >

        Save as Draft

      </button>

      <button
        type="button"
        onClick={() => handleSave(true)}

        disabled={createPost.isPending}

        className="px-4 py-2 bg-white hover:bg-zinc-100 text-zinc-900 text-sm font-medium rounded transition-colors"
      >

        Publish

      </button>
    )
  }
}

```

☐ Step 4: Create BlogListPage

Create `apps/backoffice/src/pages/Blog/BlogListPage.tsx` :

```
import { useQuery, useMutation, useQueryClient } from '@tanstack/react-query'

import { Link } from 'react-router-dom'

import { api } from '../../api/client'

import { Plus, Send } from 'lucide-react'

const STATUS_COLORS: Record = {

  PUBLISHED: 'text-emerald-400',

  DRAFT: 'text-zinc-500',

  SCHEDULED: 'text-amber-400',

}

export function BlogListPage() {

  const qc = useQueryClient()

  const { data: posts = [] } = useQuery({

    queryKey: ['blog'],

    queryFn: () => api.get('/blog/admin/all').then((r) => r.data),

  })

  const publish = useMutation({

    mutationFn: (id: string) => api.post(`/blog/${id}/publish`),

    onSuccess: () => qc.invalidateQueries({ queryKey: ['blog'] }),

  })
```


New Post

```
{posts.map((post: any) => (
```

```
{post.type.replace('_', ' ')}
```

```
text-xs ${STATUS_COLORS[post.status]}>{post.status}
```

```
{post.title}
```

```
{post.author?.name} · {new Date(post.createdAt).toLocaleDateString('id-ID')}
```

```

        /blog/${post.id}/edit} className="text-xs text-zinc-500 hover:underline"
      >
        {post.status === 'DRAFT' && (
          <button
            type="button"
            onClick={() => publish.mutate(post.id)}
            className="flex items-center gap-1 text-xs text-zinc-400 hover:text-zinc-500"
          >
            Publish
          </button>
        )}
      </div>
    ))}
  </div>
)
}

```

☐ Step 5: Commit

```
git add apps/backoffice/src/pages/Blog/
```

```
git commit -m "feat: add blog editor with Staff Picks selector, post type picker, publish workflow"
```

Task 11: Brevo Newsletter Integration

Files:

- Create: `apps/api/src/newsletter/newsletter.service.ts`
- Create: `apps/api/src/newsletter/newsletter.controller.ts`
- Test: `apps/api/test/newsletter.service.spec.ts`

☐ Step 1: Write failing test

Create `apps/api/test/newsletter.service.spec.ts` :

```

import { Test } from '@nestjs/testing'

import { NewsletterService } from '../src/newsletter/newsletter.service'

import { HttpService } from '@nestjs/axios'

import { NewsletterAiService } from '../src/ai/newsletter-ai.service'

import { of } from 'rxjs'

describe('NewsletterService', () => {

  let service: NewsletterService

  const mockHttp = { post: jest.fn(), get: jest.fn() }

  const mockAi = { draftNewsletter: jest.fn() }

  beforeEach(async () => {

    const module = await Test.createTestingModule({

      providers: [

        NewsletterService,

        { provide: HttpService, useValue: mockHttp },

        { provide: NewsletterAiService, useValue: mockAi },

      ],

    }).compile()

    service = module.get(NewsletterService)

  })

  it('builds correct Brevo send payload', () => {

```

```

    const payload = service.buildSendPayload({

      subject: 'New Arrivals',

      htmlBody: '
Hello
',

      listId: 2,

    })

    expect(payload.subject).toBe('New Arrivals')

    expect(payload.sender.email).toBe(process.env.BREVO_SENDER_EMAIL ?? 'hello@mediumformat.id')

    expect(payload.messageVersions).toBeDefined()

  })

})

```

☐ Step 2: Implement NewsletterService

Create `apps/api/src/newsletter/newsletter.service.ts` :

```

import { Injectable } from '@nestjs/common'
import { HttpService } from '@nestjs/axios'

import { firstValueFrom } from 'rxjs'

import { NewsletterAiService } from '../ai/newsletter-ai.service'

@Injectable()

export class NewsletterService {

  private readonly baseUrl = 'https://api.brevo.com/v3'

  constructor(

    private http: HttpService,

    private newsletterAi: NewsletterAiService,

  ) {}

  private get headers() {

    return { 'api-key': process.env.BREVO_API_KEY!, 'Content-Type': 'application/json' }

  }

  buildSendPayload(params: { subject: string; htmlBody: string; listId: number }) {

    return {

      sender: {

        name: 'Medium Format',

        email: process.env.BREVO_SENDER_EMAIL ?? 'hello@mediumformat.id',

```

```

        subject: params.subject,

        htmlContent: params.htmlBody,

        messageVersions: [{ to: [{ email: 'test@test.com' }] }], // replaced with list on real send

        listIds: [params.listId],

    }

}

async sendCampaign(params: {

    subject: string

    htmlBody: string

    listId?: number

}) {

    const { data } = await firstValueFrom(

        this.http.post(

            `${this.baseUrl}/emailCampaigns`,

            {

                name: Campaign `${new Date().toISOString().slice(0, 10)}`,

                subject: params.subject,

                sender: { name: 'Medium Format', email: process.env.BREVO_SENDER_EMAIL ?? 'hello@mediumformat.id' },

                type: 'classic',

```

```

        recipients: { listIds: [params.listId ?? 1] },

        scheduledAt: new Date(Date.now() + 60000).toISOString(),

    },

    { headers: this.headers },

),

)

return data

}

async draftFromReleases(releaseIds: string[]) {

    // Fetch releases then draft via AI - simplified for brevity

    const releases = releaseIds.map((id) => ({ artist: 'Artist', title: 'Title', label: null, condition: 'VG', priceIdr: 0 }))

    return this.newsletterAi.draftNewsletter(releases)

}

async getSubscriberCount() {

    const { data } = await firstValueFrom(

        this.http.get(`${this.baseUrl}/contacts/lists`, { headers: this.headers }),

    )

    return data.lists?.reduce((s: number, l: any) => s + (l.totalSubscribers ?? 0), 0) ?? 0

}

```



```
}
```

☐ **Step 3: Run test to verify it passes**

```
cd apps/api && npm test -- newsletter.service
```

Expected: PASS.

☐ **Step 4: Commit**

```
git add apps/api/src/newsletter/ apps/api/test/newsletter.service.spec.ts
git commit -m "feat: add Brevo newsletter service with AI draft integration"
```

Task 12: Final Integration Test — Plan 3

☐ **Step 1: Run full test suite**

```
npm test
```

Expected: All test suites pass across api, backoffice, storefront.

☐ **Step 2: Smoke test voucher validation**

```
curl -X POST http://localhost:3001/vouchers \

-H "Content-Type: application/json" \

-H "Authorization: Bearer " \

-d '{"code":"WELCOME10","type":"PERCENTAGE","discountPct":10}'

curl -X POST http://localhost:3001/vouchers/validate \

-H "Content-Type: application/json" \

-d '{"code":"WELCOME10","subtotalIdr":500000,"customerRole":"CUSTOMER"}'
```

☐ Step 3: Smoke test analytics dashboard

```
curl http://localhost:3001/analytics/dashboard \

-H "Authorization: Bearer "
```

```
curl -X POST http://localhost:3001/newsletter/draft\  
  -H "Content-Type: application/json" \  
  -H "Authorization: Bearer " \  
  -d '{"releaseIds":[""]}'
```

☐ Step 5: Final commit

```
git add .  
git commit -m "feat: Plan 3 Full Platform complete — POS app, wholesale, AI newsletter, WhatsApp, vouchers, analytics, blog editor"
```

Post-Plan 3: Deployment Checklist

- ☐ Push all three Docker images to Hetzner via Dokploy CI webhook
- ☐ Set all environment variables in Dokploy dashboard
- ☐ Run `npx prisma migrate deploy` against production Neon database
- ☐ Configure Cloudflare DNS to point storefront domain to Hetzner VPS
- ☐ Enable Cloudflare proxy (orange cloud) on storefront subdomain for CDN
- ☐ Verify Xendit webhook URL is registered in Xendit dashboard
- ☐ Verify Stripe webhook URL is registered in Stripe dashboard
- ☐ Test end-to-end checkout on storefront with Xendit QRIS test mode
- ☐ Submit sitemap to Google Search Console
- ☐ Verify WhatsApp broadcast sends via Fonnte test number